SAP BTP Developer's Guide

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SAP Business Technology Platform (SAP BTP) | Cloud

Public

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SAP BTP Developer's Guide

The SAP BTP Developer's Guide is the starting point for developing a business application on SAP BTP. It contains **recommendations** and **best practices** that give you an overview of what you should consider when working on development projects on SAP BTP. It also contains links to step-by-step instructions when required.

Is This Guide for You?

If you're an **architect or a development project lead**, this guide helps you to plan the architecture of your application and the services on SAP BTP you can use to achieve your goal.

If you're a **developer**, this guide helps you to define the correct methodologies and tools for your development project.

If you're an **SAP partner**, this guide helps you to set up SAP BTP for developing and running applications and services for your customers. See Build and Run Multitenant SaaS Applications for Partners Who Are Independent Software Vendors.

Note

If you're an **administrator**, you have to set up the correct organizational structure to create an account and security model. In this case, use the SAP BTP Administrator's Guide.

Note

This guide is targeted at customers who want to run and use applications in a production environment. If you're an SAP BTP trial user, you might still find that some information in this guide is useful. Check out the following page for more details about trial accounts: Trial Accounts and Free Tier. Please note that the services available in the trial version differ from the ones in the enterprise version.

Prerequisites

If you're new to SAP BTP, read:

- SAP Business Technology Platform

  SAP BTP is a cloud platform that brings together application development, automation, data management, analytics and planning, integration, and AI capabilities into one unified environment optimized for SAP applications. It offers:

  - An intuitive, modern development environment for both, professional IT and citizen and business developers,
  - Built-in data models, integrations, workflows, application templates, and AI business services
  - Self-service data discovery, modeling, planning, and analytics for business users in a governed environment

- SAP BTP Guidance Framework

  This is the central access point for architects, developers, and administrators to build and run enterprise-grade solutions on SAP BTP. It comprises decision guides, reference architectures, methodologies, recommendations, and DevOps principles.

- Basic Platform Concepts

  Contains regions, runtimes, accounts, members, quotas, capabilities and links to in-depth explanations.

- Getting Started Checklist

Explains your responsibilities and SAP's responsibilities when it comes to application lifecycle management.

- SAP BTP Administrator's Guide

  Helps you plan and set up your landscape and your lifecycle management for running applications on SAP Business Technology Platform (SAP BTP). It's part of the SAP BTP Guidance Framework and contains recommendations for planning development projects – from setting up the correct organizational structure to creating an account and security model, to operating applications.

- SAP Discovery Center

  Helps you turn your data into business value with SAP BTP. In the SAP Discovery Center you can find end-to-end scenarios, called missions, the Service Catalog of SAP BTP, and reference architectures.

- SAP Architecture Center

  Provides solution reference architectures, helping businesses adopt SAP solutions to turn data into valuable business insights.

How to Use This Guide

Use the SAP BTP Developer's Guide to help you implement business applications on SAP BTP. This guide is part of the SAP BTP Guidance Framework and explains the building blocks for developing, delivering, and integrating business applications.

Understand the Available Technology

Understanding the development use cases, recommended services, and the continuous integration and continuous delivery (CI/CD) principles offered by SAP Business Technology Platform (BTP) is essential to streamline application development, deployment and integration within the SAP ecosystem.

See Understanding Available Technology.

Select Runtime and Programming Model

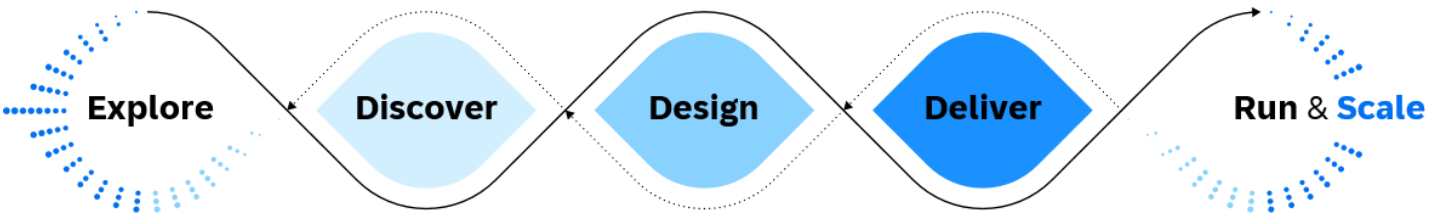
Choosing a suitable runtime and the corresponding programming model is crucial to optimize application development, deployment and management on SAP BTP. There are different strengths and benefits you can explore before deciding:

- SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP
- SAP BTP ABAP Environment with ABAP Cloud

Go Through the Development Phases

Development projects for business applications have similar characteristics. Standardized development guidance is driving developer efficiency.

Based on the experiences of successful business applications, this guide condenses best practices and technologies that can be safely recommended to you. These recommendations are structured following the design-led development phases:



The focus of the SAP BTP Developer's Guide is on the design, deliver and run and scale phases. However, you can also find information related to the explore and discover phases.

- **Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes**

  The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It supports Java (with Spring Boot), JavaScript, and TypeScript (with Node.js), which are some of the most widely adopted languages. CAP guides developers along a path of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

  - Explore

    Identify the business opportunity and set up the roles and responsibilities in your team.

  - Discover

  - Design

    - User Experience Design

      Design the user experience of your application to provide good usability.

    - Technology Design

      Use domain-driven design to create an application that is modularized and has shorter innovation cycles and a clearer focus.

  - Deliver

    - Set Up

      Before you begin developing your applications, make sure your landscape setup is appropriate for managing their lifecycles.

    - Develop

      Develop your application by following the recommendations outlined in this section about the two programming models, the numerous programming languages, the respective development tools and also about aspects such as multitenancy, security, extensibility and others.

  - Run and Scale

    Get constant feedback from your users, optimize the application based on this feedback, and operate the application using the services and capabilities SAP BTP offers.

- **Developing with ABAP Cloud in the SAP BTP ABAP Environment**

  ABAP Cloud reflects the modern way to develop ABAP. It allows you to build lifecycle-stable and cloud-ready business applications, services, and extensions.

Contribute to the SAP BTP Developer's Guide

→ Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

- **Feedback > Create issue**: Provide feedback about a documentation page. This option opens an issue on GitHub.
- **Feedback > Edit page**: Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

More information:

- Contribution Guidelines
- Introduction Video
- Introduction Blog Post

What's New for SAP BTP Developer's Guide

2026

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
SAP BTP Developer's Guide	Kyma	Terraforming Kyma Runtimes Procedure is Aligned with Terraform Conventions and Clarifying Use Cases	The entire Terraforming Kyma Runtimes procedure is now streamlined, aligning it with Terraform conventions and clarifying use cases. The updated process now distinguishes between local execution and CI/CD automation, highlights key considerations for each scenario, and includes clearer steps for seamless Kyma cluster deployment. See Terraforming Kyma Runtimes.	Info only	General Availability	Changed	Technology	Not applicable
SAP BTP Developer's Guide	ABAP	Tutorials for Partners Developing ABAP-Based Applications	Tutorials and Reference Sample for Full-Stack ABAP Cloud Multitenant SaaS Applications for Partners introduces the Music Festival Manager reference app that guides partners in building scalable multitenant SaaS solutions on SAP BTP using the ABAP RESTful Application Programming Model. The tutorials provide a structured, end-to-end	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			approach to developing, deploying, and integrating enterprise-grade applications with SAP S/4HANA Cloud. See Tutorials and Reference Sample for Full-Stack ABAP Cloud Multitenant SaaS Applications for Partners.					
SAP BTP Developer's Guide	Cloud Foundry Kyma	Establish an End-to-End Observability for Your Application	You can use an integrated monitoring across SAP BTP, combining central observability for problem detection through SAP Cloud ALM and problem resolution tasks via SAP Cloud Logging for local observability of SAP BTP application health visibility. With OpenTelemetry-based data collection, you gain consistent, real-time insights to accelerate issue resolution, strengthen reliability, and perform data-driven decisions and operations. See Establish End-to-End Observability.	Info only	General Availability	Changed	Technology	Not applicable
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	Reduce Total Cost of Ownership using SAP HANA Cloud	You can use SAP HANA Cloud, the cloud-native Database-as-a-Service that forms the database management foundation of SAP BTP, to reduce the total cost of ownership (TCO) when following a number of best practices. See Data Persistence Using USAP HANA	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			Cloud.					
SAP BTP Developer's Guide	ABAP	Update Content for ABAP Cloud in the SAP BTP ABAP Environment	The ABAP content has been enhanced with specific recommendations, and more hands-on materials. See Developing with ABAP Cloud in the SAP BTP ABAP Environment .	Info only	General Availability	Changed	Technology	Not applicable
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	UX Design and Product Compliance Standards	A new page about UX design and product compliance standards has been introduced.	Info only	General Availability	New	Technology	Not applicable
SAP BTP Developer's Guide	Cloud Foundry Kyma	SAP Partner-Related Information for CAP-Based Applications	The SAP partner information for CAP-based applications has been enhanced with total cost of ownership calculation of multitenant SaaS applications and more tutorials and sample applications. See Build and Run Multitenant SaaS Application for Partners Who Are Independent Software Vendors.	Info only	General Availability	Changed	Technology	Not applicable
SAP BTP Developer's Guide	Cloud Foundry Kyma	Security Information for CAP-Based Applications	Security information for CAP-based applications is added to the different development phases: • Explore • Discover • Design • Develop: Security Considerations for Applications • Run and Scale	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	Introduction of Strengths and Benefits and the Runtimes and the Programming Models	There is now a section explaining the strengths and benefits of SAP BTP ABAP environment with ABAP Cloud and SAP BTP, Cloud Foundry and SAP BTP, Kyma runtimes with CAP: • Strengths and Benefits of SAP BTP, Cloud Foundry and SAP BTP, Kyma runtimes with CAP • Strengths and Benefits of SAP BTP ABAP environment with ABAP Cloud	Info only	General Availability	New	Technology	Not applicable
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	Restructuring of the SAP BTP Developer's Guide	The SAP BTP Developer's Guide has been restructured to meet the needs of the technology and the user preferences. These are the main sections of the guide: • Understanding Available Technology • Strengths and Benefits of the Runtimes and Their Programming Models • Using SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP • Using SAP BTP ABAP	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			Environment with ABAP Cloud					
SAP BTP Developer's Guide	Cloud Foundry Kyma	Introduction to CI/CD	There is a new chapter that introduces you to Continuous Integration and Delivery (CI/CD), as well as SAP's solution for it. See Introducing Continuous Integration and Delivery (CI/CD).	Info only	General Availability	New	Technology	Not applicable
SAP BTP Developer's Guide	Cloud Foundry Kyma	Connect to Remote Systems	There is a new page explaining how to connect to remote systems, the different use cases you might have, and what is the best approach for each use case. See Connecting to Remote Systems.	Info only	General Availability	New	Technology	Not applicable
SAP BTP Developer's Guide	Cloud Foundry Kyma	Reference Full-Stack Multitenant SaaS Application for Partners	There is now a reference application dedicated to SAP partners that provides a recommended way to becoming a SaaS provider of multitenant applications based on the SAP BTP. See Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners.	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	Cloud Foundry Kyma	Content Tabs for Node.js and Java added to Tutorial Navigator	You can now develop a CAP application in Java following the SAP BTP Developer's Guide. There is a new content tab **Java** in the tutorials from the Develop a Full-Stack CAP Application Following SAP BTP	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			Developer's Guide group of tutorials.					
SAP BTP Developer's Guide	Kyma	Tutorials for Consuming Remote Services and Deploying in SAP BTP, Kyma Runtime Moved to Tutorial Navigator	The tutorials that show how to integrate the SAP S/4HANA Cloud Business Partner API to the Incident Management application are now moved to the Tutorial Navigator. There are two main groups depending on whether you have an SAP S/4HANA Cloud tenant, or you use a mock server: • Consume Remote Services from S/4HANA Cloud in Your Full-Stack CAP Application Following the SAP BTP Developer's Guide and Deploy in SAP BTP, Kyma Runtime • Consume Remote Services from a Mock Server in Your Full-Stack CAP Application Following the SAP BTP Developer's Guide and Deploy in SAP BTP, Kyma Runtime	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	Kyma	Automate the Provisioning of Resources Needed to Create a Kyma Cluster	Using the Terraform provider for SAP BTP, you can now automate the provisioning of resources needed to create a Kyma cluster,	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			deploy applications, run tests, and tear down the resources when they are no longer required. See Terraforming Kyma Runtimes.					
SAP BTP Developer's Guide	Kyma Cloud Foundry	New Command That Simplifies Deployment Configuration	Deployment configuration steps are now simpler. There's a new command that does multiple configuration steps automatically so that you do not have to do them manually. As a result, tutorials are now easier to follow. Depending on the runtime you're using, check the following tutorials: • For SAP BTP, Cloud Foundy runtime, see Deploy in SAP BTP, Cloud Foundry Runtime and Integrate Your Application with SAP Build Work Zone, Standard Edition. • For SAP BTP, Kyma runtime, see Deploy in SAP BTP, Kyma Runtime and Integrate Your Application Deployed in SAP BTP, Kyma Runtime with SAP Build Work Zone, Standard Edition.	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
SAP BTP Developer's Guide	Kyma Cloud Foundry	New Mission: Develop a Multitenant CAP Application	You can now develop multitenant CAP applications and deploy them in SAP BTP. The Develop a Multitenant CAP Application Following the SAP BTP Developer's Guide mission includes enabling multitenancy, deploying the application in the SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime, and subscribing to the multitenant application. See Develop a Multitenant CAP Application Following the SAP BTP Developer's Guide.	Info only	General Availability	New	Technology	Not applicable
SAP BTP Developer's Guide	Kyma Cloud Foundry	Get to Know the Architecting Multi-Region Resiliency for SAP BTP Use Cases Reference Architecture	A link to the Architecting Multi-Region Resiliency for SAP BTP Use Cases reference architecture has been added to the SAP BTP Developer's Guide. It shows the multi-region high availability architecture for SAP BTP services and the applications built on SAP BTP. See Developing Resilient Applications.	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	Kyma Cloud Foundry	SAP Event Broker Becomes SAP Cloud Application Event Hub	The name of the SAP Event Broker service has been changed to SAP Cloud Application Event Hub. See: • Business Application Services	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			• What Is SAP Cloud Application Event Hub?					
SAP BTP Developer's Guide	Kyma Cloud Foundry	New Option for Integrating with SAP Build Work Zone, Standard Edition	You can now integrate your application with SAP Build Work Zone, standard edition through Common Data Model. For more information about the Common Data Model, see About the Common Data Model and Developing HTML5 Business Solutions as Content Providers. Depending on the runtime you're using, check the following tutorials: • For SAP BTP, Cloud Foundy runtime, see Integrate Your Application with SAP Build Work Zone, Standard Edition. • For SAP BTP, Kyma runtime, see Integrate Your Application Deployed in SAP BTP, Kyma Runtime with SAP Build Work Zone, Standard Edition.	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	Change in the Structure of the SAP BTP Developer's Guide	The new structure of the SAP BTP Developer's Guide now follows the design-led development process. It also now contains the development best practices introduced in the former Best	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			Practices for SAP BTP guide, now renamed to SAP BTP Administrator's Guide. See SAP BTP Developer's Guide.					
SAP BTP Developer's Guide	Kyma Cloud Foundry	New Mission: Develop an Enterprise-Grade CAP Application	You can now develop enterprise-grade CAP applications and deploy them in SAP BTP. The Develop an Enterprise-Grade CAP Application Following the SAP BTP Developer's Guide mission includes options for tracking changes, adding audit logs, and uploading attachments. See Mission: Develop an Enterprise-Grade CAP Application Following the SAP BTP Developer's Guide.	Info only	General Availability	New	Technology	Not applicable
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	Information About Third-Party Cookies	There is now information about the usage of third-party cookies. See Development Guidance.	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	Kyma Cloud Foundry	New Graphic Showing the Order of the Tutorials and Missions for CAP	You can now use an interactive graphic to follow the tutorials and missions related to the SAP Cloud Application Programming Model (CAP) and based on the Incident Management sample application. See Tutorials for SAP Cloud Application Programming Model.	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	Cloud Foundry	Tutorials for Consuming Remote Services Moved to	The tutorials that show how to integrate the SAP S/4HANA Cloud Business Partner API to the	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
		Tutorial Navigator	Incident Management application are now moved to the Tutorial Navigator. There are two main groups depending on whether you have an SAP S/4HANA Cloud tenant, or you use a mock server: • Consume Remote Services from S/4HANA Cloud in Your Full-Stack CAP Application Following the SAP BTP Developer's Guide and Deploy in SAP BTP, Cloud Foundry Runtime • Consume Remote Services from a Mock Server in Your Full-Stack CAP Application Following the SAP BTP Developer's Guide and Deploy in SAP BTP, Cloud Foundry Runtime					
SAP BTP Developer's Guide	Kyma Cloud Foundry	Explore the Develop a Side-by-Side CAP-Based Extension Application Following the SAP BTP Developer's Guide Mission	There is a new mission published that is part of the hands-on materials for the SAP Cloud Application Programming Model (CAP). The mission explains how you can implement side-by-side SAP Cloud	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			Application Programming Model (CAP)-based extension applications and deploy them in SAP BTP and is based on the Incident Management sample application. See: • Tutorials for SAP Cloud Application Programming Model • Mission: Develop a Side-by-Side CAP-Based Extension Application Following the SAP BTP Developer's Guide					
SAP BTP Developer's Guide	Kyma Cloud Foundry	Explore the Implement Observability in a Full-Stack CAP Application Following SAP BTP Developer's Guide Mission	There is a new mission published that is part of the hands-on materials for the SAP Cloud Application Programming Model (CAP). The mission explains how you can add observability to your SAP Cloud Application Programming Model (CAP)-based applications and is based on the Incident Management sample application. See: • Tutorials for SAP Cloud Application Programming Model	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			• Mission: Implement Observability in a Full-Stack CAP Application Following SAP BTP Developer's Guide					
SAP BTP Developer's Guide	ABAP	Explore the Develop a Full-Stack RAP Application Following the SAP BTP Developer's Guide Mission	There is a new mission published that is part of the hands-on materials for ABAP Cloud: Develop a Full-Stack RAP Application Following the SAP BTP Developer's Guide. It is based on the Travel sample application of the Flight Reference Scenario that is the best practice example for developing, deploying, and operating a full-stack application using ABAP RESTful Application Programming Model (RAP) and SAP Fiori. See Tutorials for ABAP Cloud.	Info only	General Availability	Announcement	Technology	Not applicable
SAP BTP Developer's Guide	ABAP Kyma Cloud Foundry	Build Your Application from Concept to Code	The SAP BTP Developer's Guide now provides insights on how to deliver a cloud application from concept to a final product. The section **From Concept to Code** outlines a progressive approach that includes the following steps: • Identify the business problem • Deliver a cloud application from concept	Info only	General Availability	Announcement	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process
			to a final product • Understand modularization in general • Understand modularization in the context of microservice-based architecture • Learn why the domain-driven design is an adequate approach • Learn about the domain-driven design starter modeling process • Check how complex your scenario is For more information, see [From Concept to Code].					

Understanding Available Technology

SAP BTP Developer's Guide has a comprehensive set of tools and services at your disposal that allow you to keep the pace and at the same time benefit from future investments in SAP BTP. The following graphic includes both ABAP and non-ABAP runtimes and provides an overview of the architecture you can use when designing and building your applications.

The overall suite of applications consists of multiple business modules that are either implemented for SAP Cloud Application Programming (CAP) or ABAP Cloud. Each business module consists of one or multiple self-contained services following a three-tier architecture with presentation, logic, and persistence layer. In both architecture styles, SAP Fiori and SAPUI5 is used to implement front end artifacts. These front end artifacts are consuming its data using OData for transactional and Information Access (InA) for analytical applications.

The application logic for CAP-based applications is implemented in Node.js, Java and Typescript and is deployed in SAP BTP, Cloud Foundry runtime or SAP BTP, Kyma runtime. The application logic for ABAP Cloud is implemented in ABAP and deployed in the SAP BTP, ABAP environment. In both cases, SAP HANA Cloud is used to store relational business data.

Additional enterprise qualities are reached by integrating the business modules with complementing SAP BTP application services such as SAP Build Work Zone as a central entry point, SAP Datasphere as a cross-application data warehouse or Identity

Authentication for identity authentication among others.

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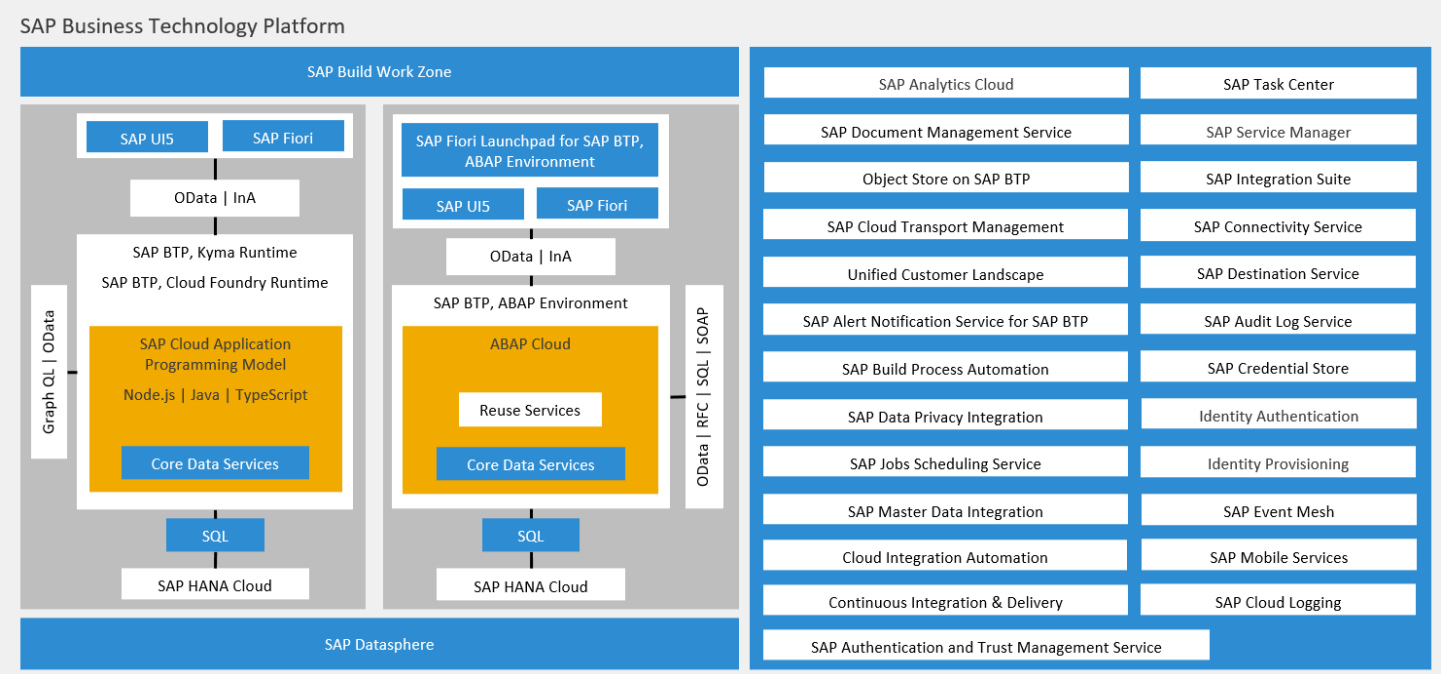


Please note that image maps are not interactive in PDF outputs.

Capabilities in SAP BTP and Their Respective Services

Capability	Cloud Application Programming	ABAP Cloud
Development Tools	SAP Business Application Studio Designed and optimized for business application development in SAP ecosystems, SAP Business Application Studio enhances productivity by offering specialized tools for various scenarios, including SAP Fiori application development, SAP HANA native extensions, full-stack and mobile application development, and more. Central to the development environment is Code-OSS, the open-source foundation of Visual Studio Code, ensuring a familiar experience for developers when creating SAP-centric applications. SAP Business Application Studio streamlines the building, testing, and deployment of applications with integrated features for source control and testing. Furthermore, its Full-Stack Application Productivity Toolkit offers intuitive visual tools covering the entire development process, guaranteeing seamless integration with various SAP services and solutions.	ABAP Development Tools for Eclipse SAP Business Application Studio Use ABAP development tools for Eclipse to benefit from an efficient development environment for all ABAP-based development artifacts. Use the SAP Business Application Studio to develop the SAP Fiori parts of your ABAP-based applications.
	SAP Continuous Integration and Delivery The Continuous Integration & Continuous Delivery (CI/CD) service offers pre-	

Capability	Cloud Application Programming	ABAP Cloud
	configured pipelines that automate the building and testing of code changes developed in SAP Business Application Studio. These pipelines can be customized to meet your specific needs. By providing immediate feedback on the status of your code, the service helps to identify and fix issues early in the development process. This accelerates development and improves the quality of your code.	
Software Delivery	SAP Continuous Integration and Delivery By enabling deployment, setting the deployment target, and optionally configuring the handover to SAP Cloud Transport Management, you can use SAP Continuous Integration and Delivery to automate your delivery process. This helps to accelerate your delivery cycles.	ABAP Lifecycle Management • For customers: Develop applications as software components and deliver them via the Manage Software Components application. See Software Components. A hidden Git repository is automatically managed per software component using gCTS for transport management. This process can be automated with CI/CD pipelines including steps like test automation using ABAP Test Cockpit. See Automate the Software Lifecycle Management Process. SAP Cloud Transport Management can be used to optionally integrate with higher level change management processes. See How to Export Using SAP Cloud Transport Management. • For partners: In addition to the customer scenario, products can be built based on software components with the help of the Landscape Portal to set up multitenant SaaS applications or to offer installable products like SDKs for other customers and partners. See Landscape Portal.
	SAP Cloud Transport Management SAP Cloud Transport Management offers a standardized, enterprise-ready change management process. For changes in SAP BTP, Cloud Foundry runtime, the corresponding pipeline in SAP Continuous Integration and Delivery includes out-of-the-box transports in the Cloud Transport Management service. While we strongly recommend using the Continuous Integration and Delivery service, you can also use SAP Cloud Transport Management to transport your changes without a pipeline.	
Persistence	SAP HANA Cloud Use SAP HANA Cloud service for a relational storage. Use HANA Data Lake Files as Object Storage.	SAP HANA Cloud SAP BTP, ABAP environment comes with an own ABAP-managed SAP HANA Cloud instance. Static resizing of the SAP HANA Cloud instance is supported. ABAP dictionary tables can be tagged to make use of the SAP HANA Native Storage

Capability	Cloud Application Programming	ABAP Cloud
	Consider compliance features like such as Audit Log, Cryptography or Customer Managed Keys that SAP HANA Cloud offers.	Extensions; also, database indexes (like unique secondary indexes and fuzzy search indexes) and Dynamic View Caches can be defined. Furthermore, simple Database Partitioning based on primary keys is possible. Access to the SAP HANA instance data is only supported via the ABAP layer, for example, by using ABAP SQL, natively via ABAP-managed database procedures, and the ABAP SQL Service for external clients.
Programming Model	SAP Cloud Application Programming Model (CAP) Use SAP Cloud Application Programming Model as programming model for non-ABAP applications. Go-to frameworks for business application development. It supports the most widely adopted languages, which are: Java (with Spring Boot), JavaScript and TypeScript (with Node.js). SAP Cloud Application Programming Model guarantees to run against specific versions of Node.js and Java. Remember to plan your application to run for at least 5 years if not more. We recommend that you choose SAP BTP services over homegrown services.	ABAP RESTful Application Programming Model (RAP) Use ABAP RESTful Application Programming Model as a programming model within ABAP Cloud. With ABAP RESTful Application Programming Model, you can develop services for all types of SAP Fiori applications as well as publishing Web APIs.
Runtime	SAP BTP, Cloud Foundry Runtime Use SAP BTP, Cloud Foundry runtime as a runtime for CAP-based applications. SAP BTP, Kyma Runtime Use SAP BTP, Kyma runtime as a runtime for CAP-based applications.	SAP BTP, ABAP Environment Use SAP BTP, ABAP environment for ABAP-based applications. SAP BTP, ABAP environment delivers and enforces the ABAP Cloud development model and is based on Kubernetes. The abstraction of containers and clusters is managed by the SAP BTP, ABAP environment infrastructure.
Client Library	SAP Cloud SDK CAP is using SAP Cloud SDK behind the scenes. SAP Cloud SDK provides client libraries for consuming OData/OpenAPI services, Destination service, and Connectivity service that extend SAP solutions and other OData/OpenAPI services on SAP Business Technology Platform. These clients allow you to concentrate on your unique business logic, simplifying development and ensuring seamless integration.	Service Consumption Model Use the Service Consumption Model to generate local ABAP APIs to consume remote OData, SOAP and RFC services.

Capability	Cloud Application Programming	ABAP Cloud
Connectivity	SAP Connectivity Service Use Connectivity service to securely access on-premise systems and such hosted in virtual private cloud which are exposed via the Cloud Connector. Using the service channels feature of the Cloud Connector, it is also possible to access cloud systems such as databases from on-premise and virtual private cloud networks. For Kubernetes-based workloads, you can utilize the Connectivity Proxy to ensure secure technical connectivity from the cloud to your on-premise and virtual private cloud systems. Within the Kyma runtime, the Connectivity Proxy comes as a managed offering.	SAP Connectivity Service The Connectivity service is automatically integrated with the BTP ABAP tenants to securely access on-premise systems and such hosted in virtual private cloud which are exposed via the Cloud Connector.
	SAP Destination Service Use Destination service to store and retrieve technical connection properties to the target systems. It automates the process of retrieving OAuth access tokens to the configured target systems. In addition, the Destination service generates and is able to renew X.509 client certificates issued by the SAP Cloud PKI.	Communication Management SAP BTP, ABAP Environment provides a Communication Management to integrate the custom applications with other systems to enable data exchange. An event consumption and event provisioning is also natively supported and integrated into the ABAP RESTful Application Programming Model (RAP). In addition, the SAP Destination Service can also be used to re-use destinations for CAP applications. See SAP Destination Service.
	Transparent Proxy for Kubernetes Use transparent proxy to enhance developers' user experience and simplify the consumption of target systems by automating the retrieval of destinations from the Destination service, as well as integration with the connectivity proxy and authentication with target systems. Within the SAP BTP, Kyma runtime, the transparent proxy comes as a managed offering.	
Application Programming Interface	SAP Business Accelerator Hub Follow API guidelines and use the SAP Business Accelerator Hub.	
User Interface (Web)	SAP Fiori Elements SAPUI5 Use SAP Fiori elements for OData V4 to benefit from a presentation of a common UI and UX. For more flexibility, use SAP Fiori element's Flexible Programming Model with or without SAPUI5 Freestyle. Personalization and theming are automatically supported. SAP Fiori elements and SAPUI5 help you present one consistent solution experience to your customers, and you benefit from the upcoming designs and UX improvements automatically.	

Capability	Cloud Application Programming	ABAP Cloud
	Remember that even the simplest UI components or more complex ones like geographic maps have compliance requirements like accessibility and theming.	
User Interface (Mobile)	SAP Mobile Development Kit SAP BTP SDK for Android SAP BTP SDK for iOS Use SAP MDK, BTP SDK for Android or BTP SDK for iOS for mobile applications. These SDKs help you present one consistent solution experience to your customers. Mobile application development is massively accelerated as SAP mobile services and its SDKs generate the complete synchronization and authentication layer. Additional supported features like logging, tracing, crash reporting bring extra value to you. After deployment the Mobile services give administrators all the necessary tools and services to operate a mobile solution. Features like push, offline, mobile specific security and more simplify the operation.	
Central Entry Point	SAP Build Work Zone, standard edition SAP Build Work Zone, advanced edition SAP HTML5 Application Repository Service for SAP BTP Enable the central launchpad to offer a unified end-user experience. Customers want to create a personalized view on the applications you produce. Make sure your application can be added to a central launchpad, regardless of where it runs.	SAP Fiori Launchpad for SAP BTP, ABAP Environment SAP Build Work Zone, standard edition SAP Build Work Zone, advanced edition SAP HTML5 Application Repository Service for SAP BTP SAP BTP, ABAP Environment comes with a dedicated central entry point the SAP Fiori launchpad for SAP BTP, ABAP Environment. This central entry point offers a unified end-user experience for ABAP-based applications. It is also used as the entry point for all SAP Fiori applications to administer the ABAP system. SAP Build Work Zone, standard edition offers a unified end-user experience across several systems for federation scenarios.
Integration	SAP Event Mesh SAP Cloud Application Event Hub SAP Integration Suite, advanced event mesh SAP Event Mesh can be used to distribute events between selected SAP cloud and on-premise applications and third-party applications. SAP Integration Suite, advanced event mesh is currently not supported by CAP and ABAP Cloud. SAP Cloud Application Event Hub allows CAP-based applications to consume business events from supported SAP cloud applications. Currently, SAP Cloud Application Event Hub is not supported by ABAP Cloud.	
	SAP Master Data Integration SAP Master Data Integration service acts as the central master data hub. It uses the integration models of SAP One Domain Model as the basis for master data replication. SAP	

Capability	Cloud Application Programming	ABAP Cloud
	will extend the support of SAP One Domain Model in all SAP cloud applications over time to integrate them. When out-of-the-box integration with SAP Master Data Integration is not available, SAP Integration Suite can be used to integrate with SAP ERP Central Component, SAP S/4HANA, and non-SAP applications.	
	SAP Integration Suite Use the Cloud Integration capability of SAP Integration Suite for mediated data or process integration, especially if complex integration flows, transformations or dedicated protocols are required. Besides mediated application-to-application integration, Cloud Integration usage is recommended for business-to-business and business-to-governance processes. The API Management capability of SAP Integration Suite allows you to easily enable your developer community with governed, secure, and policy-compliant access to all your APIs, events, and integrations. Graph is a recent addition to the API Management capability of SAP Integration Suite. Graph is a powerful semantic API mediation, combining and exposing the data from diverse systems in a complex enterprise landscape as connected data graphs. The data graphs are accessed via a single, curated, and uniform data-as-a-service API and make API consumption much easier.	
	Cloud Integration Automation Cloud Integration Automation service provides you a guided workflow to integrate SAP cloud solutions to on-premise and other SAP cloud solutions. The guided workflow contains instructions for manual and automated tasks to enable an easy and quick integration configuration setup.	
Observability	SAP Cloud ALM SAP Cloud ALM helps you to implement and operate intelligent cloud and hybrid business solutions. For your CAP-based applications on SAP BTP, SAP Cloud ALM provides a central, personalized and unified operations user experience. Specifically for Java and Node.js custom-built applications in SAP BTP, Cloud Foundry runtime, there are the Data Collection Instrumentation Libraries based on Open Telemetry designed to enable the data collection infrastructure in SAP Cloud ALM. With these libraries, you can collect data for many observability use cases, such as: • Real User Monitoring • Health Monitoring • Integration and Exception Monitoring	SAP Cloud ALM All ABAP cloud applications are supported by SAP Cloud ALM for central observability. SAP Cloud ALM provides the following use cases: • Real User Monitoring • Health Monitoring • Synthetic User Monitoring • Integration Monitoring • Job and Automation Monitoring • Data Forwarding to SAP Focused RUN For Health Monitoring, you could extend the delivered content with your own custom metrics. SAP BTP, ABAP Environment strictly distinguishes between platform monitoring and application monitoring. The platform monitoring, like availability monitoring, is ensured by the service itself. For application monitoring, which is in your responsibility, respective tools are offered. Use the Technical Monitoring Cockpit to
	SAP Alert Notification service for SAP BTP Local expert tool that allows you to subscribe to events from the platform – such as from used services, from your	

Capability	Cloud Application Programming	ABAP Cloud
	custom-built apps, or from the infrastructure – and to consume them via your channel of choice (such as by receiving notifications via email or in your preferred chat application). Can be integrated into central alerting of SAP Cloud ALM. SAP Cloud Logging SAP Cloud Logging allows you to analyze your SAP BTP workloads in great detail regarding performance, errors, usage, and other characteristics. It covers processing of logs, metrics, traces across SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime with flexible storage, alerting, and dashboarding.	analyze and optimize the application on-stack: • System workload • Resource consumption and capacity • Detailed statistics captured for single requests • Outbound communication • SQL statements
Customer Landscape Discovery	Unified Customer Landscape Use Unified Customer Landscape service for customer landscape management. There are different ways to add systems to the **System Landscape** page in the SAP BTP cockpit: manually or automatically. If a system of your solution is associated with your global account or through a subscription in SAP BTP cockpit associated with a given subaccount, it will appear in the list automatically. Otherwise, you have to add your system manually. Systems are added to the list in one of the following ways: • Auto-Discovered An auto-discovered system is a system (associated with the given global account) that has been discovered and added automatically to the list based on information of the existing system landscape. Any SAP system of the supported system types that is associated with the same customer ID, with which your global account in SAP BTP is associated, will be added automatically in the system landscape list. • Subaccount/<my-subaccount> Specifies that the system has been added through a subscription in SAP BTP cockpit associated with a given subaccount. The subscription has been discovered and added	Landscape Portal Landscape Portal is offered to manage all the systems within your global account in SAP BTP. It shows the list of the ABAP systems, it enables the control of system hibernation to reduce costs and it offers features to control the system lifecycle, for example, to nominate systems for the pre-upgrade prior to the standard upgrade of a quarterly ABAP platform release.

Capability	Cloud Application Programming	ABAP Cloud
	automatically through the subaccount. • Manually-Added Specifies that the system has been added to the list manually by the global account administrator, using the **Add System** button and completing the wizard. The system has been associated with the global account in SAP BTP.	
Extensibility and Integration	Side-by-Side Extensibility with Unified Customer Landscape: • Register Systems • Create Formations • SAP SuccessFactors Extensibility Service • SAP S/4HANA Cloud Extensibility Service Use Unified Customer Landscape service that offers customer landscape management capabilities for your SAP S/4HANA, SAP Ariba, SAP SuccessFactors, and other SAP and third-party systems in one single experience. In the SAP BTP cockpit, you get a comprehensive overview of all your systems associated with your customer ID. These systems can be registered or auto discovered. They are conveniently listed as a record in the **Systems** list in the **System Landscape** page in the SAP BTP cockpit. Moreover, Unified Customer Landscape lets you integrate one or more systems in a common business case by including these systems in a formation.	On-Stack Extensibility by Extending SaaS Applications In addition to the standard side-by-side approach for core applications such as SAP S/4HANA, SAP BTP, ABAP Environment also offers two extensibility options within the product itself: • With developer extensibility, you can extend partner solutions which are installed in your customer system, for example by adding custom fields, custom nodes and business logic. • With key user extensibility, you can extend a multitenancy SaaS application offered by a partner. The supported extensibility features to extend SaaS applications are UI adaptations, custom fields and custom logic (implement Business Add-Ins). You can create business configuration objects. See Business Configuration. You can create and set up integration scenarios by maintaining communication arrangements.
Data Privacy	SAP Data Privacy Integration Use Data Privacy Integration service to make your SAP BTP application compliant with the corporate Data Privacy and Protection standards. Integrate with the Data Privacy Integration service to support cross consumable Data Privacy and Protection features for our customers and support cross Data Privacy and Protection compliance in end-to-end processes.	
Workflow	SAP Task Center SAP Task Center service enables integration with SAP applications to provide a single entry point for end users to access all their assigned tasks. The tasks can be accessed by end users through the SAP Task Center Web application. SAP Task Center helps you integrate tasks into a central solution.	

Capability	Cloud Application Programming	ABAP Cloud
	Use SAP Task Center as a unified inbox for tasks across multiple applications with integrated user experience. Tasks from multiple SAP solutions are gathered in one list and ready to be processed in just one click, shortening the completion time for business-critical tasks. For example, business users can process all their tasks from the connected systems, without the need to switch and log in separately into different inboxes. SAP Build Process Automation SAP Build Process Automation provides a simpler and faster way to adapt, improve, and innovate business processes with drag-and-drop simplicity. The solution combines workflow management, SAP Intelligent Robotic Process Automation (SAP Intelligent RPA) functionality, decision management, process visibility, and embedded AI capabilities into one intuitive low-code experience. You can start automation projects with hundreds of process content packages, SAP Intelligent RPA bots and connectors designed specifically to enhance the capabilities of the SAP solutions you are using. It provides a unified development experience for users of all skill levels enabling fusion teams of business experts and developers to work together and solve challenges faster.	
Job Scheduling	SAP Job Scheduling Service SAP Job Scheduling service allows you to define and manage jobs that run once or on a recurring schedule. Use this runtime-agnostic service to schedule action endpoints in your application or long-running processes using Cloud Foundry tasks. Use REST APIs to schedule jobs, including long-running jobs asynchronously, and create multiple schedule formats for simple and complex recurring schedules. Manage jobs and tasks and manage schedules with a web-based user interface.	Application Jobs Job scheduling is integrated into the product. Application Jobs can be defined, implemented and monitored.
Application Logs		Application Logs SAP Task Center helps you integrate tasks into a central Application Logging is integrated into the product. Application Logs can be defined, implemented, and monitored.
Analytics	SAP Analytics Cloud Use embedded analytics scenarios in your application including dashboards, multi-dimensional reports and KPIs.	SAP Analytics Cloud Use SAP Analytics Cloud on top of Information Access (InA)-enabled Core Data Service analytical models. Furthermore, Dragonfly-based multi-dimensional reporting can be used to integrate Embedded Analytics functionality in a component-based and SAP Fiori-native way.
	SAP Datasphere For analytics across applications use SAP Datasphere.	SAP Datasphere Via the ABAP SQL Service it is possible to integrate with SAP Datasphere. Furthermore, the ABAP SQL Service in

Capability	Cloud Application Programming	ABAP Cloud
		combination with ABAP ODBC Driver or the SAP HANA Cloud ABAP SDA Adapter allows data federation via external clients respectively via SAP HANA Cloud-based SAP BTP applications.
Document Management	SAP Document Management Service Document Management Service, Integration Option lets you build document management capabilities for your business applications using the integration component or the easy-to-use, reusable UI component. Document Management Service, Application Option is a standalone, ready-to-use web application that provides document management capabilities for your enterprise content.	
	Object Store on SAP BTP Object Store service on SAP BTP lets you store and manage objects, which involves creation, upload, download, and deletion. This service is specific to the Infrastructure-as-a-Service layer such as Azure Blob Storage, Amazon Web Services, and Google Cloud Platform.	
Low Code/No Code	SAP Build SAP Build is a low-code offering to accelerate development and automation. It enables you to create enterprise apps (SAP Build Apps), automate processes (SAP Build Process Automation), and design business sites (SAP Build Work Zone) without writing code. Accelerate development with prebuilt connectors and business content for SAP and third-party systems to integrate seamlessly. SAP Build fosters collaboration between business and development teams with built-in governance and lifecycle management.	
Service Management	SAP Service Manager SAP Service Manager service allows you to consume platform services in any connected runtime environment, track the creation and management of service instances, and share services and service instances between different environments.	
Authentication	SAP Authentication and Trust Management Service The SAP Authentication and Trust Management service lets you manage user authorizations and trust to identity providers. Identity providers are the user base for applications. We recommend that you use an Identity Authentication tenant, an SAP on-premise system, or a custom corporate identity provider. User authorizations are managed using technical roles at the application level, which can be aggregated into business-level role collections for large-scale cloud scenarios.	Access Management User authorizations are managed and can be aggregated into business roles locally in SAP BTP, ABAP environment.

Capability	Cloud Application Programming	ABAP Cloud
	Identity Authentication The Identity Authentication service provides you with controlled cloud-based access to business processes, applications, and data. It simplifies your user experience through authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options.	Identity Authentication The Identity Authentication service provides you with controlled cloud-based access to business processes, applications, and data. It simplifies your user experience through authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options. Technical users for system-to-system communication are managed locally in the SAP BTP, ABAP environment. SAP BTP, ABAP environment supports mTLS and basic authentication as authentication options for technical users.
Audit Logging	SAP Audit Log Service SAP Audit Log is a core, security, and compliance-based SAP BTP service to provide means for audit purposes. The default and advanced capabilities of Audit Log Service are available for SAP BTP applications and services.	Manage Security Audit Log Security audit logging is provided automatically by the SAP BTP, ABAP Environment. It can be configured by the Manage Security Audit Log administration application. Read Access Logging Read Access Logging (RAL) is used to monitor and log read access to sensitive data. This data may be categorized as sensitive by law, by external company policy, or by internal company policy.
Security	SAP Credential Store SAP Credential Store service provides a repository for passwords, keys and keyrings for applications that are running on SAP BTP. It enables the applications to retrieve credentials and use them for authentication to external services, or to perform cryptographic operations and TLS communication. SAP Credential Store is exposed to the applications via a REST API.	Communication Management The ABAP environment offers its Communication Management. It contains a credentials store which allows the applications to perform outbound communication using the credentials for authentication to external services and TLS communication.
Identity Management	Identity Provisioning The Identity Provisioning service automates identity lifecycle processes. It helps you provision identities and their authorizations to various cloud and on-premise business applications.	Identity Provisioning The Identity Provisioning service automates identity lifecycle processes. SAP BTP, ABAP environment supports the Identity provisioning service to provision business users and their assignment to business roles.

Capability	Cloud Application Programming	ABAP Cloud
Technical Ops Automation	Terraform Provider for SAP BTP The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP.	
	SAP Automation Pilot SAP Automation Pilot allows to reduce operational efforts to run your cloud applications on SAP BTP by automating them. For this, it brings catalogs of automated commands, such as around database and application lifecycle management tasks for the different runtimes of SAP BTP. Use these commands to model and run your custom scenarios, optionally combined with custom scripts. Also, you can describe your needs to let the included Generative AI feature come up with a command for you. The created commands can be triggered manually, via scheduling or externally, such as from SAP Cloud ALM.	
Feature Toggling and Delivery	SAP Feature Flags SAP Feature Flags service is designed to help you manage and control the deployment of new features within your applications. You can activate and deactivate features dynamically without the need for redeployment, enabling a more flexible and responsive development process. SAP Feature Flags service supports gradual feature rollouts, allowing you and your team to test new functionalities with specific user groups before a full-scale launch. Additionally, it provides the capability to quickly disable features if issues arise, minimizing potential disruptions. The service integrates seamlessly with existing SAP environments, ensuring that feature management is consistent and efficient across different applications. Overall, SAP Feature Flags Service offers a practical solution for managing feature releases, enhancing the ability to deliver stable and reliable software updates.	

Related Information

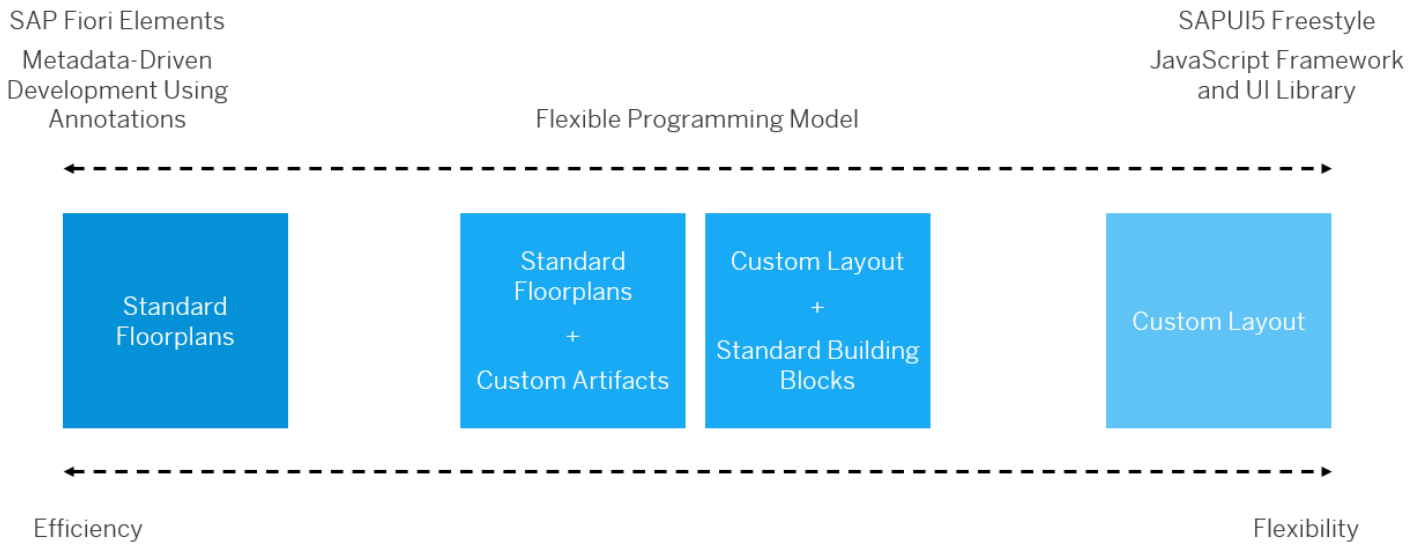
UX Design and Product Compliance Standards

UX Design and Product Compliance Standards

Applications built on SAP BTP need to provide a consistent and compliant user experience across runtimes, while giving developers flexibility to address diverse business requirements. The SAP Fiori design language is the standard for achieving this goal.

From a development perspective, there are three main recommendations for creating user interfaces:

- SAP Fiori elements
- Flexible Programming Model
- Freestyle SAPUI5



SAP Fiori Elements

SAP Fiori elements is a framework on top of SAPUI5 that accelerates the development of SAP Fiori applications by providing predefined templates and patterns based on OData services, ensuring consistent UI design and functionality without extensive coding. Provides the fastest development speed, lowest total cost of ownership, and out-of-the-box compliance with SAP Fiori design guidelines.

Floorplans

Floorplans in SAP Fiori elements are standardized UX design templates that guide the layout and behavior of applications. They include variations like List Report, Object Page, and Overview Page, each serving specific business needs and ensuring a cohesive user experience across SAP applications.

Building Blocks

Building blocks in SAP Fiori Elements are reusable UI components that encapsulate common functionalities, such as data visualization or form entry. They streamline development by enabling consistent application behavior and enhancing modular design within SAP Fiori applications.

Flexible Programming Model

A hybrid approach that extends SAP Fiori elements with custom UI logic where needed. Recommended for applications that follow standard floorplans but require selective flexibility (for example, custom actions, advanced visualizations). Balances reuse of compliant templates with targeted freedom for developers.

Extensions

Standard templates offer multiple extension points, allowing you to enhance them with custom elements.

Custom Pages

When developing custom pages, leveraging the SAP Fiori elements base component enables the use of building blocks and additional features like routing, all readily accessible.

Freestyle SAPUI5

Full control over UI logic and layout using SAPUI5 as a UI framework. Recommended only when business scenarios demand highly customized interfaces that cannot be covered by SAP Fiori elements or the flexible programming model. Requires higher development and maintenance effort but still benefits from the Horizon theme and SAP Fiori guidelines.

SAP Fiori Tools

SAP Fiori tools provide many capabilities to increase the efficiency of developing SAP Fiori applications using either SAP Fiori elements or the freestyle SAPUI5 approach. SAP Fiori tools, together with SAP Fiori elements, reduce development time, maintenance costs, and leverage the advantages of a metadata-driven UI. SAP Fiori tools include the following:

- Application Wizard for the initial creation of an application.
- Service modeler for viewing the data model.
- (SAP Fiori elements only) XML and form-based editor for maintaining of annotations.
- (SAP Fiori elements only) Application page structure and ability to configure the SAPUI5 flexibility settings.
- (SAP Fiori elements only) Guided development for implementing features.

Compliance Standard

Applications designed with the recommended UX approaches automatically inherit a broad set of SAP product compliance standards. These ensure that applications are enterprise-ready without requiring developers to implement them from scratch.

Out-of-the-Box Compliance

By using SAP Fiori elements or the flexible programming model on top of SAPUI5 with Horizon theme, applications automatically comply with:

- Accessibility (WCAG 2.2) – support for screen readers, ARIA roles, keyboard navigation.
- SAP Fiori design guidelines – consistent floorplans, navigation, and semantics.
- Responsive design – adaptive layouts for desktop, tablet, and mobile.
- Performance best practices – SAPUI5 optimizations for data loading and rendering.
- Security standards – built-in protection (XSRF, CSRF tokens), secure authentication with Identity Authentication service/SAP Authorization and Trust Management service.

To be fully compliant, applications must also:

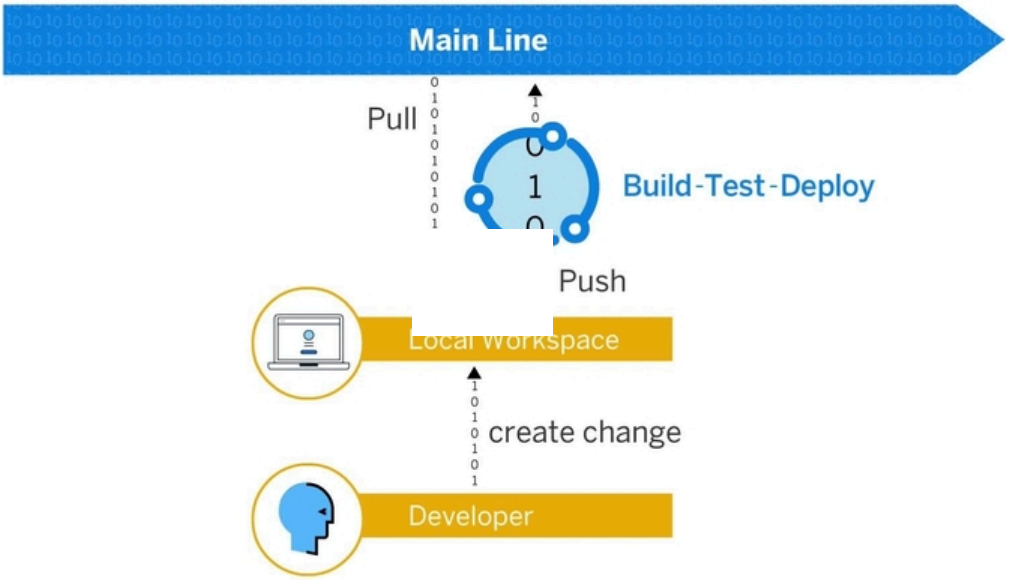
- Expose OData V4 services (CAP, ABAP Cloud, or Node.js) with proper annotations.
- Enable internationalization (i18n) for texts and labels.
- Respect SAP Fiori design guidelines when customizing UI (colors, icons, navigation).

Introducing Continuous Integration and Delivery (CI/CD)

The acronym CI/CD combines two proven DevOps concepts for automation, which are integral to application development and build on each other: **Continuous Integration (CI)** and **Continuous Delivery (CD)**.

Continuous Integration (CI)

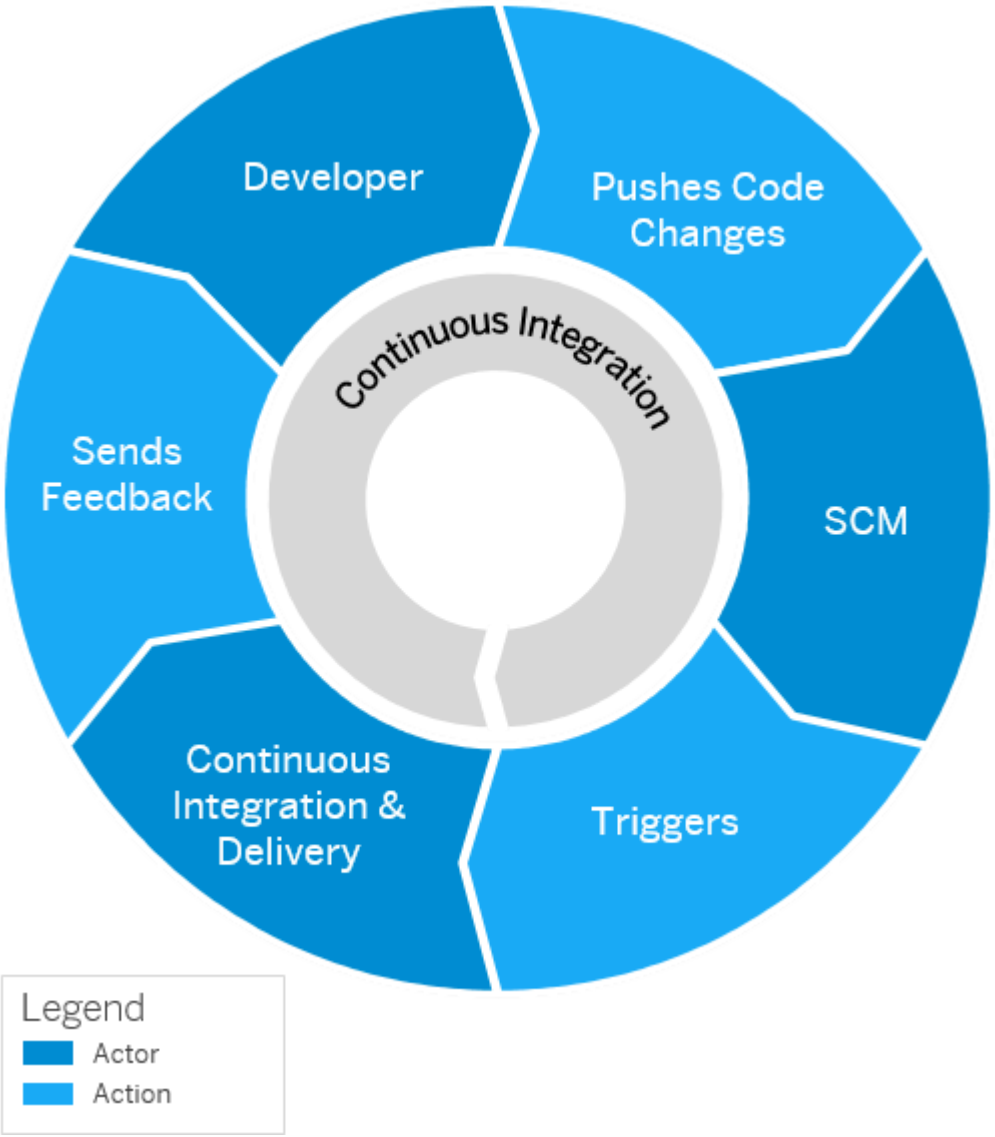
The following video provides a brief introduction to Continuous Integration:



Open this video in a new window

CI describes a software development process in which various team members frequently integrate their contributions into a single main line. Before each integration, the changes are verified through builds and automated testing. The more code you need to integrate, the harder it becomes, and the more problems may arise. Therefore, it is ideal to perform small integrations more often – in other words, continuously. This allows you to detect errors as quickly as possible and prevents integration problems from building up until the end of the development cycle.

The following figure shows the basic flow for Continuous Integration:



Basic Flow for Continuous Integration

The Continuous Integration basic flow comprises the following steps:

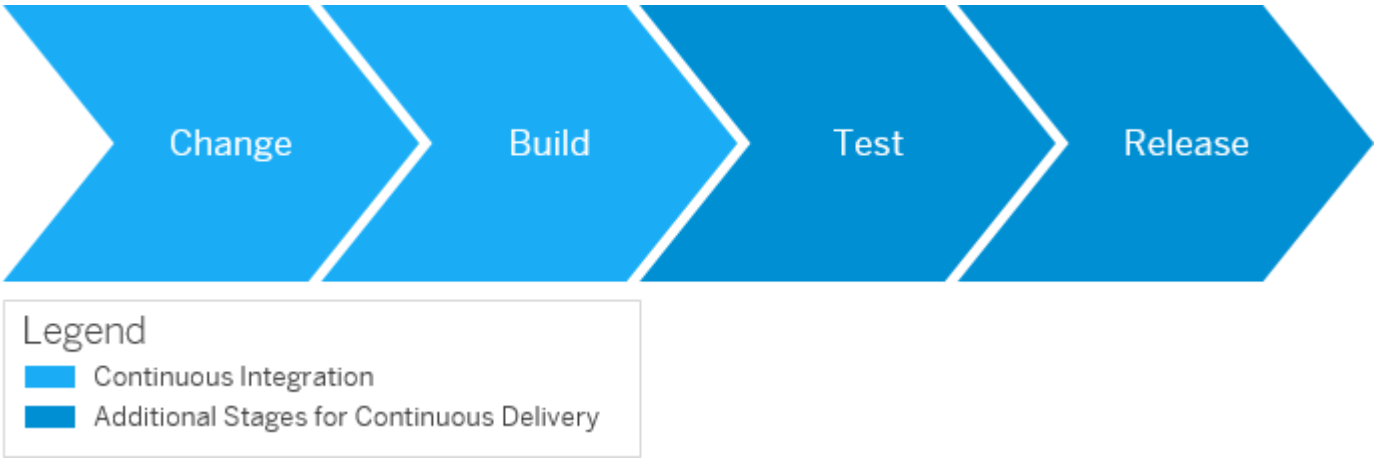
1. The Developer writes code and pushes the code changes to a repository on a central source code management system (SCM).
2. The Source Code Management System triggers the Continuous Integration (CI) server.
3. The Continuous Integration server runs automated builds and tests, then immediately sends feedback about the outcomes to the developer.

As you can see from the figure, Continuous Integration is a cyclical process. As soon as the CI server sends its feedback to the developer, the flow starts over. The developers either correct their previous code change, which must then be built and tested again, or they start working on an entirely new one.

Continuous Delivery (CD)

The concept of Continuous Delivery (CD) expands on that of Continuous Integration. It adds the aspect that any change successfully passing the tests is immediately ready for deployment to production, both from a technical and qualitative perspective.

The following figure shows the relation between Continuous Integration and Continuous Delivery:



Relation Between CI and CD

The Continuous Delivery process ensures that the most current version is successfully built, tested, and provided in a shippable format. Based on the release decision by the development team or delivery manager, it can be shipped to customers or deployed to production at any time.

Continuous Integration Principles

The Continuous Integration process, which forms the foundation of the Continuous Integration and Delivery (CI/CD) concepts, is based on several principles. For a clearer understanding, check out the following lesson from the "Discovering DevOps with SAP BTP" learning journey, where three of these principles are explained using an analogy: Explaining Continuous Integration Principles

For detailed explanations of all CI principles, refer to the following documentation: Continuous Integration Principles

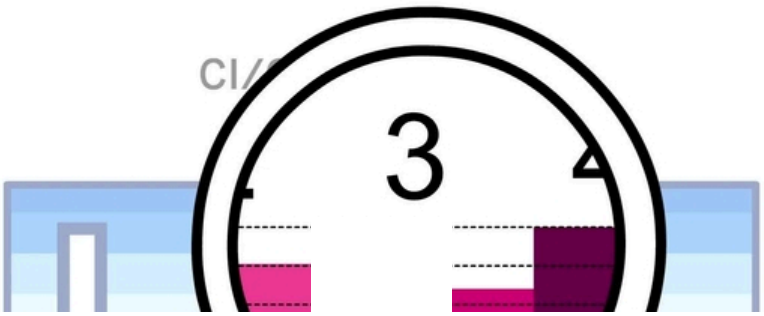
Related Information

Explaining the Continuous Integration and Delivery Process Flow
Continuous Integration and Delivery Process Flows
Explaining Actors in a CI/CD Process Flow
Actors in a CI/CD Process Flow

SAP Continuous Integration and Delivery

SAP Continuous Integration and Delivery is a service on SAP BTP that helps development teams collaborate more effectively, improve the quality of their software, and accelerate their delivery process. It provides pre-built CI/CD pipelines tailored for SAP-specific use cases, which can be connected to a source repository to automatically build, test, and deploy code changes. This helps identify and prevent integration problems early in the development process.

The following video provides a brief introduction to SAP Continuous Integration and Delivery:



Open this video in a new window

The Continuous Integration & Delivery service focuses on two key benefits: **Simplicity** and **Flexibility**.

Simplicity

SAP Continuous Integration and Delivery stands out for its ease of use, offering predefined CI/CD pipelines that can be used right out of the box. These pipelines require minimal configuration, making them accessible even for users without extensive CI/CD expertise. Additionally, the service includes all the necessary infrastructure, eliminating the need for users to manage it themselves.

Flexibility

SAP Continuous Integration and Delivery provides a high degree of flexibility, allowing users to customize pipelines with additional commands, credentials, and variables. This flexibility ensures the service not only covers standard use cases but also addresses more complex scenarios. Users can expand their pipelines with various tasks executed before or after specific stages, using additional commands. Moreover, extra credentials and variables allow for further customization of CI/CD jobs.

For a detailed overview and various demos on how to create CI/CD jobs in SAP Continuous Integration and Delivery, take a look at the following lesson in the 'Discovering DevOps with SAP BTP' learning journey: Creating a Basic CI/CD Job with SAP Continuous Integration and Delivery

Related Information

SAP Continuous Integration and Delivery
What Is SAP Continuous Integration and Delivery

Data Persistence Using SAP HANA Cloud

SAP HANA Cloud is a cloud-native Database-as-a-Service that forms the database management foundation of the SAP Business Technology Platform (SAP BTP). It provides a single, scalable, and highly performant database environment for both transactional and analytical workloads, supporting multiple data models like relational, document, geospatial, and vector data. SAP HANA Cloud enables you to build modern, intelligent applications that leverage real-time enterprise data, generative AI, and flexible data integration- all without the administrative overhead of managing hardware or backups. It seamlessly integrates with other SAP BTP services and external systems, empowering developers to innovate with secure, elastic, and fully managed data services within the broader SAP BTP ecosystem.

In the overall developer context for SAP HANA Cloud serves as the core data layer, enabling data-driven application development, analytics, and intelligent business processes, while providing cloud scalability and multi-model flexibility essential for modern enterprise application development.

See SAP HANA Cloud, SAP HANA Database.

SAP HANA Cloud Helps Reduce the Total Cost of Ownership

SAP HANA Cloud can help with reducing the total cost of ownership (TCO) when following these best practices:

1. Native Storage Extension (NSE)reduces the memory size of your SAP HANA Cloud instance. It is possible to keep data on disks, which are not frequently accessed. The Native Storage Extension advisor provides you recommendations for data being managed in-memory or in Native Storage Extension.

   SAP HANA Cloud advisors create data-driven recommendations that support database administrators in optimizing system performance and efficiency. The NSE Advisor provides a flexible interface to act on individual recommended objects to convert from column-loadable to page-loadable, or vice versa.

2. Elastic Compute Nodes (ECN)enable adding on demand computation power to your SAP HANA Cloud instance in situations with high workload, for example, for analytics and planning, where you see a high CPU-utilization and performance degradation. Elastic Compute Nodes can be disabled at any time after the high-workload phase. The Elastic Compute Nodes enabling or disabling can even be scheduled and automated, for example with the SAP Automation Pilot. We recommend the SAP-sample GitHub repository a ready to use command for Elastic Compute Nodes in SAP Automation Pilot.

   The usage monitoring with ECN Advisors is an intelligent SAP HANA Cloud component that collects statistics about the usage that is provided to the user, developer, or administrator.

3. Table Partitioning helps you to optimize the query performance in your SAP HANA Cloud instance. Tables are split into different parts (partitions) that simplify and accelerate the data operations on large tables. Partitioning is typically used in multiple-host systems, but it may also be beneficial in single-host systems.

4. Native Multi-Tenancy allows you to develop and operate a SaaS application that serves many customers at the same time within a single instance. Customer data is isolated by default and can be managed independently from each other. SAP HANA Cloud allows up to 1000 native database tenants, which sizes can vary from a few kilobytes up to terabytes.

5. SAP HANA Cloud is already available at 16GB memory in the BTP Free Tier Offering. You can use free tier to get started and are not limited to a trial period. You can also seamlessly upgrade your free tier instance to a full productive SAP HANA Cloud instance. Note that with the free tier license there are some limitations. See SAP HANA Database License.

6. Mutli-model capabilities of SAP HANA Cloud enable organizations to manage and analyze diverse types of data in a single, unified database platform. This approach simplifies the architecture, enhances the analytics, and supports AI-powered applications by integrating specialized engines and data models.

CAP Specifics in Supporting SAP HANA Cloud

SAP Cloud Application Programming Model (CAP) supports SAP HANA Cloud in local and hybrid development on SAP BTP by providing a standardized framework that facilitates consistent development of cloud-native applications. CAP uses Core Data Services (CDS) to define domain models and automatically generates the corresponding database artifacts for SAP HANA Cloud, enabling developers to build, test, and run applications locally or in a hybrid mode before deploying to the cloud environment.

With CAP, you can leverage seamless integration with SAP HANA Cloud including multitenancy, schema evolution, and SAP HANA Infrastructure Services-based deployment artifacts. CAP supports Node.js and Java and auto-configures the SAP HANA Cloud data source based on the available service bindings when deployed on SAP BTP. This allows unified development experience across local development setups (for example, SAP Business Application Studio or VS Code) and hybrid cloud scenarios where parts of the application or data reside on-premises or in other cloud environments.

Key features include:

- CAP abstracts the complexities of database deployment, service provisioning, and service bindings for SAP HANA Cloud.
- Local development with CAP can generate and emulate database artifacts before pushing to SAP HANA Cloud.
- SAP HANA Infrastructure Services artifacts required for SAP HANA Cloud are automatically generated by CAP build tools.

- CAP handles multi-target application configurations, enabling hybrid development workflows that integrate on-premise and cloud resources.

Overall, CAP acts as a developer-friendly layer that simplifies building, testing, and deploying SAP HANA Cloud applications on SAP Business Technology Platform locally, in hybrid settings, or fully cloud-native.

ABAP Specifics in Supporting SAP HANA Cloud

ABAP supports SAP HANA Cloud within the SAP BTP through the SAP BTP ABAP environment. This is a cloud-optimized ABAP development and runtime environment based on SAP HANA Cloud, designed for building modern cloud applications and cloud extensions, especially for SAP S/4HANA Cloud and other SAP solutions.

Key points about ABAP's role in SAP HANA Cloud on SAP BTP:

- The ABAP environment provides a cloud-native platform for developing, running, and extending ABAP applications using the ABAP RESTful Application Programming Model, optimized for SAP HANA Cloud.
- Developers can create side-by-side extensions that run on SAP BTP, keeping the core SAP systems clean by decoupling custom code and enabling independent release cycles.
- The ABAP environment supports seamless integration with SAP solutions including SAP S/4HANA Cloud, allowing ABAP extensions to access SAP HANA Cloud databases and services.
- Modern development tooling like ABAP Development Tools (ADT) in Eclipse and the SAP Build lobby enhance productivity, including AI-assisted coding support via Joule for generating code, documentation, and testing.
- The ABAP environment leverages the power of SAP HANA Cloud for in-memory data processing, real-time analytics, and scalable cloud-native application development.
- The ABAP environment supports secure APIs and integration patterns for communication within SAP landscapes and external systems, enabling hybrid and cloud extension scenarios.

The SAP BTP ABAP environment with SAP HANA Cloud enables you to build intelligent, cloud-ready applications and extensions while leveraging existing ABAP knowledge and integrating tightly with SAP's cloud ecosystem.

Strengths and Benefits of the Runtimes and Their Programming Models

SAP BTP provides different runtimes and development models to support a wide range of application needs. Whether you're building new cloud-native apps, extending existing SAP systems, or reusing legacy code, there's a suitable option.

- **SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime** support modern, flexible development using Java, Node.js, and microservices, with strong support for open standards and scalability.

  See SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP.

- **SAP BTP ABAP Environment** is ideal for teams with existing ABAP skills or codebases, offering a smooth path to the cloud with strong integration into SAP S/4HANA.

  See SAP BTP ABAP Environment with ABAP Cloud.

Each environment is designed to match different technical requirements, developer skill sets, and cost considerations—giving teams the flexibility to choose what fits best.

Related Information

SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP

Profiles in CAP: Development, Hybrid, and Production

CAP applications are design agnostic, and developers could focus on domain modeling and services without writing custom code. The development profile enables developers to design and test application prototypes with minimal setup. The mock variants of platform services and data enable developers to test without deploying CAP applications to SAP BTP. This helps customers to realize application prototypes in shorter cycles with low cost and later switch to hybrid/production profile by incrementally adding the required service bindings.

CAP's core strength lies in its focus on domain modeling. It allows developers to define business objects and their relationships using standardized data models, which can then be translated into services and applications.

See:

- Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes
- CAP documentation: Domain Modeling

Multiple IDEs and Runtimes

CAP applications supports custom code development using Node.js or Java. You can choose Node.js or Java as a programming language and develop applications and use your preferred development environment such as:

- CDS Command Line Interface (CLI)
- CDS Editors and IDEs
    - SAP Business Application Studio
    - Visual Studio Code
    - IntelliJ

Integration with SAP BTP Services

CAP applications benefit out of the box integration to platform services and SAP BTP application services in the SAP Build portfolio, enabling developers to focus on application logic without spending additional effort on integration. Through simple configuration and service bindings, CAP applications seamlessly consume SAP HANA Cloud for persistence, SAP Authorization and Trust Management service for identity and authorization, and the Destination and Connectivity services for calling SAP and third-party systems.

On the experience side, CAP services and SAP Fiori UIs are easily surfaced in SAP Build Work Zone, exposed as OData/REST APIs that SAP Build Apps can consume, and can trigger or drive SAP Build Process Automation workflows.

See:

- CAP Plugins and Enhancements
- CAP: cds add command

Microservice Development

CAP supports micro-service development and deployment to the SAP BTP, Cloud Foundry or SAP BTP, Kyma runtimes. You can develop loosely coupled microservices with separate lifecycles and different runtimes to achieve specific scalability requirements. Have in mind that microservices are not suitable for all use cases. Even though they have a lot of benefits, they are accompanied by complexity and performance losses. To decide whether to use microservices or not, see CAP: Late-Cut Microservices.

To deploy your CAP application as microservices, see Microservices in CAP.

Consume Services as REST/OData/BAPI

CAP applications can consume data from other services or external sources:

- OData and REST/OpenAPI services (import definitions, mock locally, bind in hybrid/production)
  - Using services: https://cap.cloud.sap/docs/guides/using-services/
  - OpenAPI guide: https://cap.cloud.sap/docs/guides/openapi/
- SAP Cloud SDK (Java/JavaScript) for calling SAP solutions (SAP S/4HANA, SAP SuccessFactors, and others). See https://sap.github.io/cloud-sdk/ .
- SAP API Business Hub for API definitions. See https://api.sap.com/.

Multitenancy

Multitenancy in CAP is about building one Software-as-a-Service (SaaS) application (provider) that securely serves many customer accounts (tenants) on SAP BTP, while keeping each tenant's identity, configuration, and data isolated. CAP provides standardized patterns and tooling that eliminate the need to build typical SaaS integration and operational components by hand.

See CAP: Multitenancy.

Extensibility

Extensibility in CAP is the ability to adapt and enhance a CAP application's data model, service APIs, and behavior without forking or modifying the core solution. It supports both provider-controlled extensions (delivered by the application team or partner) and tenant-specific extensions in SaaS scenarios.

See CAP: Extensibility.

Community

CAP applications benefit from large ecosystem of developers and community contribution. The large pool of developers with the required skills and packages from npm/Maven will accelerate development and reduce development costs. See:

- CAP Community in GitHub
- CAP in SAP Community

Multiple Databases

CAP supports multiple databases, and customers can choose PostgreSQL or SAP HANA Cloud for persistence. CAP supports SQLite and H2 for development, which can be later replaced with PostgreSQL or SAP HANA Cloud by changing the development profile to production.

- Official: SQLite (development profile) and SAP HANA/SAP HANA Cloud (hybrid and production profiles). See https://cap.cloud.sap/docs/guides/databases/.
- Community adapters exist for other SQL databases (for example PostgreSQL). See CAP Community in GitHub .

SAP BTP ABAP Environment with ABAP Cloud

SAP BTP ABAP environment is a modern cloud offering for side-by-side extensions that uses the ABAP Cloud development model for cloud-ready extensions.

Aligned Development Model

The ABAP Cloud development model is used both in SAP BTP ABAP environment and natively within SAP S/4HANA. This enables a consistent development experience across side-by-side and on-stack extensions, allowing you to use the same tools, patterns, and lifecycle processes.

See ABAP Cloud Development Model.

Reuse of Existing ABAP Code

Existing ABAP logic from legacy or on-premise systems can be adapted and reused in the SAP BTP ABAP environment using tools such as the Custom Code Migration App and the ABAP Test Cockpit (ATC). This reduces redevelopment effort and accelerates cloud adoption.

- The Custom Code Migration App analyzes legacy code, identifies non-compliant elements such as unreleased APIs or obsolete statements, and helps prioritize necessary adaptations. See Custom Code Analysis.
- The ABAP Test Cockpit validates the refactored code against ABAP Cloud rules to ensure code quality. See ABAP Test Cockpit.

Typical use cases include migrating stable custom logic, refactoring extensions to released APIs, and reusing business logic while modernizing UIs or integrations. See ABAP Integration and Connectivity.

Seamless SAP S/4HANA Cloud Integration

SAP BTP ABAP Environment follows the same identity, access, and connectivity concepts as SAP S/4HANA Cloud, including SAP Identity Authentication Service, OData, and RFC. This ensures seamless interoperability and simplifies user, role, and permission management across systems.

ABAP Cloud provides a unified integration approach: connections are configured the same way regardless of the target system (SAP BTP services, SAP S/4HANA Cloud, or approved external APIs). Using the service consumption model, ABAP automatically generates client code and data structures from service definitions, allowing developers to focus on core business logic.

The environment also includes numerous pre-built communication scenarios—connectors to SAP BTP services and SAP S/4HANA Cloud APIs—to accelerate setup and follow cloud-compliant integration practices.

See Integrating the ABAP Environment with SAP S/4HANA Cloud.

Partner Deployment Options

Mutitenancy

Multitenancy in the ABAP environment enables application providers to develop and operate ABAP solutions in a software-as-a-service (SaaS) model on SAP BTP. Multiple consumers can be provisioned within the same ABAP system. The application provider is responsible for operating the solution and covering the required SAP BTP resources, while consumers use the software-as-a-service.

See Multitenancy in the ABAP Environment.

Add-On Product as an On-Stack Option

An add-on product is a newly available software delivery option that allows partners to provide their solution as a software offering rather than a service. It is installed and operated directly in the customer's own SAP BTP ABAP environment, giving the customer full control over lifecycle management, operations, and data governance. This option is particularly suited for scenarios where customers want to run the solution in their own environment or need to perform comprehensive enhancements using both developer extensibility and key user extensibility. The add-on product is made available to customers through the SAP Store, and partners must have a valid SAP PartnerEdge, Build contract to deliver their solution as an add-on product.

Customers using SAP BTP ABAP environment can browse for available add-on products in the SAP Store, register for the solution, and deploy it directly into their system using the Landscape Portal.

Both multitenant SaaS and add-on products options are fully supported and can be selected based on the customer's and partner's specific needs.

Tool Support for Partners via Landscape Portal

The Landscape Portal acts as the central cockpit for partners to manage ABAP systems, tenants, and SaaS and add-on product operations. It offers system overviews, product and version lifecycle management, deployment pipelines, and monitoring functions. The portal also supports multitenant SaaS scenarios, enabling efficient administration across landscapes.

The Landscape Portal provides integrated tooling for partners to build, test, and deploy their products, making it a key enabler for delivering both SaaS and add-on products offerings.

See Landscape Portal.

Reuse Services

In ABAP, reuse services are standard, cross-application building blocks you can call (via function modules, classes, or frameworks) instead of reinventing common functionality. Common services like application jobs, logging, forms, emails, change documents, and workflow are provided out of the box, enabling rapid development and standardization.

These services are designed to be reused across applications, ensuring consistency, reliability, and reduced development effort.

See Released Components and Objects and Understanding Available Technology.

Stability Checks Before the Upgrade

The SAP BTP ABAP environment provides a pre-upgrade option that enables customers and partners to verify the stability of their custom applications up to four weeks before the standard release upgrade. This helps detect potential regressions and ensures compatibility in a controlled environment.

The pre-upgrade option is executed on nominated non-production systems during a planned downtime. It mirrors the standard upgrade process, including hotfixes. To use this option, the relevant systems must be registered in the Landscape Portal.

See Landscape Portal.

SAP Fiori Launchpad Integration

The SAP Fiori launchpad is fully integrated with SAP BTP ABAP Environment, enabling you to run and expose UI applications without deploying a separate front-end server. Users can launch applications directly from the same system that hosts your ABAP code, providing a single, consistent entry point.

The launchpad offers role-based access, navigation, and personalization features, allowing users to organize applications according to their workflow. Predefined groups and catalogs help structure applications, while deep linking and parameterized launches ensure smooth navigation across multiple applications.

This integration simplifies development and deployment, reduces operational complexity, and provides a seamless user experience across the ABAP environment.

See SAP Fiori Launchpad for SAP BTP ABAP Environment.

Code Quality and Security Compliance

High-quality ABAP Cloud code is ensured using ABAP Unit for dynamic tests and ABAP Test Cockpit (ATC) for static analysis.

ABAP Unit allows developers to write automated tests to verify functional correctness and prevent regressions. Test coverage should be measured before releasing transports, with high coverage helping to identify issues and improve robustness.

ABAP Test Cockpit checks code for syntax, performance, standards adherence, and ABAP Unit results. The default check variant `ABAP_CLOUD_DEVELOPMENT_DEFAULT` includes recommended checks for ABAP Cloud and executes unit tests. Variants can be customized, and ABAP Test Cockpit can be run manually or scheduled.

ABAP Test Cockpit integrates with transports, blocking release for high-priority findings. Exemptions can be requested for planned corrections or false positives.

The ABAP Test Cockpit Configurator app allows setting default variants, adjusting priorities, and defining handling of pseudo comments or pragmas.

See Tips and Tricks (ABAP Core) and Assuring Quality of ABAP Code.

Integrated Technical Monitoring Tools

The SAP BTP ABAP environment offers advanced monitoring capabilities to track system and application health, performance, and resource usage. It integrates seamlessly with CI/CD pipelines, enabling automated deployment, quality checks, and proactive issue detection. This ensures smooth operations and supports both developers and administrators in maintaining reliable applications.

See https://help.sap.com/docs/sap-btp-abap-environment/abap-environment/technical-monitoring.

Security

Security in SAP BTP ABAP environment is natively aligned with SAP roles, authentication mechanisms, and governance models. Built-in controls and standards support secure development and operations, protecting applications and data while simplifying user and role management.

See ABAP Development Tools for Eclipse: Security Guide.

Analytics and Core Data Services Support

SAP BTP ABAP environment leverages Core Data Services (CDS) to enable semantic data modeling and advanced analytics. As a developer, you can create high-performance analytical applications directly within the ABAP environment, using CDS views, annotations, and built-in analytical features without leaving the platform.

See Integrating SAP Analytics Cloud.

Full Transport and Lifecycle Management

In the SAP BTP ABAP environment, ABAP code can be transported and versioned using gCTS and abapGit:

- gCTS is the modern, recommended mechanism for moving objects between ABAP systems within a global account and runs under the hood of the Manage Software Components application.
- abapGit is an open-source Git client, supports use cases such as migrating on-premise code to the cloud, transferring code between cloud systems or to on-premise, exporting code when a system is decommissioned, and enabling distributed development or testing scenarios.

Both tools help maintain code quality, track changes, and support collaborative development across ABAP landscapes.

See Transport Mechanisms.

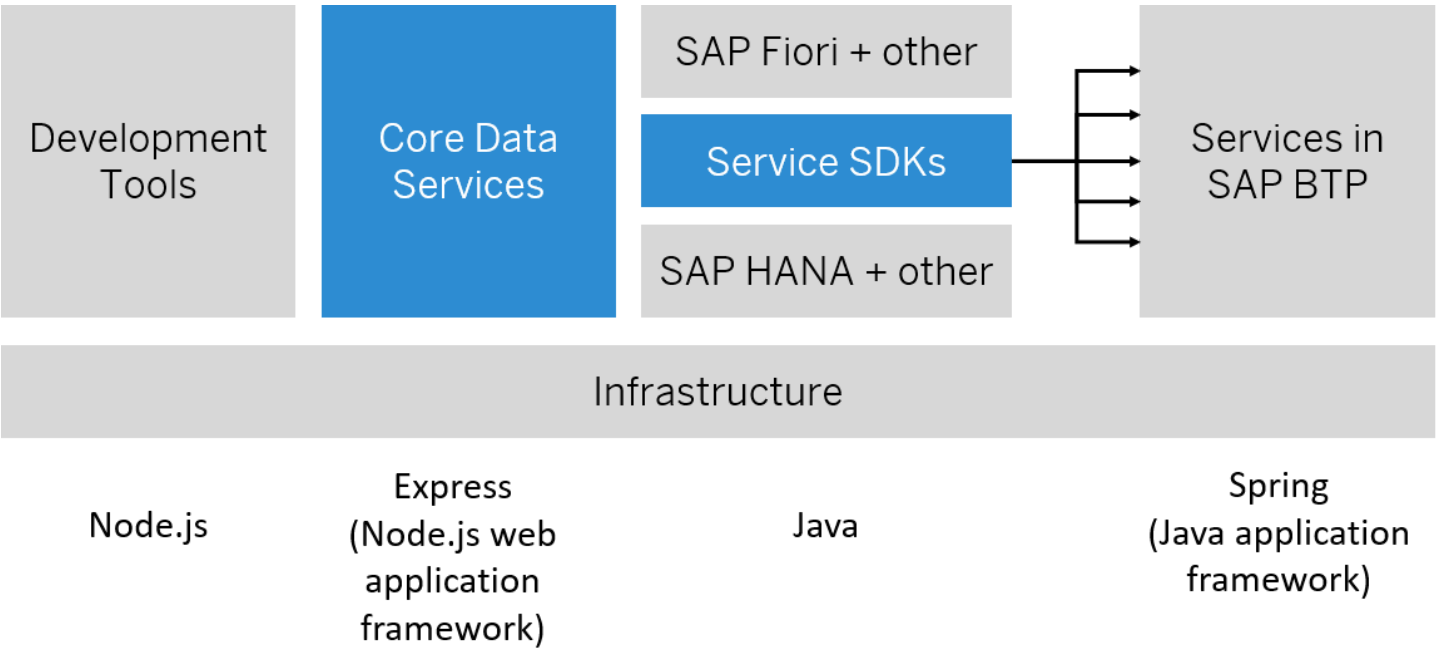
Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes

Overview

The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It guides developers along a path of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

CAP-based projects benefit from a primary focus on domain. Instead of delving into overly technical disciplines, CAP focuses on accelerated development and safeguarding investments in a world of rapidly changing cloud technologies.

The following graphic shows that the CAP framework features a mix of proven and broadly adopted open-source and SAP technologies:



On top of open-source technologies, CAP mainly adds:

- Core Data Services (CDS) as its universal modeling language for both domain models and service definitions.
- Service SDKs and runtimes for Node.js and Java, offering libraries to implement and consume services as well as generic provider implementations serving many requests automatically.

These are the CAP development profiles you can use:

- Development profile: use mocked services for local testing and development.

  You can begin building SAP Cloud Application Programming (CAP) applications locally without needing a subscription or initial setup in SAP BTP. This approach offers several advantages during the early stages of development:

  - Rapid Prototyping

    You can quickly design and test domain models and services locally, enabling faster iteration and feedback cycles.

  - Deferred Integration

    You can postpone the integration with SAP BTP services until later stages, allowing you to focus on core functionality first.

  - Mocked Platform Services

    The CAP development profile includes mocked variants of some SAP BTP services dealing with authentication, database, messaging, and application gateway. These mocks allow applications to run locally without requiring the application to be deployed to SAP BTP.

- Hybrid profile: local development and testing using selected real services

  Over time, you can add things gradually, only when they're needed. For example, you can move ahead to running your applications in close-to-productive setups for integration tests and delivery, without any change in models or code:

  - Run locally, but bind to real services (for example, SAP HANA Cloud, and the Destination service) using cds bind or a local default-env.json that simulates Cloud Foundry or Kyma service bindings.
  - Great for integration tests (for example, real SAP HANA Cloud schema, real authorization flows) before you deploy.

- Production profile: use SAP BTP services and deploy the application to SAP BTP.

Run in SAP BTP, Cloud Foundry or SAP BTP, Kyma runtime with platform-managed service bindings and production-grade settings.

- Real service instances (SAP HANA Cloud, Destination services, and others) are bound by the platform.
- Production configuration for logging, security, scaling, and multitenancy.
- Database migrations done via cds deploy (for example, to SAP HANA Cloud).

See Grow as You Go in the CAP documentation.

Related Information

Strengths and Benefits of the Runtimes and Their Programming Models
Explore
Discover
Design
Deliver
Run and Scale
Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners
Tutorials for SAP Cloud Application Programming Model

Explore

Developing a new application should always serve the purpose of your business, for example making a process more efficient and hence saving costs, making users more effective so they get better results, or supporting a business innovation which gives you a competitive advantage and allows you to make more money.

The explore phase is where the business opportunity is identified at a fairly high level. Typically key stakeholders are interviewed, or use case exploration workshops are conducted. The methods used differ, depending on whether you are looking to optimize existing processes or whether you are looking to innovate your business.

If you are looking to optimize existing processes, then you need to identify the business problem.

Identify the Business Problem

A business problem represents a gap between the desired and actual outcome of a task. It can be any hurdle, situation, or variation that prevents a business from achieving its objectives. A business problem may not have an immediately obvious resolution, and may be masked or hidden by perception and assumption. To define a true problem, you need to remove the factors that obscure the real reason behind the problem.

To solve a problem for an internal or an external customer when using a product, you need to ask questions that can help you understand the customer's needs, expectations, pain points, and feedback. Some examples of such questions are:

- What are the tasks that customers are trying to achieve with a product?
- How do the customers use the product in their daily workflow?
- What are the main challenges or frustrations that the customers face when using the product?
- How can you improve the product to better serve the customers' needs?
- How satisfied are the customers with the product and the customer service?
- What are the benefits or value that the customers get from the product?

- How do the customers measure the success or impact of the product on their goals?

Create a Governance Model

One of the first and most important steps of your journey to the cloud is to establish an appropriate organizational setup and corresponding governance model. A clear and well-thought-out organizational setup makes it easier for your employees to adopt agile processes. See Creating a Governance Model in the SAP BTP Administrator's Guide.

Security Considerations in the Explore Phase

Addressing security in the explore phase when you're shaping the project vision with stakeholders lays a strong foundation for building secure, resilient applications. This early focus helps protect sensitive data, ensures compliance with regulations, and aligns security with business objectives from day one.

At this stage, you're not writing code yet, but you're making critical decisions that influence the security posture of your application. It's the right time to ask key questions, identify potential risks, and plan how to consider security aspects throughout the development lifecycle.

For example, a customer support team designing a ticketing system might identify risks such as data breaches if sensitive incident details or personal information are not properly handled. By proactively addressing these concerns now, you can avoid costly rework later and ensure compliance with data protection laws like the GDPR or HIPAA.

The following table summarizes the security guidelines you need to be familiar with:

Early-Stage Security Guidelines for Application Development

Security Guideline	Description
Define a Secure Software Development and Operations Lifecycle (Secure SDOL) Framework	Establish a secure software development lifecycle (SDLC) framework, using standards like OWASP to guide all phases. See The Secure Software Development and Operations Lifecycle (secure SDOL) at SAP.
Engage and Train	Engage stakeholders to clarify their security expectations and ensure that development teams are trained on secure development principles. See Internal Wiki Page: SDOL-001: Security Training and Role Assignment .
Identify High-Level Security Risks	Conduct a preliminary risk assessment to identify high-level security risks based on the application use case. (For example, compliance, reputation).
Assess Potential Threats	Use threat modeling to identify potential threats relevant to your industry or app type, like SQL injection or insecure authentication. See OWASP Threat Modeling Project .
Plan for Compliance	Determine regulations (for example, GDPR, HIPAA) that apply to your application data handling and storage needs.

Starting your application project with these steps helps ensure that security is considered from the beginning, making it easier to build a safe and compliant application.

Discover

Now that the business opportunity has been identified by the explore phase, you can start to discover the concrete needs of your users as well as getting an understanding of the available technology for your landscape.

Identifying Required Use Cases

Once you have an understanding of the business opportunity you want to address, you next will need to understand in detail what your end user's tasks and challenges are. You need a holistic view of the business roles of your application's users, their jobs to be done, the information they need to get them done, along with their needs and wishes.

To get the relevant insights, you will need to visit your customers onsite or remotely, speak to everyone involved, and if onsite, observe users doing their daily work. You should gain a clear understanding of all the business roles involved, their typical tasks and pain points, and the sequence of activities.

The goal of the discover phase is to gain a thorough understanding of the user's needs, independent of any existing application or technology. In other words, which information do they need? When do they need it? What business outcomes do they need to achieve?

By the end of the discover phase, you should know what needs to be achieved – without knowing how you want to achieve it, that is addressed in the design phase.

For more information about user research during the Discover phase, have a look at the User Research Resources. The methods outlined there contain the most commonly practiced user research methods at SAP.

Security Considerations in the Discover Phase

Addressing security during the discover phase—when you're gathering detailed requirements and exploring technical options—ensures that protection is a core part of your application's foundation. This phase is about understanding user needs, assessing feasibility, and building early prototypes, making it the ideal time to identify potential risks and plan for secure implementation.

At this stage, you define your application's features and technical requirements. For example, a retail company planning an e-commerce application must ensure secure payment APIs are in place to protect transaction data—an early decision that directly impacts the system's architecture. By evaluating third-party components, identifying data flows, and conducting early security testing on prototypes, you can uncover vulnerabilities before development begins. This proactive approach saves time, reduces cost, and supports compliance with regulations like the GDPR.

The following table summarizes the security guidelines you need to be familiar with:

Security in the Discover Phase: From Requirements to Risk Reduction

Security Guideline	Description
Map Data Flows for Security	Identify and document how sensitive data moves through the system, including user inputs, API calls, and database interactions, to identify potential security risks. You can use OWASP Threat Dragon to create data flow diagrams for this purpose.
Establish Secure Architecture Principles	Define architectural guidelines to help ensure that the system is built with security in mind, such as microservice isolation, the principle of least privilege, and multi-layered security approach (defense-in-depth).
Validate Third-Party Service Security	Review the security certifications and practices of third-party services (for example, APIs, libraries) you integrate, and align with SAP BTP Security recommendations.

Security Guideline	Description
Incorporate Security in Prototypes	Include basic security features, like input sanitization or mock authentication, in your prototypes to test their effectiveness early.
Plan for Data Security During User Research	Protect user data collected during research (for example, interviews, observations) by anonymizing inputs and securing storage.
Understand Boundary Conditions	Understand boundary conditions for the new application, like integration into the existing system landscape (regarding IAM, Audit log).

By implementing these guidelines in the discover phase, you improve the security of your application, laying the groundwork for a resilient and compliant application. See Setting Up Your Security and Compliance Model.

Design

Good user experience (UX) goes beyond aesthetics and can have substantial business benefits, both financially and in overall efficiency. A great UX can improve productivity, enhance data quality, reduce training costs, and decrease the number of change requests.

A positive UX boosts user satisfaction, customer loyalty, and overall application adoption. Users are more likely to embrace an application that is intuitive and easy to use.

However, achieving a good UX involves overcoming several common challenges. Users often struggle with finding information, dealing with scattered data, excessive clicks, poor data entry interfaces, and unhelpful error messages. Beyond design, bad performance and unhelpful data visualization can also frustrate users.

The essential approach to designing a good application is design thinking, which emphasizes understanding users' needs through observation and feedback. Iteration is key: quickly sketch prototypes, gather user feedback, and refine your designs before moving to development. This ensures that the final product truly meets users' needs and optimizes resources effectively.

For organizations working with multiple applications, consistency is crucial. Users expect uniform visual design and interactions, easy access and navigation between applications, and common functions readily available. To achieve this, a design system is necessary. It ensures consistent user experience across various channels such as desktop, mobile, and so on.

In large-scale development, you also need robust frameworks, libraries, and user experience integration runtimes to manage hundreds of applications efficiently. Proper authorization tools are essential to control access for an extensive user base.

The SAP Fiori design system along with its enabling UI tools and technologies gives you the above: a comprehensive design system for web and mobile, along with a suite of development tools and UI frameworks, enabling SAP to provide a consistent UX with good usability across all SAP products, and enabling you to do the same for your applications. Using SAP's design system enables you to bring together business value, modern technology and excellent UX, thus benefitting business and users alike.

Find out more about user experience design and technology design in the next sections:

- User Experience Design
- Technology Design

Typical Development Use Case Patterns

We recommend implementing the following use case patterns on SAP BTP:

- Automate processes across backend systems

  This pattern provides low code/no code capabilities to automate processes and is meant for business experts and developers.

- Build web and mobile applications

  This pattern provides mobile-native and web development capabilities and is meant for business experts and developers.

- Develop full-stack applications

  This pattern is meant for professional developers and has the following flavors:

  - Full-stack single-tenant applications

    This pattern is meant for customers and implementation partners enabling them to develop full-stack applications on SAP BTP.

  - Full-stack multitenant Software-as-a-Service (SaaS) applications for partners that are independent software vendors

    This pattern is meant for SAP partners enabling them to develop full-stack SaaS applications on the SAP BTP and to distribute these applications to their customers. Partners can easily onboard multiple customers (tenants) onto a single application with strictly separated data. This approach dramatically reduces the total cost of ownership at cloud scale.

    The Poetry Slam Manager partner reference application on GitHub provides a comprehensive example to build, deploy, and run such a scalable SaaS solution. See Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application 'Poetry Slam Manager' .

  - Tenant-specific extensions for partners and customers to add or modify features in multitenant SaaS solutions, such as the Poetry Slam Manager, without affecting the core product used by all tenants.

    The Partner Reference Application Extension "Catering Management" demonstrates a robust framework for delivering highly-tailored functionalities for specific customers and industries, using SAP BTP and CAP extensibility as a foundation. See Tutorials and Reference Sample for Custom Extension of a Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application 'Catering Management' .

CAP Design Principles

SAP Cloud Application Programming Model (CAP) focuses on the domain of the application, by capturing domain knowledge and intent instead of imperative coding. This means:

- Close collaboration of developers and domain experts in domain modeling.
- Out-of-the-box implementations for best practices and recurring tasks.
- Platform-agnostic approach to avoid lock-ins, hence protecting investments.

Agnostic Design

CAP avoids technology lock-ins through higher-level concepts and APIs, which abstract low-level platform features and protocols to a large extent. In particular, this applies to:

- Platform-specific deployment approaches and techniques
- Platform-specific identity providers and authentication strategies
- Onboarding and offboarding of tenants in Software-as-a-Service (SaaS) solutions and tenant isolation

- Synchronous protocols like REST, OData, or GraphQL
- Asynchronous channels and brokers like SAP Event Mesh, Message Queue, or Kafka
- Different database technologies including SQL and NoSQL

These abstractions allow CAP to quickly adapt to new emerging technologies or platforms, without affecting the application code.

Open and Opinionated Design

While CAP certainly gives opinionated guidance, it does this without sacrificing openness and flexibility. You as a developer stay in control of which tools or technologies to choose, or which architecture patterns to follow.

- All abstractions follow a glass-box pattern that allows unrestricted access to lower-level things, if necessary.
- Best Practices served out of the box with generic solutions for many recurring tasks.
- Out-of-the-box support for SAP Fiori and SAP HANA.
- Dedicated tools support provided in SAP Business Application Studio, and Visual Studio Code or Eclipse.

Domain-Driven Design

CAP uses Core Data Services as its ubiquitous modelling language, with Core Data Service Aspects separating core domain aspects from generic aspects. Core Data Service's human-readable nature fosters collaboration of developers and domain experts.

As Core Data Service models are used to fuel generic providers — the database as well as application services — CAP ensures that the models are applied in the implementation. And as coding is minimized you can more easily refine and revise their models, without having to refactor large boilerplate code bases.

Security Considerations in the Design Phase

In the design phase, you define the user experience and technical architecture of your application. This includes applying domain-driven design principles and following the best practices of the programming model you have chosen, CAP or ABAP Cloud, to create modular, user-centric solutions. Prioritizing security at this stage ensures your application is built on a resilient architecture that protects sensitive data and supports regulatory compliance.

Design decisions shape how users interact with your application and how its internal components communicate. For example, a retail company designing an e-commerce platform might implement a secure checkout process that enforces user authentication and encrypts payment data. By embedding security practices early—such as defining role-based access controls, securing APIs, and validating input—you reduce the risk of vulnerabilities being introduced into the system.

Proactively addressing security in the design phase not only streamlines compliance with regulations like the GDPR but also builds user trust and reduces the cost of future remediation.

The following table summarizes the security guidelines you need to be familiar with:

Designing Secure Applications from the Ground Up

Security Guideline	Description
Design Secure User Interfaces	Incorporate security features like secure authentication and input validation when designing the UI using SAP Fiori.

Security Guideline	Description
Define Access Control Models	Securely authenticate users and technical clients with methods such as OpenID Connect and OAuth. Design role-based access controls (RBAC) and attribute-based access control (ABAC) in domain models to restrict data access.
Secure API and Service Interactions	Design APIs with authentication (for example, OAuth) and encryption (for example, TLS) to protect data exchanges.
Plan for Secure Extensibility	Ensure extensions (for example, custom logic) are isolated and validated to prevent introducing vulnerabilities.
Validate Domain Models for Security	Review Core Data Services (CDS) models to ensure sensitive data is protected and access is restricted.

These guidelines help you to implement an application that integrates security into its UI, architecture, and domain models, supporting the development of a compliant and trustworthy solution.

See:

- The Secure Software Development and Operations Lifecycle (secure SDOL) at SAP
- SAP BTP Guidance Framework

User Experience Design

Once you have a sufficient understanding of your users' needs, by successfully completing the Discover phase, you can start designing the user experience of your application.

User experience goes beyond the user interface, and covers all aspects of the interactions a user has with the application: getting the right information shown at the right time, allowing the user to get their work done with only a few clicks needed, providing a fast and accessible solution, supporting all the international languages your users use in their systems.

A product with good **usability** gives users a good user experience.

Usability is defined by the degree to which these aspects are provided to a user to get their tasks or jobs done:

- Effectiveness
- Efficiency
- Ease of learning (Intuitiveness)
- Error tolerance
- Satisfaction

Consistency contributes to ease of learning and efficiency.

When providing users with applications to complement or extend your SAP landscape, you will need to ensure that your applications have a similar look and feel to SAP's standard applications, and hence provide a consistent user experience for your users. This helps to make the applications intuitive to use (ease of learning) and helps with efficiency, since they look and behave the same as other applications users are already familiar with. For this reason, we recommend that you follow the SAP Fiori for Web Design Guidelines. In particular, follow the Best Practices for Designing SAP Fiori Apps.

In addition to consistency, ensure that your application is accessible, by following the guidance provided by Accessibility in SAP Fiori. Also, be sure to provide language support for all relevant languages as well as providing suitable user assistance.

When designing your application, make sure that you continue with your user research, so that you can get quick feedback about design ideas and artifacts. Iterate often, with the motto “Fail fast, fail early” – in other words, sketch out solutions, get feedback, try again, and keep on iterating until you are confident that your design is what your users need.

For more information about user research during the design phase, have a look at the User Research Resources related to the Design phase. The methods outlined there contain the most commonly practiced user research methods at SAP.

Technology Design

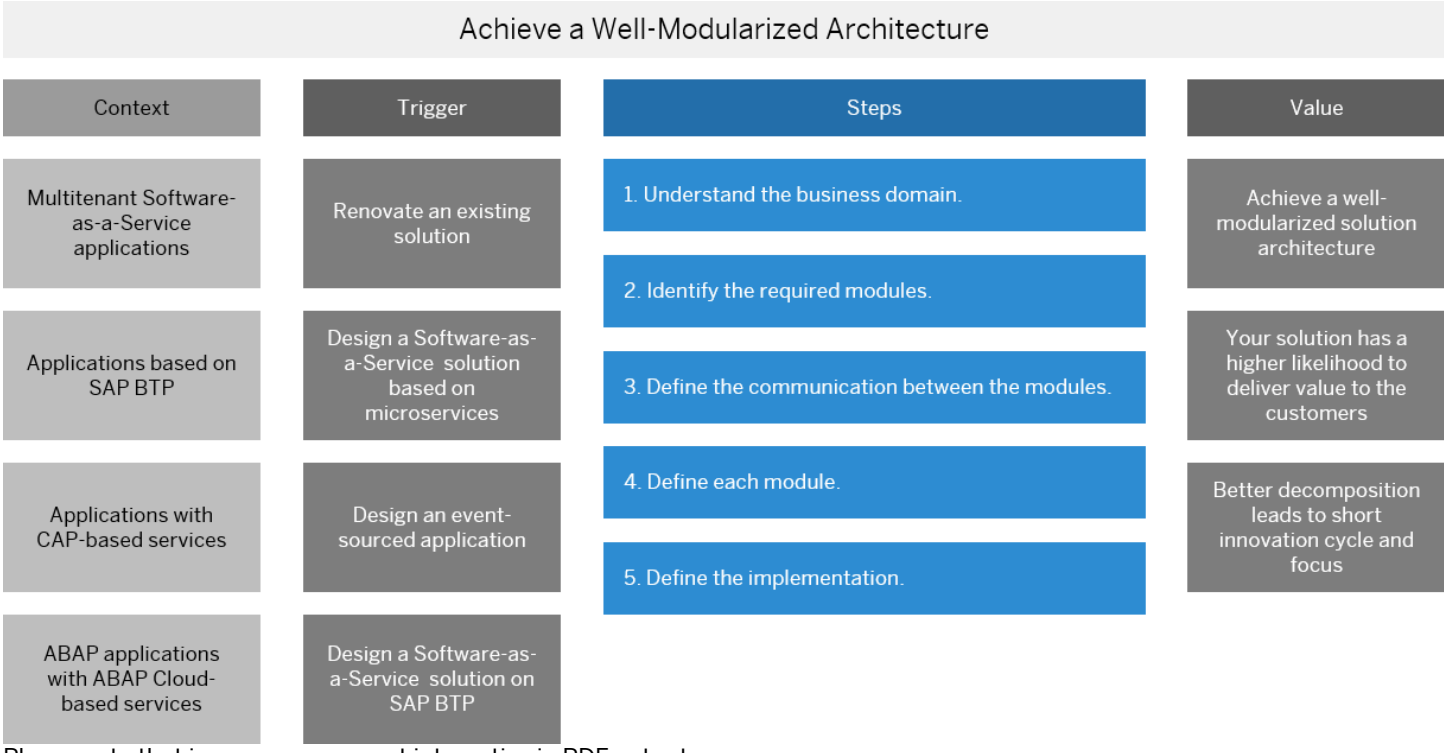
Before starting to implement an application on SAP BTP, it's better if you do some planning and research beforehand. See:

- Explore
- Discover

Deliver a Cloud Application from Concept to a Final Product

The columns in the following graphic show for which use cases using a well-modularized architecture suits best, what is the motivation, the steps and the benefits.

The image is interactive: you can click the column headings to take you to the relevant part of the relevant section below.



Please note that image maps are not interactive in PDF outputs.

Context

Let's assume the following scenario: you and your team are tasked to deliver a multitenant application on SAP BTP or SAP S/4HANA Cloud Public Edition and you want to base the application either on SAP Cloud Application Programming Model (CAP) or on ABAP Cloud.

Trigger

Use the following triggers as a motivation to engage in sophisticated design process:

- Renovating, evolving or reimplementing an existing solution

  This involves transforming a legacy system into a modern, more efficient and scalable solution. Legacy systems often pose difficulties in terms of maintainability, integration with newer technologies, and can be a bottleneck for a business's growth. This transformation process could involve refactoring code, adopting new architecture patterns like microservices, or completely reimplementing the software with a new technology stack.

- Designing a new Software-as-a-Service solution based on microservices

  Software-as-a-Service is a model where software is provided over the internet on a subscription basis. When designing a new Software-as-a-Service solution based on microservices, the system is decomposed into a collection of services that are independently deployable, scalable, and can be developed by different teams. Each microservice corresponds to a specific business capability and can be developed, deployed and scaled independently. This architectural style promotes flexibility and scalability, and it can speed up the development process by enabling parallel development across teams. It's important to design these services around the business capabilities.

- Considering more asynchronous behavior and event-sourced applications

  Asynchronous behavior in an application refers to operations that can be executed independently and don't block the flow of the program. This makes the application more responsive and can improve system-wide performance. Event sourcing is an architectural pattern that stores changes to the application state as a sequence of events, rather than just storing the current state. This can provide a full audit trail, simplify debugging by allowing the state of an application at any point in time to be recreated, and can facilitate asynchronous, message-driven architectures. By integrating these into the system, it can enhance the system's scalability and reliability.

- Designing a Software-as-a-Service application on SAP BTP

  When designing a Software-as-a-Service application on SAP BTP, the application is built to be delivered over the internet and leverages the services provided by SAP BTP. This can include data storage services, AI and machine learning capabilities, analytics and more. Using such a platform can speed up development time, as many of the underlying infrastructure considerations and services are taken care of by the platform.

Steps

Follow these steps to ensure smooth implementation of your application:

1. Understand the business domain.

   Understanding the business domain is of the utmost importance. Hence, before starting your development, make sure that you understand the business opportunity outlined in the Explore phase, and that you have completed the Discover phase so that you have a thorough understanding of the user needs and required use cases. This will ensure that you create software that is useful, flexible, and capable of evolving alongside the business.

2. Identify the required modules.

   Identifying the required modules means breaking down the overall system into smaller, manageable, and functionally cohesive units. Each module is designed to encapsulate a specific portion of the domain model and is typically aligned with a specific sub-domain or business capability. By identifying the modules, developers can ensure separation of concerns and make the system more maintainable. Each module should be able to operate independently, reducing dependencies and allowing for parallel development and easier testing.

3. Define the communication between the modules.

   Once the modules are identified, the next step is to define how they will interact with each other. This involves establishing interfaces, contracts, or APIs for communication. This is often guided by the concept of context mapping, which helps to define the relationships and interactions between different related contexts. Communication between modules should be

carefully managed to preserve the autonomy of the modules. The communication could be synchronous or asynchronous, and could be based on various patterns like request/response, events, or commands.

4. Plan the implementation.

   After you have an understanding of the business domain, have identified the required modules, and defined how they will communicate, the next step is to plan the implementation. This is the moment where you decide which technologies, patterns, and practices to use to build each module. This can include selecting the programming languages, databases, message brokers, and other tools that will be used. It also involves choosing the architectural patterns (like layered architecture, hexagonal architecture, or microservices) and design patterns that will be applied. The implementation should be guided by the principles of the domain model, encapsulating the business logic in entities, value objects, aggregates, and domain events. Additionally, the implementation should consider non-functional requirements like scalability, performance, security, and maintainability.

Value

If you follow the steps mentioned in the previous section, you will achieve the following benefits:

- Well-modularized architecture

  In the context of software development, a well-modularized architecture refers to the design of a system where the different functionalities and responsibilities are partitioned into distinct modules. Each module encapsulates a specific subset of the system's behavior and can be developed, tested, and maintained independently of the others. Modules are often designed to be loosely coupled, meaning that they interact with each other through well-defined interfaces, but their internals are hidden from each other. This encapsulation increases the maintainability of the system because changes inside one module don't impact the others. It also improves the scalability of the system, as individual modules can be scaled up or down based on their individual requirements. Finally, modularization can increase the speed of development because different teams can work on different modules at the same time without interfering with each other.

- Higher likelihood to deliver value to your customers

  The focus is on understanding the business domain and creating a software model that reflects and serves that domain. By aligning the software closely with the business needs, there's a higher likelihood of delivering value to the customer. This comes from a deep understanding of the business, its rules, and its needs. This understanding guides the software development process, ensuring that the software solves the right problems and provides the features that the business actually needs. As a result, the software is more likely to be valuable to the end users, who are the ones interacting with the business domain on a daily basis.

- Better decomposition that would lead to shorter innovation cycles and clearer focus

  Decomposition in software design is the process of breaking down a complex system into smaller, manageable parts or components. Better decomposition, often achieved through practices like domain-driven design and microservices architecture, allows for clearer focus on individual parts of the system and can lead to shorter innovation cycles. With a well-decomposed system, teams can focus on specific components or services, understand them thoroughly, and make improvements or innovations without having to understand the entire system. This can significantly shorten the time it takes to go from an idea to a working feature, because changes are localized to specific parts of the system and don't require broad, system-wide understanding or modifications. Moreover, better decomposition can lead to clearer focus because each component or service can be designed around a specific business capability or domain concept. This allows the team working on that component to focus on the specific business problems and user needs related to that capability or concept, rather than being distracted by other parts of the system. This kind of focus can lead to better solutions and more valuable software.

Understand Modularization in General

Modularization is a general programming concept where you separate program functions into independent pieces, called modules. These modules then act like building blocks, with each block containing all the necessary parts to execute a piece of the

functionality. When the modules are put together, they make up the executable program.

A modular code is:

- **Easier to maintain**

  It's easier to read, understand, debug, and modify code that is modularized. It also reduces the risk of introducing errors or conflicts when changing or adding new features. A good modularization also enables reuse of existing modules across different products, which saves time and effort.

- **Easier to test**

  Each module can be tested individually and independently from the rest of the system. This also facilitates automated testing and continuous integration, which improve the quality and reliability of the software product.

- **Easier to scale**

  Each module can be deployed, updated, or replaced separately, without affecting the whole system. This also allows for parallel development and faster delivery of new features or bug fixes. A good modularization also supports distributed and cloud-based architectures, which enhance the performance and availability of the software product.

- **More reliable**

  Each module has a clear and well-defined interface and responsibility. This also reduces the coupling and dependency between modules, which minimizes the impact of failures or errors in one module on the rest of the system. A good modularization also enables fault tolerance and recovery mechanisms, which improve the resilience and robustness of the software product.

- **More adaptable**

  Each module can be easily modified or extended to meet changing requirements or customer needs. This also increases the flexibility and diversity of the software product, as different modules can be combined or configured in various ways.

- **Easier to evolve**

  Each module can be evolved independently or collaboratively, without affecting the overall functionality. This also promotes innovation and creativity, as new modules can be created or reused to generate novel solutions or features.

Understand Modularization in the Context of Microservice-Based Architecture

The right modularization is very important in the context of microservices, because microservices are essentially a form of modular software development. Microservices aim to create small, focused, and independent services that communicate through well-defined interfaces and protocols. Each service is responsible for a specific functionality or domain, and can be developed, deployed, and scaled independently from other services. This way, microservices enable a high degree of modularity, flexibility, and agility in software development.

However, not all modularizations are equally effective or beneficial for microservices. A poor modularization can lead to problems such as:

- **High coupling and low cohesion**

  If the services are not well-defined and separated, they may depend on each other too much, or have too many responsibilities. This can result in tight coupling between services, which makes them hard to change, test, or reuse. It can also result in low cohesion within services, which makes them complex, inconsistent, or redundant. High coupling and low cohesion reduce the maintainability, testability, and reliability of microservices.

- **Service sprawl and fragmentation**

If the services are too fine-grained and numerous, they may become difficult to manage, monitor, and coordinate. This can result in service sprawl, which increases the operational complexity and overhead of microservices. It can also result in service fragmentation, which reduces the performance and availability of microservices. Service sprawl and fragmentation reduce the scalability and adaptability of microservices.

- **Lack of alignment and consistency**

  If the services are not aligned with the business domain and goals, they may not deliver the expected value or quality, which reduces the relevance and effectiveness of microservices. It can also result in lack of consistency, which reduces the usability and interoperability of microservices. Lack of alignment and consistency reduce the evolution and innovation of microservices.

The right modularization is essential if you want to benefit from using microservices. For example:

- **Maintain modules easily**

  The right modularization helps you create simple, consistent, and reusable services that are easy to read, understand, debug, and modify. It also enables the isolation and decoupling of services, which reduces the risk of errors or conflicts when changing or adding new features.

- **Test each service individually and independently from the rest of the system**

  The right modularization enables you to use automated testing and continuous integration, which improve the quality and reliability of the software product.

- **Deploy, update, or replace each service separately, without affecting the whole system**

  The right modularization supports distributed and cloud-based architectures, which enhance the performance and availability of the software product.

- **Have clear and well-defined interface and responsibility of each service**

  The right modularization reduces the coupling and dependency between services, which minimize the impact of failures or errors in one service on the rest of the system. It also enables fault tolerance and recovery mechanisms, which improve the resilience and robustness of the software product.

- **Modify or extend each service to meet changing requirements or customer needs**

  The right modularization increases the flexibility and diversity of the software product, as different services can be combined or configured in various ways.

- **Evolve each service independently or collaboratively, without affecting the overall system functionality**

  The right modularization promotes innovation and creativity, as new services can be created or reused to generate novel solutions or features.

Learn Why Domain-Driven Design Is an Adequate Approach

Domain-driven design is a major software design approach, focusing on modeling software to match a domain according to input from that domain's experts. Under domain-driven design, the structure and language of software code (class names, class methods, class variables) should match the business domain.

Domain-driven design helps achieving the traits of a good modularization by:

- **Placing the project's primary focus on the core domain and domain logic**

  Domain-driven design emphasizes close collaboration between domain experts and developers to create software that accurately models the problem domain. By focusing on the domain's core concepts, relationships, and behavior, domain-driven design enables you to create more robust, flexible, and maintainable systems.

- **Building complex designs on the model of the domain**

  Domain-driven design helps you to define the boundaries of each module by identifying the core concepts, aggregates, and their associated business logic. This enables teams to develop and maintain these modules independently, promoting scalability and modularization within the overall system.

- **Initiating a creative collaboration between technical and domain experts to iteratively refine a conceptual model that addresses particular domain problems**

  Domain-driven design fosters a common language shared by domain experts, users, and developers: the ubiquitous language. The ubiquitous language is used in the domain model and for describing system requirements. This enhances the communication and understanding among the stakeholders, and facilitates the adaptation and evolution of the software product.

Learn About the Domain-Driven Design Starter Modeling Process

The domain-driven design starter modeling process is a process for learning and applying domain-driven design in practice. It covers eight steps from aligning with the business model to coding the domain model. It is flexible and iterative, and suitable for beginners who want to master domain-driven design. See the Domain-Driven Design Starter Modeling Process GitHub repository that is part of the Domain-Driven Design Crew GitHub organization.

Check How Complex Your Scenario Is

Check which requirements your scenario has and sum up the points for the respective requirement. If your score is 7 or higher, you should evaluate how to leverage the concept and tools of the domain-driven design in your project.

We also recommend that you follow the domain-driven design modeling process.

See Domain-Driven Design Starter Modeling Process .

Scenarios Ordered by Complexity

Scenario Requirements	Points	Comments
For applications that work with the four functions: create, read, update, delete (CRUD), and only perform simple database queries, no special design is required. A simple database tool with a user interface suffices.	0	A simple application does have the complexity to benefit elaborated design methods.
You have a simple application if it has up to 30 use cases. You can build it with a CAP application that works with the four functions: create, read, update, delete (CRUD).	1	Just for reference, this means your application does not have more than 30 functions.

Scenario Requirements	Points	Comments
Your application has between 30 and 40 use cases.	2	Make sure you don't underestimate any hidden complexities. However, having many methods or functions doesn't always mean high complexity. They might be simple and well-structured. It's when a class or module has many responsibilities, indicated by methods or functions that perform widely varying tasks, that complexity becomes a challenge.
Your application may not be complex now, but it could grow later.	3	Talk to domain experts to discuss the complexity of your application. The complexity increases with the number of features that the stakeholders need.
Your application will change a lot over time, and you don't know what kind of changes will happen.	4	You require a sophisticated design method to manage the complexity of refactoring your model over time.
You are building an application in a new and exciting domain that nobody has done before. This means it could be complex and challenging.	5	You need to collaborate with domain experts and try different models to make your application work.

Related Information

Domain-Driven Design Crew GitHub Organization
Domain-Driven Design Community at SAP

Deliver

Now that you have completed your design for the required user experience of your application along with the appropriate application architecture, it's time to deliver what you have designed. In other words, set up the development and test landscape, develop and test your application and finally release it to the productive environment. This applies to the first release of the application. From then on, you could apply a similar approach for subsequent releases, or preferably, move to a mode of continuous feature delivery, whereby additional features are researched with users, designed, developed and then deployed once ready, without having to be bundled in a new release of the entire application. See:

- You start by **setting up the landscape** in SAP BTP. See Set Up.
- Then, you continue with the development, choosing the **programming language, runtimes, tools** and others. Have in mind that depending on the programming model, there are some specifics. See:
  - Develop
- There are **sample applications** presented in tutorials and end-to-end scenarios, called missions:
  - Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners
  - Tutorials for SAP Cloud Application Programming Model

Set Up

Configuring the landscape setup in its essence is an administrative task described in detail in the SAP BTP Administrator's Guide.

Set Up Your Account Model

The hierarchical structure between global accounts, directories, and subaccounts lets you define an account model that accurately fits your business and development needs. See Setting Up Your Account Model in the SAP BTP Administrator's Guide.

Set Up the SAP BTP Infrastructure Automatically Using the Terraform Provider

The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP. By leveraging this provider, you can simplify and streamline the deployment and maintenance of SAP BTP services and applications. See Automated SAP BTP Infrastructure Setup.

Manage Users and Members

User accounts enable users to log on to SAP BTP and access subaccounts and use services according to the permissions given to them. In this context, it's important to understand the difference between the two types of users that we refer to: platform users and business users.

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot applications and services on SAP BTP.

Business users use the applications that are deployed in SAP BTP. For example, the end users of SaaS applications or services.

See User and Member Management in the SAP Business Technology Platform documentation.

Develop

When using SAP Cloud Application Programming Model (CAP), you can choose the right runtime of your use case: SAP BTP, Cloud Foundry runtime or SAP BTP, Kyma runtime. To make an informed decision, see:

- SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP
- Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Metering:

- Cloud Foundry runtime: Metering starts when you start using the runtime memory, for example, when you deploy an application.

  If your global account uses the consumption-based commercial model and you want to understand how the consumption of SAP BTP, Cloud Foundry runtime is calculated, see Consumption Monitoring in the SAP BTP, Cloud Foundry runtime documentation.

- Kyma runtime: Metering starts when you enable the Kyma runtime. This triggers the creation of a dedicated Kubernetes cluster, and the cluster is metered from the start, before you have deployed any applications.

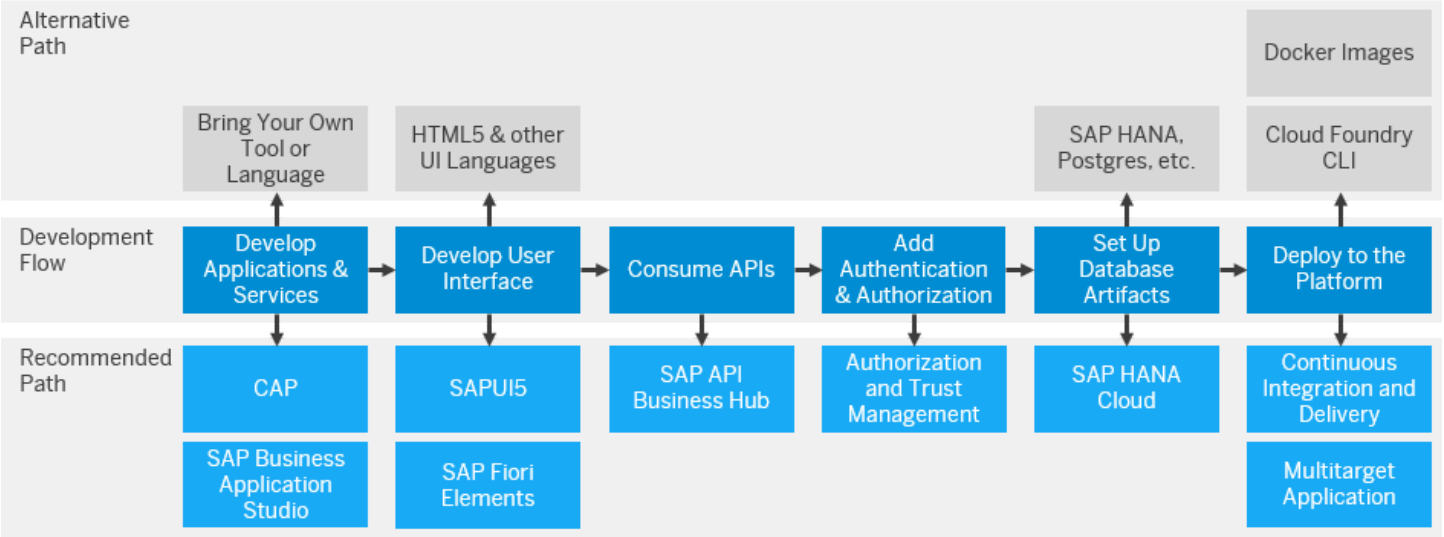
For Cloud Foundry , we recommend that you create multitarget applications for managing dependencies more easily. For more information, see Using Multitarget Applications to Manage Dependencies.

For best practices, guidelines and step-by-step instructions, see Development.

Development Options Overview

The following graphic is designed to help you find the information you need for your programming purposes. The bottom row represents the tools, frameworks, services, and deployment options recommended by SAP. If you want full flexibility you can also bring your own development tools and languages, as shown in the top row.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

Development Guidelines

To ensure consistency and foster collaboration between developers, we recommend that you define guidelines that include standards for programming principles, code styles, and naming conventions.

SAP provides the following official guidelines:

- SAPUI5 Guidelines
- SAP Fiori Design Guidelines

SAP BTP and Third-Party Cookies

The deprecation of third-party cookies by browser manufacturers might impact applications running on SAP BTP. For example, logon scenarios in your application might fail due to session management and authentication issues.

Third-party cookies are set by domains other than the one a user is directly visiting. They are commonly used for online advertising and tracking user behavior across multiple websites. In SAP BTP, SAP uses third-party cookies for service management and single sign-on.

Prepare your applications for the third-party cookie deprecation and test them. See Preparing and Testing Your Solution for Third-Party Cookie Deprecation.

All the available solutions are listed in the SAP Note 3409306 - Removal of Third-Party Cookies in Google Chrome and Microsoft Edge Browser.

Tutorials and End-to-End Scenarios

There are tutorials and end-to-end scenarios developed to show you how to apply the recommended path for the Cloud Application Programming Model (CAP). All these tutorials and scenarios are based on the Incident Management sample

application. For a full list of the tutorials and their use cases, see Tutorials for SAP Cloud Application Programming Model.

Related Information

Programming Languages, Tools, and APIs
Security Considerations for Applications
Developing Resilient Applications
Using Multitarget Applications to Manage Dependencies
Extending Existing SAP Solutions Using SAP BTP
Performing UI, Usability, and Unit Tests
Connecting to Remote Systems

Programming Languages, Tools, and APIs

SAP BTP provides numerous tools for your development project when using SAP BTP, Cloud Foundry runtime or SAP BTP, Kyma runtime with CAP.

The following subsections give you more information about:

- Programming Languages
- Continuous Integration and Delivery (CI/CD)
- Tools Available for SAP BTP Multi-Cloud Foundation
- APIs

Programming Languages

SAP BTP supports many different programming languages; the availability of each depends on the development runtime you're using.

Environment	More Information
Cloud Foundry runtime	Development in the Cloud Foundry Environment
Kyma runtime	Development in the Kyma Environment

Related Information

SAP BTP, Serverless Runtime - Development

Continuous Integration and Delivery (CI/CD)

Configure and run predefined continuous integration and delivery (CI/CD) pipelines that automatically build, test, and deploy your code changes to speed up your development and delivery cycles.

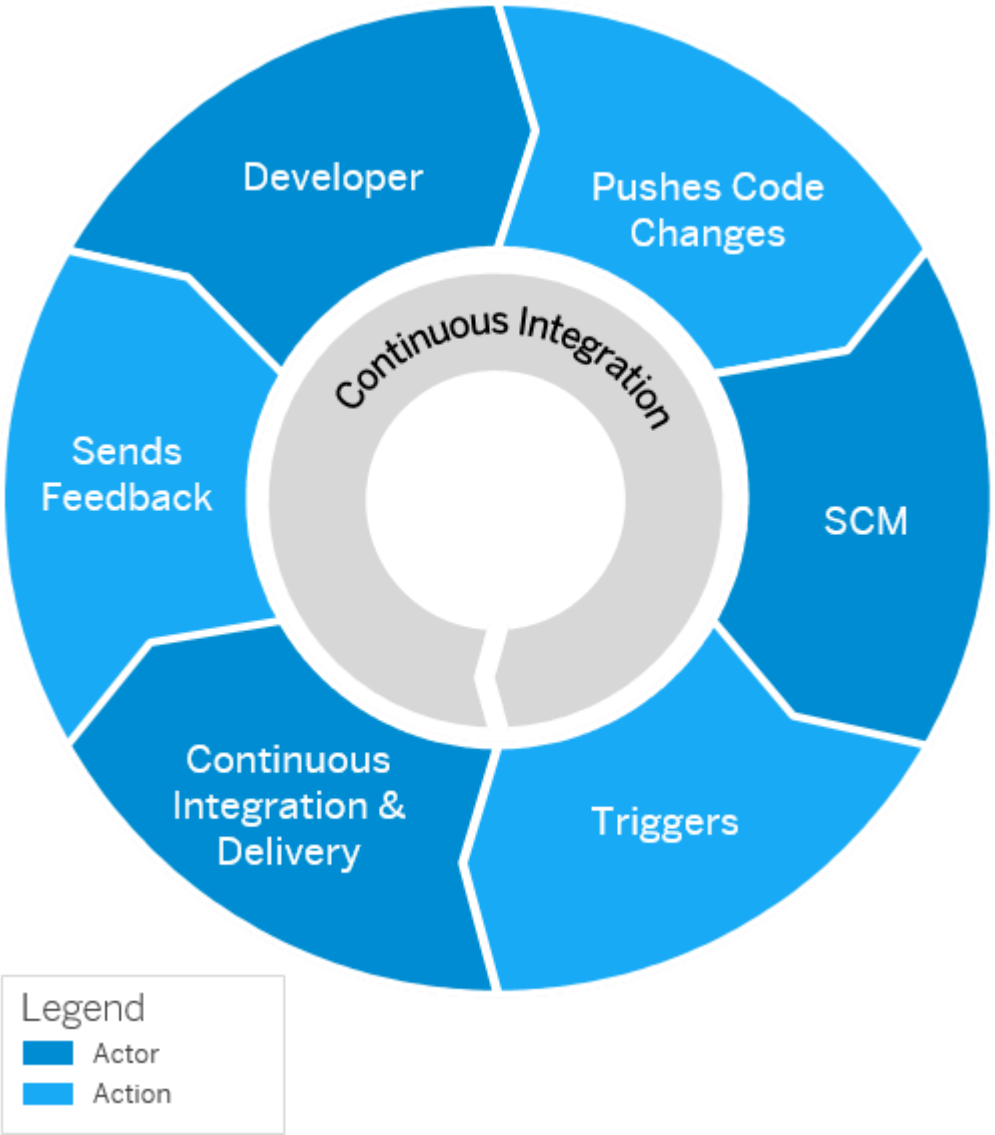
i Note

For links to all SAP solutions for CI/CD, blog posts, presentations, and tutorials, have a look at our Continuous Integration and Delivery by SAP overview.

Continuous integration (CI) describes a software development process, in which various team members integrate their contributions frequently into a single main line. Before each integration, the changes are verified through builds and automated testing. Thereby, you can detect errors as quickly as possible and prevent integration problems before completing the development.

The **continuous delivery (CD)** concept expands on the one of continuous integration. It adds the aspect that any change that has successfully passed the tests is immediately ready to be deployed to production, both from a technical and a qualitative point of view.

The following graphic shows the basic flow for continuous integration and delivery:



Continuous Integration Basic Flow

For more information about the continuous integration and continuous delivery concepts, see What Are Continuous Integration and Continuous Delivery?.

Use

Depending on your use case, you can choose between different CI/CD pipelines to help you implement continuous integration and delivery in your software development.

SAP Continuous Integration and Delivery lets you configure and run predefined pipelines for the development of the following applications:

- Configure a Cloud Foundry Environment Job

  Create an SAP Continuous Integration and Delivery job for SAP Fiori and SAP Cloud Application Programming Model projects in the Cloud Foundry environment.

- Configure an SAP Fiori for the ABAP Platform Job

  Create an SAP Continuous Integration and Delivery job for SAP Fiori projects for the ABAP platform.

- Configure an SAP Integration Suite Artifacts Job

  Create an SAP Continuous Integration and Delivery job for SAP Integration Suite artifacts.

To learn more about the CI/CD pipelines supported by SAP Continuous Integration and Delivery and the stages each pipeline can comprise, see Configuring Jobs.

Get Started with CI/CD

SAP Continuous Integration and Delivery provides an easy, UI-guided way to set up the service and configure and run your pipelines, without hosting your own Jenkins instance.

To set up SAP Continuous Integration and Delivery:

1. Enable the service in the SAP BTP cockpit.
2. Assign either the **Administrator** or **Developer** role to your user.
3. Enable the API usage to connect SAP Continuous Integration and Delivery to other services, if necessary.

To configure SAP Continuous Integration and Delivery:

ℹ Note

Only administrators of SAP Continuous Integration and Delivery can configure the service.

1. Configure credentials for connecting SAP Continuous Integration and Delivery to other services (for example, GitHub, GitLab, Bitbucket Server, or Azure Repos to clone your sources, and SAP BTP to deploy your built application).
2. Add your repository.

Now you can create and modify your CI/CD jobs and monitor their outcome. If you want to automate your builds, you can configure a webhook between your repository and the service. You can create and modify timed triggers for your jobs, if necessary.

For more information, see Initial Setup and Configuration.

Automate the Provisioning of Resources Needed to Create a Kyma Cluster

Integrate the Terraform module, which uses the Terraform provider for SAP BTP, into CI/CD pipelines, so you can automate the provisioning of resources needed to create a Kyma cluster, deploy applications, run tests, and tear down the resources when they are no longer required. As there is no need to have resources running continuously, this approach reduces operational costs.

See Terraforming Kyma Runtimes.

Learn and Get Certified

Depending on your learning goals and level of expertise, you can choose from the following offerings:

- Exploring DevOps with SAP BTP

  Learn how to efficiently apply DevOps practices to your development projects on SAP BTP.

- Efficient DevOps with SAP

  This openSAP course introduces general DevOps approaches and key principles. Learn about the basic CI/CD principles (week 1) and how to deliver cloud applications using CI/CD (week 3).

- Continuous Integration and Delivery Introduction Guide

  The CI/CD Introduction Guide provides you with basic knowledge for setting up and implementing continuous integration and delivery processes. It gives an overview of the concepts and principles of CI/CD, explains both procedures and their relation, and helps you plan your own CI/CD process.

Terraforming Kyma Runtimes

By integrating the Terraform module into CI/CD pipelines, you can automate the provisioning of SAP BTP resources required to create a Kyma cluster and tear them down when no longer required. Such an approach helps you reduce operational costs.

For this purpose, we recommend you to use the Terraform module for Kyma , which helps you to automatically manage the required SAP BTP resources for creating and using the SAP BTP, Kyma runtime in a continuous integration scenario.

- Terraform provider for SAP BTP
- The Terraform module for Kyma

⚠ Caution

It is important to distinguish the Terraform module from the Kyma modules. The Terraform module is a development tool. The word "module" in this context comes from the Terraform provider for SAP BTP naming convention.

Prerequisites

- The CI/CD pipeline must have access to the Terraform or OpenTofu command line interface (CLI). See Install Terraform and the Terraform CLI or Installing OpenTofu .
- The administrative access to SAP BTP global account or subaccount, depending on the scenario.

ℹ Note

It is recommended to have a technical user in place for the CI/CD pipeline. It is required that the technical user is recognized as a platform user. It must be invited to a custom SAP Cloud Identity Services tenant that is trusted on the SAP BTP global account level. The technical user must have the appropriate role collection assigned for the module to work. See the following section for more details.

Supported Scenarios

There are two main use cases that are supported by the Terraform module:

- Deploying Kyma runtime in an existing subaccount. This requires that the technical user executing the Terraform script in the CI/CD pipeline has the **Subaccount Administrator** role collection assigned in the targeted subaccount. The module creates all required resources in the given subaccount. See an example in GitHub .

- Deploying Kyma runtime into a new subaccount. This requires that the technical user executing the Terraform script in the CI/CD pipeline has the **Global Account Administrator** role collection assigned in the targeted SAP BTP global account. The module creates all required resources, including a new subaccount. See an example in GitHub .

Procedure

The module requires a set of input parameters , among which a technical user's credentials must be provided. Using a credential management system, such as HashiCorp Vault , is highly recommended.

The module produces a set of outputs that provide information about the created Kyma runtime, such as subaccount ID, environment instance ID, and API server URL.

Provided that the *<store_kubeconfig_locally>* and *<store_cacert_locally>* variables are set to `true`, the module may also produce the following additional sensitive files, useful for accessing Kyma runtime using kubectl:

- `kubeconfig.yaml`
- `CA.crt`

1. To use the module, create a dedicated folder in your project repository, for example, `tf`, create the `*.tf` files to define main resources, variables, providers, and outputs, following the naming convention for clarity, and `terraform.tfvars` files for passing variables. The structure should look like this:

```
+-- tf
|   +-- main.tf
|   +-- terraform.tfvars
|   +-- provider.tf
|   +-- variables.tf
|   +-- outputs.tf
```

2. In `variables.tf`, define the inputs needed for the resources defined in `main.tf` and providers defined in `provider.tf`.

```
###
# SAP BTP provider configuration
###
variable "BTP_GLOBAL_ACCOUNT" {
  type        = string
  description = "Subdomain of the SAP BTP global account"
}

variable "BTP_BOT_USER" {
  type        = string
  description = "Technical username"
}

variable "BTP_BOT_PASSWORD" {
  type        = string
  description = "Technical user password"
  sensitive   = true
}

variable "BTP_CUSTOM_IAS_TENANT" {
  type        = string
  description = "Custom IAS tenant"
  default     = "custom-tenant"
}

variable "BTP_BACKEND_URL" {
  type        = string
  description = "BTP backend URL"
  default     = "https://cli.btp.cloud.sap"
}
```

```
###
# Kyma module
###
variable "BTP_NEW_SUBACCOUNT_NAME" {
  type        = string
  description = "Name of the subaccount to be created"
  default     = null
}

variable "BTP_NEW_SUBACCOUNT_REGION" {
  type        = string
  description = "Subaccount region name"
  default     = null
}

variable "BTP_KYMA_PLAN" {
  type        = string
  description = "Kyma plan name"
  default     = "azure"
}

variable "BTP_KYMA_REGION" {
  type        = string
  description = "Kyma region"
  default     = "westeurope"
}

variable "BTP_KYMA_CUSTOM_ADMINISTRATORS" {
  type    = list(string)
  default = []
}
```

3. In the `terraform.tfvars` file, provide values for the variables that are necessary for the Kyma module. For example:

```
BTP_NEW_SUBACCOUNT_NAME        = "my-subaccount"
BTP_NEW_SUBACCOUNT_REGION      = "..." # for example 'eu20'
BTP_GLOBAL_ACCOUNT             = "..."
BTP_BOT_USER                   = "..."
BTP_BOT_PASSWORD               = "..."
BTP_BACKEND_URL                = "..." # for example 'https://cli.btp.cloud.sap'
BTP_CUSTOM_IAS_TENANT          = "..."
BTP_KYMA_PLAN                  = "..." # for example 'azure'
BTP_KYMA_REGION                = "westeurope"
BTP_KYMA_CUSTOM_ADMINISTRATORS = ["..."] # list of emails of users that should be gransted `c
```

ℹ Note

Check the .terraform.tfvars.example for reference.

4. In `provider.tf`, add providers that are required by the Kyma Terraform module. See Terraform provider for SAP BTP , and terracurl .

```
terraform {
  required_providers {
    btp = {
      source  = "SAP/btp"
      version = "~> 1.18.1"
    }
    terracurl = {
      source  = "devops-rob/terracurl"
      version = "~> 2.1.0"
    }
  }
}

provider "btp" {
  globalaccount  = var.BTP_GLOBAL_ACCOUNT
  cli_server_url = var.BTP_BACKEND_URL
  idp            = var.BTP_CUSTOM_IAS_TENANT
  username       = var.BTP_BOT_USER
  password       = var.BTP_BOT_PASSWORD
```

```
}

provider "terracurl" {}
```

5. In `main.tf`, include the Kyma module as a child module.

```
module "kyma" {
  source                    = "git::https://github.com/kyma-project/terraform-btp-kyma-enviro

  BTP_NEW_SUBACCOUNT_NAME        = var.BTP_NEW_SUBACCOUNT_NAME
  BTP_NEW_SUBACCOUNT_REGION      = var.BTP_NEW_SUBACCOUNT_REGION
  BTP_KYMA_PLAN                  = var.BTP_KYMA_PLAN
  BTP_KYMA_REGION                = var.BTP_KYMA_REGION
  BTP_KYMA_CUSTOM_ADMINISTRATORS = var.BTP_KYMA_CUSTOM_ADMINISTRATORS
  store_kubeconfig_locally       = true # enable to produce kubeconfig.yaml file for initial

  # check inputs section what else can be configured
}
```

6. In `outputs.tf`, define the outputs readable by Terraform CLI.

```
output "subaccount_id" {
  description = "The subaccount ID where the Kyma environment is created."
  value = module.kyma.subaccount_id
}

output "environment_instance_id" {
  description = "The Kyma environment instance ID."
  value = module.kyma.environment_instance_id
}

output "apiserver_url" {
  description = "The API server URL of the Kyma cluster."
  value       = module.kyma.apiserver_url
}
```

7. Run the Terraform CLI in the `tf` folder.

```
cd tf
terraform init
terraform plan -out=plan.out
```

8. Validate that the planned changes are correct, and then confirm the `apply` operation.

```
terraform apply plan.out # add -auto-approve in CI/CD
```

This creates a new Kyma runtime in your SAP BTP account.

Result

As a result of applying the configuration, you have Kyma environment up and running. You can read the environment instance ID, the Kubernetes API server URL, and the subaccount ID of the Kyma cluster from the Terraform outputs, for example:

```
terraform output apiserver_url
terraform output environment_instance_id
terraform output subaccount_id
```

To enable immediate access to the cluster in headless scenarios, the module can be used with the following additional flags:

```
store_cacert_locally           = true
store_kubeconfig_locally       = true
```

With that, the module produces the `kubeconfig.yaml` and `CA.crt` sensitive files. With these files, you can access Kyma runtime using kubectl, so be thoughtful not to publish them in any way.

Additionally, the module comes with a Bash and a PowerShell scripts that you can use to extract cluster ID and domain values from the Kyma cluster using kubectl. The scripts read the kubeconfig file from the KUBECONFIG environment variable, extract the cluster ID and domain, and write them to local files called `domain.txt` and `cluster_id.txt`.

Next Steps

Remote State Storage

Terraform maintains a state file that maps your configuration to the actual infrastructure, such as SAP BTP resources. With that, it detects drift and safely computes changes. By default, Terraform writes the state in the local filesystem in the working directory, which may be fine for local executions, but doesn't work for concurrent runs initiated in CI/CD pipelines.

For CI/CD pipelines that create and maintain resources, configure a remote backend to store state centrally, for example: Terraform Cloud, AWS S3, GCS, Azure Blob Storage. See Backend block configuration overview for more information on remote backends for state storage.

Destroying Resources

To destroy all resources, run the *destroy* command:

```
terraform destroy # -auto-approve when called from CI/CD pipeline
```

> ⚠ **Caution**
>
> The *destroy* command removes all infrastructure resources mapped in the state files.
>
> Always ensure you know which configuration you are about to destroy, and avoid `-auto-approve` when executing locally.

Tools Available for SAP BTP Multi-Cloud Foundation

SAP BTP includes many tools to help you develop and manage applications. The availability of some of the tools depend on the runtime that you are running on.

Tool	Description
Account Administration in the Cockpit	The SAP BTP cockpit is the web-based administration interface of SAP BTP and provides access to a number of functions for configuring and managing applications, services, and subaccounts. Use the cockpit to manage resources, services, security, monitor application metrics, and perform actions on cloud applications. Used for account administration in SAP BTP.
SAP Business Application Studio	Designed and optimized for business application development in SAP ecosystems, SAP Business Application Studio enhances productivity by offering specialized tools for various scenarios, including SAP Fiori application development, SAP HANA native extensions, full-stack and mobile application development, and more.

Tool	Description
	Central to the development environment is Code-OSS, the open-source foundation of Visual Studio Code, ensuring a familiar experience for developers when creating SAP-centric applications. SAP Business Application Studio streamlines the building, testing, and deployment of applications with integrated features for source control and testing. Furthermore, its Full-Stack Application Productivity Toolkit offers intuitive visual tools covering the entire development process, guaranteeing seamless integration with various SAP services and solutions. Used for: • SAP Fiori development for ABAP and non-ABAP • CAP full-stack application development
SAP Build	SAP Build enables everyone, no matter the skill level, to create and augment enterprise applications, process automations, and business sites with drag-and-drop simplicity. Used for low-code, no-code development.
Command Line Interface for Cloud Foundry	The Cloud Foundry command line interface enables you to work with the Cloud Foundry runtime to deploy and manage your applications. Used for managing subaccounts in the SAP BTP, Cloud Foundry runtime.
Cloud Connector	The Cloud Connector serves as the link between on-demand applications in SAP BTP. This is the browser-based and existing on-premise systems. You can control the resources available for the cloud applications in those systems. Used for connectivity between SAP BTP applications and on-premise systems.
Terraform provider for SAP BTP	The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP.
kubectl	The Kubernetes command line tool to communicate with a Kubernetes cluster's control plane, using the Kubernetes API. Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.
kubelogin	A `kubectl` plugin for Kubernetes OpenID Connect (OIDC) authentication. Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.
Helm	The package manager for Kubernetes, used for installing and managing Kubernetes applications in form of Helm charts. Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.
Paketo (Pack)	Pack is a tool maintained by the Cloud Native Buildpacks project to support the use of buildpacks. Pack lets you build container

Tool	Description
	images, which are collaboratively maintained making it easier to maintain and update. Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.
Docker Desktop	Docker Desktop is an application that enables you to manage (build, push, pull, and run) container images on your desktop and a docker-compatible command line interface. Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.
SAP Cloud SDK	SAP Cloud SDK provides a layer of abstractions for features of SAP BTP such as logging, multitenancy, and connectivity. It also includes project templates for different execution environments and implementations. Used for programming in Java and JavaScript.
Service-Specific Tools	The services that run on SAP BTP can come with service-specific tools. For an overview of the services and their tools, see the SAP Discovery Center.

APIs

Discover and consume APIs to manage, build, and extend the core capabilities of SAP BTP.

An Application Programming Interface or API is an **interface provided by an application for interacting with other applications**. APIs specify how software programs are able to exchange information with each other, even if designed and run by different organizations. They facilitate interaction by selectively exposing certain functionality, allowing different apps, websites, and devices to communicate effectively with each other. More importantly, APIs enable businesses to reach beyond regular business channels and share data, content, and services directly to both B2B (business to business) and B2C (business to consumer) clients, making UI development easy.

API Offerings on SAP BTP

SAP BTP enables you to consume APIs and publish your own ones through the following offerings:

Offering	Description
SAP BTP on SAP Business Accelerator Hub	The SAP Business Accelerator Hub provides you with one central repository for browsing and accessing APIs from SAP and selected partners. Test APIs and try out mock data in your systems. It is also the official place where REST and OData REST API references are published.
API Management Capability within SAP Integration Suite	With the API Management capability, you can build, manage, publish, and monetize your own APIs within one secure and scalable environment.
Software Development Kits (SDKs) For an overview of the available SDKs, see Tools.	The SDKs available for SAP BTP offer APIs to, for example, accelerate enterprise app development.

Related Information

SAP API Style Guide

Security Considerations for Applications

When building applications, use the security features of SAP BTP, such as protection from web attacks. We recommend that your developers configure and deploy application-based security artifacts containing authorizations, and administrators assign these authorizations using the cockpit. SAP BTP offers platform roles that help you ensure a segregation of duties, such as between app development and administration.

It's likely that data protection and privacy influence your architecture and the functions of your application. Consider any implications as early as possible in your development process. Security monitoring is done with audit logging. SAP BTP writes logs for security-relevant events and the written log files are digitally signed to ensure their integrity.

See the following for more information about security best practices:

- CAP Security Guide
- Giving Access Rights to Platform Users
- SAP BTP Security Recommendations

You implement, deploy, and validate your application to ensure it meets both functional and performance requirements. Prioritizing secure deployment and configuration at this stage helps you launch your application with strong protections in place—minimizing vulnerabilities and aligning with enterprise-grade security standards.

Delivering your application marks the transition from development to production, where your application becomes accessible to users. This is a critical point for enforcing security controls. For example, a retail company deploying an e-commerce application must ensure secure payment processing and defend against runtime threats such as:

- SQL injection
- Cross-site scripting (XSS)
- Cross-Site Request Forgery (CSRF)

By applying secure deployment practices—such as hardening configurations, validating inputs, and enabling runtime protection—you reduce the risk of exploitation and ensure compliance with data protection regulations like the GDPR.

By implementing built-in security features offered by the SAP Cloud Application Programming Model (CAP), you can significantly enhance your application defense. This includes leveraging:

- Parameterized queries (for example, cds.select, .where) to prevent SQL injection.
- CDS constraints (for exampe, @assert.format) for robust input validation.
- CSRF protection for UI applications, such as checkout forms.

Furthermore, CAP's built-in authentication and authorization mechanisms protect user access. Together, these practices help reduce vulnerabilities, support GDPR compliance, and keep user trust.

The following table summarizes the security guidelines you need to be familiar with:

Delivering Secure Applications Deployed to SAP BTP

Security Guideline	Description
Secure Environment Configuration	Configure the SAP BTP runtimes (for example, SAP BTP, Cloud Foundry runtime, SAP BTP, Kyma runtime with secure settings, such as restricted network access. See: • SAP BTP Cloud Foundry Runtime • Secure Development in the Kyma Environment | SAP Help Portal
Validate Security During Testing	Conduct security tests (for example, penetration testing) during validation to identify and fix vulnerabilities before go-live.
Implement Secure Deployment Pipelines	Use CI/CD pipelines with security checks (for example, code scanning, dependency validation) to ensure safe deployments. See: • Introducing Continuous Integration and Delivery (CI/CD) • Get Started with an SAP Cloud Application Programming Model Project in SAP Continuous Integration and Delivery | SAP Tutorials
Secure Application Secrets Management	Use SAP Credential Store to securely manage secrets like API keys or database credentials. See SAP Credential Store.

Following these guidelines will significantly improve the deployment of your application to SAP BTP and enable rigorous validation processes. This approach is critical for safeguarding against potential threats, mitigating vulnerabilities, and ensuring compliance with rule-based standards. For more practical examples on how to implement security in CAP, see CAP Security Guide | capire.

Developing Resilient Applications

Our best practices about resilient application design help you to make your applications running on SAP BTP stable and highly available.

There are many different principles and patterns you can use to make your software resilient. However, it isn't always easy to find the combination that best fits your applications. For an overview of the various options you have when developing applications and detailed information about the individual patterns you can use, as well as examples for applied resilience principles and how to develop your own resilient apps on SAP BTP, read Developing Resilient Apps on SAP BTP.

For examples on how to use selected SAP BTP services in a multi-region setup, see Route Multi-Region Traffic to SAP BTP Services Intelligently .

You can also use the Architecting Multi-Region Resiliency for SAP BTP Use Cases reference architecture in the SAP Discovery Center. It shows the multi-region high availability architecture for SAP BTP services and the applications built on SAP BTP.

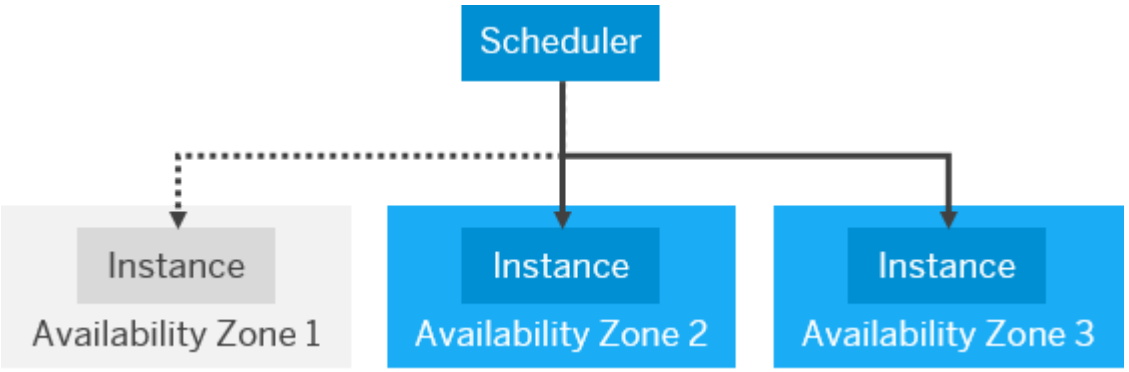
Developing Resilient Applications in Cloud Foundry

In SAP BTP, Cloud Foundry environment, you can use the availability zones: Benefit from the high availability mechanisms in Cloud Foundry by setting up your applications with multiple instances. You do not have to do anything explicitly to distribute the instances across the different availability zones - this is handled by the platform. For more information, see Availability Zones in the Cloud Foundry Environment.

Setting up your application with multiple instances might have an impact on your applications' capability for handling requests: In case of an availability zone failure in a 3-availability-zone-deployment, at least one third of the application instances are affected

and unavailable until Cloud Foundry can reschedule these instances on a healthy virtual machine.

During that period, the remaining instances in the healthy availability zones have to carry the request load.



Distribution During an Availability Zone Failure

Developing Resilient Applications in Kyma

Learn how to develop stable Kubernetes and microservice-based applications, as well as Istio-related resiliency tips with Develop Resilient Applications in the Kyma Runtime.

Related Information

Resilience, High Availability, and Disaster Recovery (Cloud Foundry, ABAP and Kyma Environments)

Using Multitarget Applications to Manage Dependencies

One challenge of moving into the cloud is deploying applications that consist of multiple interconnected components.

In Cloud Foundry, we recommend that you develop multitarget applications that let you package those components into one bundle, and deploy and manage them all at once. Cloud applications often come with a lot of heterogeneity, which is one of the key strengths of cloud development, allowing for agility, resilience, and the rapid development of new features. However, it also increases the complexity of cloud applications, which:

- Usually consist of multiple interdependent software modules
- Are written in different programming languages using multiple development tools
- Might involve different products
- May be deployed to multiple target runtimes

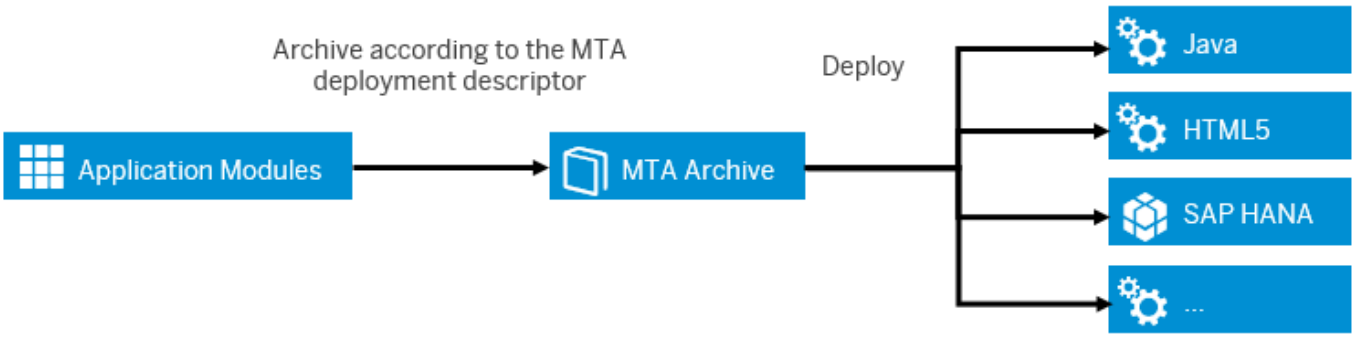
A combined lifecycle lets you deploy all parts together, automatically, and in the right order, and manage the configuration of the complete solution. You can achieve such a combined lifecycle by developing multitarget applications. Each multitarget application has the following characteristics:

- One archive file that includes all modules and a description of the dependencies
- Can be delivered, transported, linked to SAP software components, and deployed
- The process can be automated in a continuous integration pipeline

The multitarget application archive contains all required application types and configurations, as well as a deployment descriptor file. It is intended to be used as a generic artifact that can be deployed and managed on several SAP BTP subaccounts. For

example, you can use one multitarget application archive in your development subaccount and reuse it in your production subaccounts.

As all interdependencies are part of the archive file, it's easy to pass multitarget applications from development to operations. All required information for deployment is provided during the development process. Due to the benefits provided by applying the multitarget application approach, it is also part of the SAP Cloud Application Programming Model.



→ Recommendation

The approach isn't mandatory for applications that are running on SAP BTP – you can also develop without applying it. Without the multitarget application approach, you'll need to manually deploy your application artifacts, for example by triggering the deployment from SAP Business Application Studio or manually uploading artifacts via SAP BTP cockpit.

We recommend that you use multitarget applications in the following cases:

- You're developing a business application composed of several different parts – apps, services, content, and others – that you want to manage as a single unit.
- Your business application has dependencies to external resources, such as backing services (database, messaging, and so on), APIs, and configurations from other applications.
- Your business application has a certain default configuration, for example memory, disk, number of individual app instances, environment variables, service plans, and others.

For more conceptual information about multitarget applications and detailed step-by-step instructions, see Multitarget Applications in the Cloud Foundry Environment.

There are several options to create multitarget application archives:

- If you use SAP Business Application Studio, see MTA Development.
- If you use SAP Web IDE Full-Stack, you can use multitarget application templates for Cloud Foundry applications, where the descriptor file is maintained automatically, for example, whenever you add a new module in the SAP Web IDE.
- If you have development modules from other sources, you can use the multitarget application archive builder, a Java-based command-line tool that builds modules and packages them into a deployable multitarget application archive, together with a deployment descriptor. It is available for download from SAP Development Tools (see https://tools.hana.ondemand.com/#cloud).
- If you use Continuous Integration and Delivery (CI/CD), you can let the pipeline perform an automated build of your changes pushed to your central source code repository. The outcome can also be an MTA archive. For more information, see Delivering Applications.

Extending Existing SAP Solutions Using SAP BTP

SAP BTP offers services, tools, and capabilities to develop, extend, or integrate business applications in the cloud. You can implement additional workflows or modules on top of existing solutions, which lets you benefit from out-of-the-box security, inherited data access governance, user interface embedding, and so on.

Overview

All standard SAP solutions are offered with customizing capabilities. Additionally, customers often have their own requirements for innovative or industry-specific extensions and the extension capability of SAP BTP can help them build, deploy, and operate their new functionalities easily and securely.

You can use SAP BTP to extend standard SAP solutions without disrupting their performance and core processes. When building extension applications, you can also benefit from the automation of the integration between SAP BTP and the extended SAP solutions.

There are three main extension types you can use to extend your SAP solution:

- In-app extensions: allows you to extend the back end and the UI directly in your SAP solution. You can use in-app extensions only if the SAP solutions that you want to extend provide such a possibility. For example, you can use in-app extensions for extending SAP SuccessFactors, and SAP S/4HANA Cloud. See:
    - SAP S/4HANA Cloud In-App Extensibility
    - SAP SuccessFactors Extension Center: In-App Extensions
- Side-by-side extensions: allows you to integrate with your SAP solution additional business functionality deployed in SAP BTP.

  The advantages of the side-by-side extensions can be grouped in these categories:
    - Business: you can include the access of new user groups, you can extend the scope with custom solutions, you can complement data and business services from different back ends.
    - Technology: more developers are enabled, your digital core remains focused, you can add innovative solutions by leveraging new technology services in the cloud.
    - Operations: you can share the extensions

When planning your extension, consider these recommendations:

- Always start simple to check the result quickly. Later on, you can grow the complexity of your extension slowly.
- Keep your digital core focused and bring to SAP BTP only the data that makes sense.
- Define what you want to achieve in the beginning and look for the easiest way to do that. For example, if you want to add a field in SAP S/4HANA, it's better to create an in-app extension, than to implement an extension application on SAP BTP.
- Ensure the availability of suitable APIs and events, because you will need them when developing and extension application on SAP BTP.

In SAP BTP, you have these options to extend your SAP solution:

- Extensions with automated configurations in the Cloud Foundry and Kyma environments: applicable for SAP S/4HANA Cloud, and SAP SuccessFactors.
- Extensions with automated configurations in the Kyma environment: applicable for SAP S/4HANA Cloud, SAP SuccessFactors, SAP Cloud for Customer, SAP Commerce Cloud, and SAP Field Service Management.

Extensions with Automated Configurations

SAP BTP provides a standard way for extending SAP solutions and developing event-driven extensions and applications. This framework includes:

- Simplified, standardized and unified extensibility and configuration for the SAP solutions.
- Central catalog per customer for all connected SAP systems where data such as APIs, events, credentials and other is stored. You can benefit from business services and actionable events across end-to-end business processes.

If you have to group the systems of different SAP solutions in the same business case, you can set up the connectivity between all these systems and SAP BTP in a single formation.

The following SAP solutions currently support the automated configurations:

- SAP S/4HANA Cloud
- SAP SuccessFactors
- SAP Cloud for Customer
- SAP Commerce Cloud
- SAP Field Service Management

Use Cases

The extension use cases include but are not limited to:

- Extending the user interface of your SAP cloud solution.

  You can complement an existing SAP solution with a new or improved SAP Fiori user interface without adding any major logic or data.

  You can also provide necessary information and tools on mobile devices.

  For example:

  - Building completely new user interfaces that can be seamlessly integrated in the SAP SuccessFactors Employee Central home page.
  - Adding new functionality to SAP SuccessFactors by connecting it to other SAP solutions, such as SAP Commerce Cloud, SAP Cloud for Customer, SAP Field Service Management, SAP S/4HANA Cloud.
  - Building completely new SAP Fiori-based user interfaces that can be seamlessly integrated in the SAP Fiori launchpad.

- Enhancing the functionality of your SAP cloud solution by connecting it with other SAP solutions.

  For example:

  - Connecting SAP S/4HANA Cloud with other SAP solutions, such as SAP Commerce Cloud, SAP Cloud for Customer, SAP Field Service Management, SAP SuccessFactors.
  - Connecting SAP SuccessFactors by connecting it to other SAP solutions, such as SAP Commerce Cloud, SAP Cloud for Customer, SAP Field Service Management, SAP S/4HANA Cloud.

- Enhancing core business processes with analytics and machine learning capabilities.
- Extend data insights by consolidating and combining data in one central place.
- Building an extensions ecosystem, by creating and operating SaaS applications and selling them to multiple customers.

For example, as an SAP Partner, you can build a multi-tenant SaaS extension application and provide it to your customers via:

- SAP App Center
- SAP BTP cockpit

Resources

The following resources are available for your extension scenarios. The links to various documents guide you through the configuration tasks that you need to perform to enable the SAP BTP for developing extension applications for your existing SAP solutions, and the learning journeys are collections of links to additional resources.

- SAP S/4HANA Cloud
  - Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment (SAP Help Portal)
  - Extend SAP S/4HANA Cloud on SAP Business Technology Platform (Mission in Discovery Center)
  - SAP Extensibility Explorer for SAP S/4HANA Cloud
  - Tutorials about the integration of SAP BTP ABAP Environment with SAP S/4HANA Cloud
- SAP Customer Experience (CX) products
  - Extending SAP Customer Experience Products in the Kyma Environment (SAP Help Portal)
  - Create Extensions in the Kyma Environment Using Functions (SAP Help Portal)
- SAP SuccessFactors
  - Extending SAP SuccessFactors in the Cloud Foundry and Kyma Environment (SAP Help Portal)
  - Extend SAP SuccessFactors on SAP BTP (Mission in Discovery Center)
  - Build Resilient Cloud Native Applications (Mission in Discovery Center)

Related Information

Extensions
Extension Architecture Guide
SAP Application Extension Methodology Overview

Performing UI, Usability, and Unit Tests

You should always conduct careful testing to ensure that your application is of high quality. Create a release candidate to propagate throughout your landscape.

For example, consider testing the user interface, usability, and performance, as well as running general unit tests. Developers should use unit tests to ensure that software modules behave as expected. Unit tests are the smallest tests, and they detect issues fast. Taking good care of unit tests usually leads to better maintainable and understandable code. If you use the Continuous Integration process, you should consider automated tests as part of the pipeline.

SAPUI5

SAPUI5 supports QUnit testing. We recommend that you write unit tests as small as possible, ensuring that you only test one single thing at a time. This will not only make debugging easier, but also maintaining the tests in the long run.

For an overview of available testing options for SAPUI5 developments, see Testing and Performance Measurement.

For SAPUI5 and SAP Fiori testing, please see the demokit for SAP UI5: https://sapui5.hana.ondemand.com/#/topic.

Testing in SAP Web IDE

SAP Web IDE empowers developers to test and evaluate the functionality and performance of their apps:

- Use the application preview to test functionality, design and performance before deploying it. There are different configuration options available, such as SAP UI5 runtime version, URL parameters. See Running Applications in Development Mode.
- Run your application using a client mock server. See Run Applications with Mock Data.
- Execute unit tests for SAPUI5 (Qunit) and Java (JUnit). See Developing Application Tests and Test a Java Module.
- Execute SAPUI5 integration tests based on OPA5. See Performing Integration Tests.

Connecting to Remote Systems

Overview

The connectivity in SAP BTP provides robust features with which you can establish reliable communication channels, ensuring data flows smoothly and securely across different environments. This capability is essential for enterprise applications, typically complex and needing to consume or push data to a variety of sources or destinations, including systems that are directly accessible, hosted in public or private clouds, or located on customer premises. This connectivity empowers businesses to extend, innovate, and optimize their digital landscapes effectively.

For Kubernetes setups, the SAP BTP transparent proxy serves as the entry point to leverage central and secure connection configuration management (SAP Destination service) along with hybrid technical connectivity (SAP Connectivity service). It provides a unified and seamless connection experience to target systems.

Services and Components

SAP Connectivity Service

The Connectivity service offers a set of services and software components which enable:

- Cloud applications to make secure connections to systems hosted on premise or in the private cloud.
- The establishment of secure communication channels from the customer's on-premise or private cloud network to cloud applications, keeping full control and auditability of the resources are exposed to the cloud.

The secure exposure of the private systems is ensured by the Cloud Connector. This way, no network ports need to be exposed on the on-premise or private cloud side. On the side of the cloud applications, the secure interconnectivity is ensured by the connectivity proxy, offered as Software-as-a-Service or Software-as-a-Product, depending on the application hosting environment.

Various communication protocols towards the target system are supported.

Cloud Connector

The Cloud Connector serves as a secure link between cloud applications and on-premise systems, allowing controlled access to them. It functions as a reverse invoke proxy within a secured network, providing fine-grained control over both accessible on-premise resources and cloud applications that use it.

The Cloud Connector serves as a secure link between cloud applications and systems hosted in public or private clouds, allowing controlled access to them. It functions as a reverse invoke proxy within a secured network, providing fine-grained control over both accessible on-premise or private cloud resources and cloud applications using it. As a distributed and integral part of the Connectivity service, it pairs with a region in SAP BTP. The SAP BTP domain model (subaccounts) and the optional Cloud Connector Location ID are used to identify a particular Cloud Connector.

Connectivity Proxy

The connectivity proxy serves as a cloud counterpart to which a Cloud Connector technically establishes a secure link. The connectivity proxy is designed to support various runtimes, ensuring secure connectivity.

- In SAP BTP, Cloud Foundry runtime, the connectivity proxy is offered as a central and shared feature as part of the Connectivity service.
- In the SAP BTP, Kyma runtime, it's offered as a naively-integrated Kyma Module, simplifying the deployment and configuration.
- In native Kubernetes environments, the connectivity proxy is offered as a Kubernetes component. The connectivity proxy, as a distributed and integral part of Connectivity service, must be paired with a subaccount in SAP BTP to grant access to the Cloud Connectors paired with that subaccount. See Connectivity Proxy for Kubernetes.
- In SAP BTP, ABAP environment, as well as SAP S/4HANA Cloud and SAP Integrated Business Planning systems, it's offered as a local integrated service within the respective product.

SAP Destination Service

The Destination service lets you find the technical information that is required to access a remote service or system from your cloud application. Use the Destination service to:

- Manage routing and authentication details, as well as custom scenario-specific parameters. This information is maintained as a destination.
- Perform authentication flows based on the configured details, and return the result to the consumer of the service as part of the regular look-up request, that is the **Find a Destination** REST API.
- Connect to an on-premise or private cloud system. In this case, the Destination service goes in combination with the Connectivity service.
- Connect to any other remote application, service or system. In this case, the Destination service goes without the Connectivity service.

SAP BTP Transparent Proxy

The transparent proxy streamlines how your Kubernetes workloads connect to remote systems using destinations in SAP BTP. It simplifies the authentication process, while ensuring effective user propagation and seamless integration with the connectivity proxy.

By leveraging Kubernetes services, the transparent proxy exposes target systems directly on the local network, allowing easy, secure, and unified access to solutions, applications, or systems that are remotely accessible via the Internet, as well as to private systems that are securely exposed using the Cloud Connector, with access provided through the connectivity proxy.

In the SAP BTP, Kyma runtime, the transparent proxy is available as a natively integrated Kyma Module.

Recommendations

- Use the transparent proxy to simplify and streamline your Kubernetes workloads' connectivity with remote systems using destinations in SAP BTP. It enhances developer experience by simplifying authentication and automating destination retrieval, while ensuring seamless, secure integration with the connectivity proxy.
- Use the Destination service to store and manage your connection configuration data (including credentials, certificates, URL, headers, queries, and so on) at design time, and automate the access token retrieval process at runtime for your application.
- Use the Connectivity service, more specifically Cloud Connector and the connectivity proxy, to establish a secure connection between cloud applications and on-premise or private cloud systems.

Use Cases

The connectivity use cases include but are not limited to:

Cloud to Cloud

Connecting Kubernetes Applications to Cloud Systems

You can connect to a cloud service or system from your Kubernetes environment. For example:

- Developing an application that requires consumption of an SAP HANA database.
- Adding new functionality that requires the consumption of an SAP BTP service.
- Creating new user interfaces that require consumption of an OData service.
- Adding new functionality that requires integration with third-party APIs.
- An application requires data exchange or service consumption via a RESTful API.

Cloud to On- Premise or Private Cloud

Connecting Cloud Foundry, Kyma, or Kubernetes Applications to On-Premise or Private Cloud Systems

You can connect to an on-premise or private cloud system from your SAP BTP, Cloud Foundry runtime, SAP BTP, Kyma runtime, or any other Kubernetes environment. For example:

- Creating new user interfaces that require consumption of an OData service.
- Developing an application that requires data exchange or service consumption via a RESTful API.
- Extending an application that requires consumption of APIs in an ABAP system via RFC.
- Developing an application that requires interaction with a mail server for e-mail operations.
- Developing an application that requires interaction with a relational or non-relational database (for example, SAP HANA, MySQL, PostgreSQL, MongoDB).
- Implementing a new functionality in an application that requires user authentication and directory services integration using an LDAP server.
- Developing an application that needs to transfer files to and from an FTP server.
- Enabling single sign-on (SSO) by forwarding the identity of cloud users to a remote system or service (user propagation).

On-Premise to Cloud (Service Channels)

You can connect to a cloud service from your on-premise or private cloud network using service channels of the Cloud Connector. For example:

- Configure an RFC connection from your on-premise or private cloud system to SAP S/4HANA Cloud.

  For scenarios that need to connect from on-premise or private cloud systems to the SAP BTP, ABAP environment using RFC, you can establish a connection to an ABAP Cloud tenant host.

- Configure a connection to a service in a Kubernetes cluster.

  Create a service channel to establish a connection to a service in a Kubernetes cluster that is not directly exposed to external access.

Related Information

Consuming the Connectivity Service
Consuming the Destination Service
Transparent Proxy for Kubernetes
Connectivity Proxy for Kubernetes
Cloud Connector
Using the Transparent Proxy
Connect DB Tools to SAP HANA via Service Channels
Configure a Service Channel for RFC
Configure a Service Channel for a Kubernetes Cluster

Run and Scale

There are two things you have to take into account to ensure your application is working properly, and bringing value:

- Get constant feedback from your users and optimize the application based on this feedback.
- Monitor and operate the application using the services and capabilities SAP BTP offers.

  See Establish End-to-End Observability.

Security Considerations in the Run and Scale Phase

In the run and scale phase, you operate, monitor, and scale your application to ensure high availability, performance, and adaptability to growing user demands. Integrating security at this stage is essential to protect against evolving threats, maintain regulatory compliance, and support secure, scalable operations.

This phase ensures your application remains responsive and resilient as usage increases. For example, a retail company operating an e-commerce platform must defend against runtime threats such as unauthorized access or suspicious activity, while also ensuring secure auto-scaling during peak shopping periods.

By leveraging built-in security features of the SAP Cloud Application Programming Model (CAP) and ABAP Cloud, you can significantly strengthen your application defenses. Key practices include:

- **Continuous threat monitoring** using SAP BTP observability tools to detect and respond to anomalies in real time.
- Applicable for CAP: **Secure auto-scaling** with CAP mechanisms that dynamically manage resources while preserving tenant isolation and enforcing security policies.

Implementing these practices helps you mitigate operational risks, comply with standards like GDPR and PCI DSS, and deliver a secure, reliable user experience at scale.

The following table summarizes the security guidelines you need to be familiar with:

Secure Operations at Scale

Security Guideline	Description
Implement Continuous Threat Monitoring	Set up monitoring tools to detect and respond to runtime threats, such as unauthorized access or anomalies. See: • Discovering Observability and Monitoring in SAP BTP, Cloud Foundry • SAP Discovery Center Mission - Implement Observability in a Full-Stack CAP Application Following SAP BTP Developer's Guide
Manage Security Patches and Updates	Regularly apply security patches to client applications and dependencies that integrate with SAP BTP to address vulnerabilities.
Secure Auto-Scaling Mechanisms	The CAP ensures secure multitenancy and supports secure auto-scaling mechanisms for multitenant SaaS applications by providing robust tenant isolation and resource management out of the box. See: • Secure Multitenancy | capire • What Is Application Autoscaler? | SAP Help Portal
Conduct Periodic Security Audits	Perform audits to verify compliance and identify gaps in security configurations or policies.
Protect User-Generated Data Securely	Securely collect and store all forms of user-generated data—whether it originates from surveys, in-app feedback forms, comments, uploads, or integrated systems like ticketing and support platforms, to protect sensitive information and ensure consistent compliance with privacy laws across all data collection points.

Through adopting these guidelines, you significantly strengthen the security, compliance, and resilience of your SAP BTP application as it operates and scales in production. For more resources, see SAP BTP Security Recommendations | SAP Help Portal.

Establish End-to-End Observability

Introduction to Observability in SAP BTP

Observability in SAP Business Technology Platform (SAP BTP) empowers development and operations teams to proactively monitor, analyze, and optimize application performance using telemetry data - logs, metrics, and traces. It is structured into two complementary layers that together provide comprehensive visibility across your SAP landscape:

- Central Observability, delivered via SAP Cloud ALM, offers a unified, strategic view across all SAP services and customer-built applications. It enables efficient issue detection through integrated monitoring, alerting, and business service

management - helping teams maintain service reliability and performance.

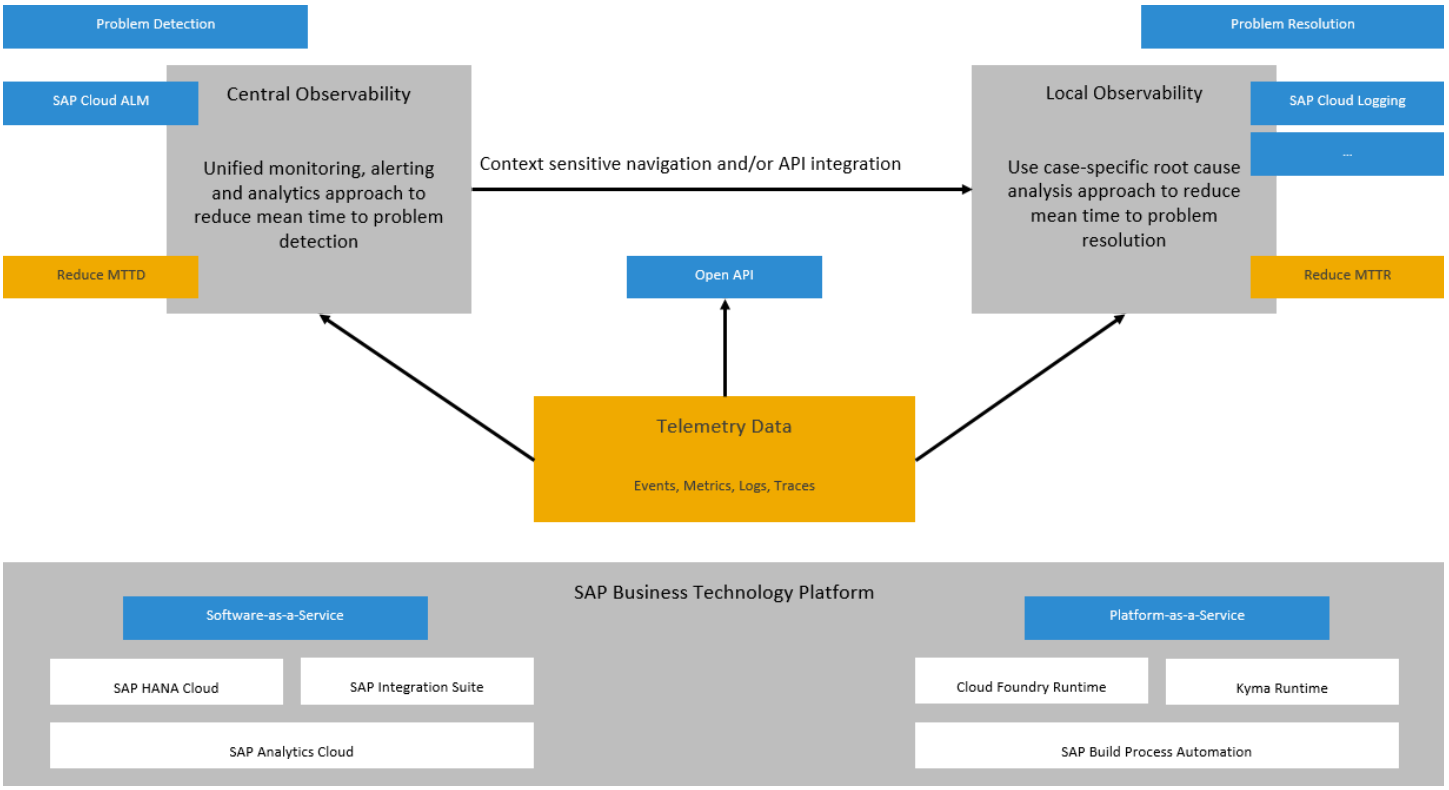
- Local Observability, enabled through tools such as SAP Cloud Logging, provides granular, technical insights into customer-built applications. It supports deep-dive analysis of runtime telemetry signals, facilitating rapid troubleshooting and performance optimization at the component level.

Both layers are interconnected through context-sensitive navigation and API integration, ensuring seamless transitions between high-level oversight and detailed diagnostics. Telemetry data—covering logs, metrics, and traces—flows from SAP BTP runtimes and services into these observability layers via OpenTelemetry-based instrumentation. This architecture reduces mean time to detect (MTTD) and mean time to resolve (MTTR), strengthening operational resilience and enabling faster, data-driven decision-making.

In addition, many SAP products provide embedded local monitoring capabilities for product-specific insights with minimal configuration. These capabilities are integrated into cross-navigation workflows from SAP Cloud ALM, allowing users to drill down from central dashboards into detailed product-level diagnostics for immediate action.

To complement these observability layers, the SAP Alert Notification service delivers real-time notifications for critical events across SAP BTP. It ensures that alerts generated in SAP solutions, central or local observability tools, can be routed to preferred communication channels - such as email, SMS, or third-party collaboration platforms - enabling proactive incident response and reducing operational risk.

SAP BTP spans Software-as-a-Service (SaaS) solutions like SAP S4/HANA, SAP HANA Cloud, SAP Analytics Cloud and more, and Platform-as-a-Service (PaaS) environments such as Cloud Foundry and Kyma. The BTP Observability capabilities integrate across these layers, ensuring consistent monitoring for both SAP-delivered and customer-built applications.



OpenTelemetry in SAP BTP: Strategy and Customer Value

OpenTelemetry, the industry-standard for cloud-native observability, ensures consistent and scalable monitoring across distributed systems. It provides a unified set APIs, and SDKs to instrument applications and collect telemetry data - logs, metrics, and traces - across distributed systems. SAP has adopted OpenTelemetry to ensure consistent, scalable, and vendor-neutral observability across SAP-built and customer-developed applications on SAP BTP.

By integrating OpenTelemetry into SAP BTP runtimes, SAP enables customers to extend built-in observability with their own instrumentation. This allows them to expose custom telemetry data for the workloads they manage, supporting deeper insights into application behavior and performance.

SAP's strategy centers on:

- Standardized instrumentation: SAP and customer applications utilize the same OpenTelemetry-based Data Collection Libraries to provide telemetry signals consistently and openly.
- Open API & integration: OpenTelemetry-compliant inbound APIs allow customers to connect non-SAP data sources, and outbound APIs enable integration with external observability tools.
- Seamless data flow: SAP-built applications send telemetry data via a managed Data Collector Runtime, while customer-built applications transmit data directly to SAP Cloud ALM.

This approach empowers customers to build observability into their operational workflows. It supports a plethora of use cases around day-2 operations, incident management, and problem resolution. Customers benefit from SAP Cloud ALM's centralized monitoring and analytics capabilities. They can leverage deep-dive insights in local monitoring tools like SAP Cloud Logging, which support slicing and dicing of OpenTelemetry-based telemetry signals. These tools also offer flexibility to build their own operational workflows on top.

Use SAP Cloud ALM as a Centralized Operation Platform

SAP Cloud ALM, where ALM stands for application lifecycle management, is included in your SAP Cloud Service subscription containing Enterprise Support, cloud editions, in SAP Enterprise Support and in Product Support for Large Enterprises.

SAP Cloud ALM helps you to implement and operate intelligent cloud and hybrid business solutions. You benefit from an out-of-the-box, native cloud solution running on SAP BTP, designed as the central entry point to manage your complete SAP landscape with content-driven guided implementation and highly automated operations.

With the operations solutions in SAP Cloud ALM, you can handle both SAP BTP-based SaaS applications and your own applications.

SAP Cloud ALM for operations provides a personalized and unified operations user experience. You can use purpose-driven different monitoring, analysis, and management use cases optimized for certain operations aspects. SAP Cloud ALM for operations empowers you to understand the health of your SAP business solutions and acts as collaboration platform between lines of business and IT.

Under Supported Solutions, you see which monitoring capabilities are available with SAP Cloud ALM. Under **SAP Solution**, filter for SAP BTP and check the supported services from the operations area.

Benefit from the powerful **Intelligent Event Processing**, embedded analytics, and intelligence functionality with a large variety of inbound and outbound integration. **Synthetic User Monitoring** monitors the performance and availability of web applications from a client-side perspective based on Selenium-based scripts. **Business Service Management** consolidates all operational events at business service level and shows them in a well-arranged event calendar.

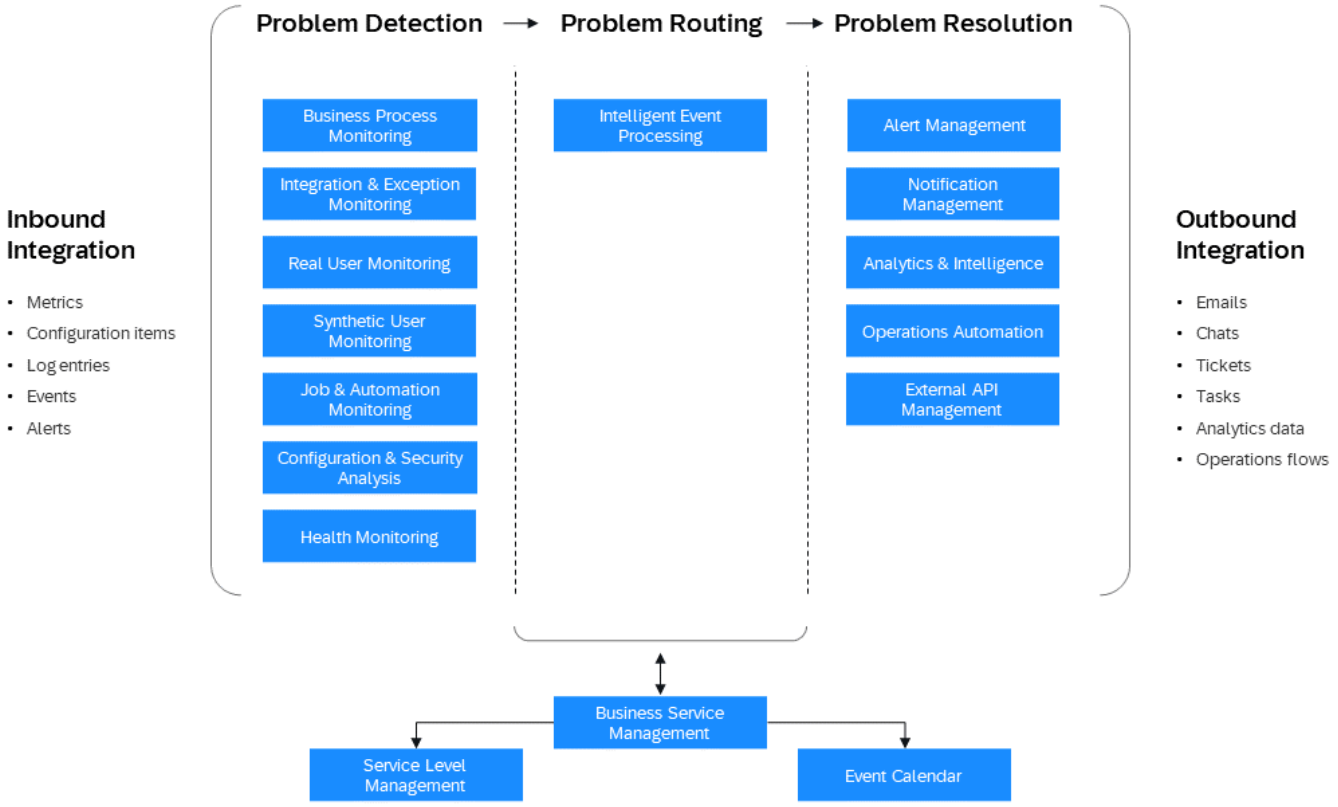
Specifically for Java and Node.js custom-built applications in SAP BTP, Cloud Foundry runtime, there are the Data Collection Instrumentation Libraries based on OpenTelemetry designed to enable the data collection infrastructure in SAP Cloud ALM. With these libraries, you can collect data for the following use cases:

- **Real User Monitoring**: Leverage both front-end and back-end measurements on single request level.
- **Health Monitoring**: Gain insights on application health based on technical metrics.
- **Integration and Exception Monitoring**: Collect and react on exceptions raised by the application.

- **Job and Automation Monitoring**: Monitor SAP Job Scheduling Service jobs.

SAP Cloud ALM is the central place to integrate and access the monitoring of your existing IT service management processes, such as ticketing systems.

Inbound and outbound APIs enable you to integrate data from your own applications. For more information, see the API Guide for SAP Cloud ALM.



See:

- Implement SAP Cloud ALM via SAP for Me
- Application help for SAP Cloud ALM for Operations
- SAP Cloud ALM for operations (SAP Support Portal)
- Expert Portal for SAP Cloud ALM Operations (SAP Support Portal), with detailed configuration instructions

Observability in Action: SAP Cloud Logging for Problem Resolution and Analytics

SAP Cloud Logging is an instance-based observability service built on OpenSearch, designed to collect, store, and analyze logs, metrics, and traces from SAP BTP runtimes (such as Cloud Foundry, and Kyma) but also arbitrary other contexts (support via API-based data ingestion). It supports OpenTelemetry standards and is integrated by SAP Cloud ALM, enabling developers and DevOps teams to gain deep operational insights for problem resolution tasks or build tailored observability workflows.

Telemetry signals can be leveraged for the following use cases for Java and Node.js custom-built applications in SAP BTP, Cloud Foundry and Kyma runtime:

- **Log Analytics** - Perform full-text search, filtering, and visualization of structured and unstructured log data to accelerate root cause analysis and detect anomalies across distributed systems.

- **Distributed System Monitoring** - Understand request flows across microservices and correlate them with the Four Golden Signals (latency, traffic, errors, saturation) to provide end-to-end visibility for scaling and optimization decisions. Use correlation identifiers and access logs to analyze service interactions, dependencies, and system impact through dashboards and service maps. Monitor at different granularity levels, prioritize critical requests, and understand complete system call chains.
- **Technology Monitoring** - Visualize runtime-specific metrics for Java and Node.js applications, including garbage collection, memory usage, and exceptions. Leverage OpenTelemetry signals to assess workload behavior and system health.
- **Custom Telemetry, Alerting & Dashboards** - Instrument applications with service-specific logs, metrics, traces using OpenTelemetry and semantic annotations. Build custom dashboards and alerting rules to support proactive monitoring and incident response tailored to your business needs.

SAP Cloud Logging empowers teams with a robust observability foundation, enabling faster troubleshooting and resilient operations across SAP BTP workloads. Combining open standards, deep integration, and advanced analytics enables teams to move from raw telemetry to actionable insights - supporting resilient operations, faster troubleshooting, and more intelligent decision-making.

SAP Cloud Logging is built on top of OpenSearch. SAP proudly supports the OpenSearch Software Foundation as a premium member, actively helping to grow and maintain the open-source project. By engaging with this Linux Foundation initiative, SAP aims to nurture a thriving community and contribute to the future of innovative search solutions.

See:

- Mission: Implement Observability in a Full-Stack CAP Application
- SAP Cloud Logging at SAP Discovery Center
- SAP Cloud Logging Product Page on SAP Help Portal
- What Is SAP Cloud Logging?

Build and Run Multitenant SaaS Applications for Partners Who Are Independent Software Vendors

If you are an independent software vendor, SAP Business Technology Platform (SAP BTP) offers architecture patterns and development practices for building multitenant SaaS applications, along with operational guidance and tutorials. Utilizing the following resources help you to streamline the development process and ensure scalable and efficient application deployment.

- Gain access to a single, comprehensive, and structured resource for partners building multitenant SaaS solutions on SAP BTP. This guide outlines the recommended path, helping you navigate every step - from the initial idea to successful monetization - and makes it easier to fully leverage SAP BTP's capabilities, including its services, scalability, and go-to-market opportunities. See End-to-End Guide for Multitenant SaaS Solutions on SAP BTP on SAP Partner Portal.
- Explore the partner reference application. It provides a comprehensive example to help you build, deploy, and run scalable SaaS solutions. See Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners ***and*** Partner Reference Application 'Poetry Slam Manager' on GitHub.
- Explore the partner reference application extension that provides tutorials and a comprehensive example for customer-specific extensions for multitenant SaaS applications. See Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application Extension 'Catering Management' on GitHub.
- Estimate your cost-to-serve by calculating the total cost of ownership for multi-tenant SaaS applications built on SAP BTP. See Total cost of ownership calculation of multi-tenant SaaS applications built on SAP BTP on SAP Partner Portal.

Related Information

Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners

Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners

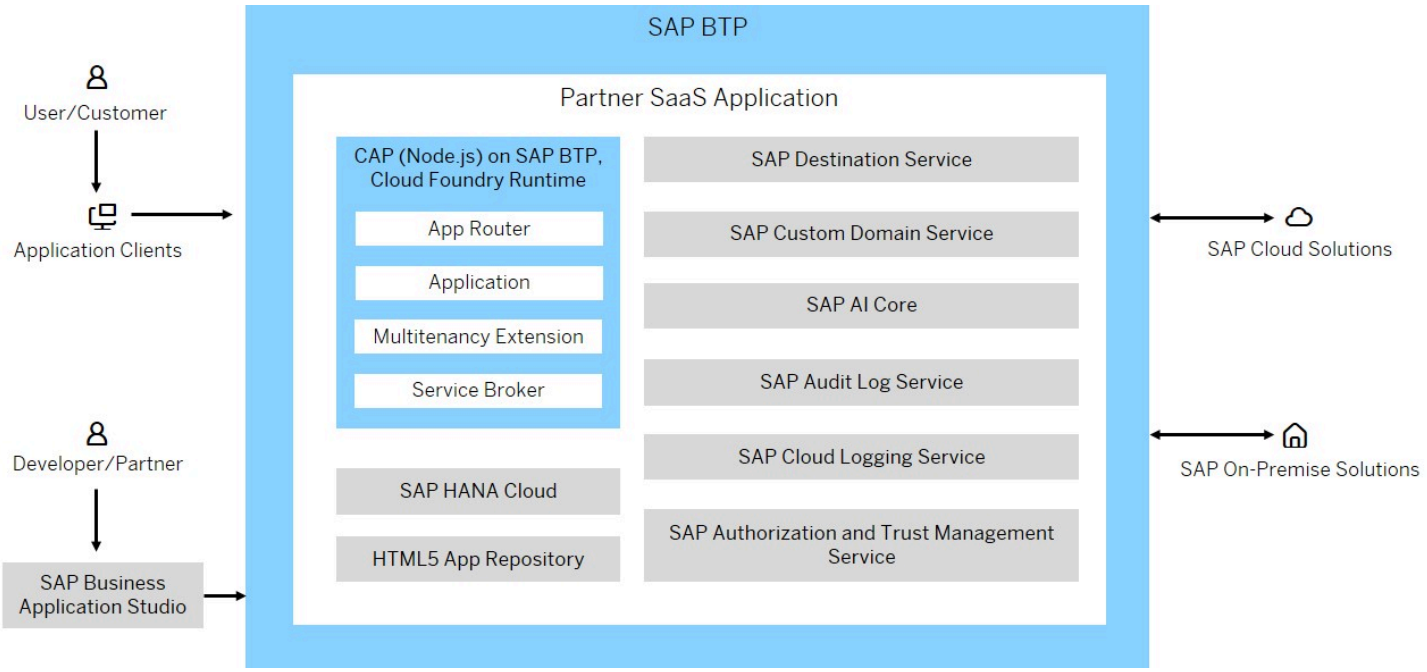
Full-Stack Multitenant SaaS Applications for Partners

The Poetry Slam Manager application is a reference for SAP partners that provides you with a recommended way to becoming a SaaS provider of multitenant applications based on the SAP BTP. See Partner Reference Application 'Poetry Slam Manager' on GitHub.

The guidance covers building, running, and integrating scalable full-stack cloud applications using the SAP Cloud Application Programming Model (CAP) and adheres to the development recommendations set out in the SAP BTP Developer's Guide. It includes an ERP-agnostic design that lets you deliver your application as a side-by-side extension to consumers using any SAP solution, such as SAP S/4HANA Cloud, SAP Business One, and SAP Business ByDesign.

By using services in SAP BTP and the SAP Cloud Application Programming Model (CAP), your application meets SAP standards for enterprise-class business solutions and empowers you to focus on your application domain.

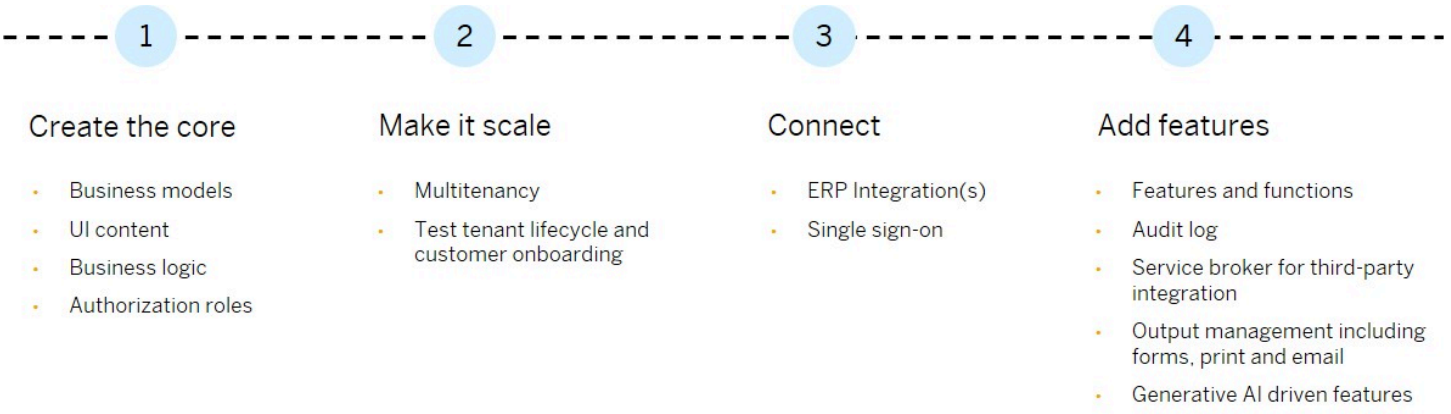
The reference application and its tutorials cover a wide range of services and components in SAP BTP. You can find a list of all services in the bill of materials of the partner reference application on GitHub. See Bill of Materials on GitHub.



The tutorials in the repository follow an incremental development approach, guiding users through the development journey. They cover:

1. Developing the core application with a focus on business models and logic.
2. Enhancing the application to a multi-customer (multitenant) solution.
3. Integrating with various ERP backends.
4. Adding more features to the application.
5. Extending the application.

To cater to customer-specific and industry-specific requirements, an extensibility feature is provided as part of incremental development approach, guiding users through extension journey. See Partner Reference Application Extension .



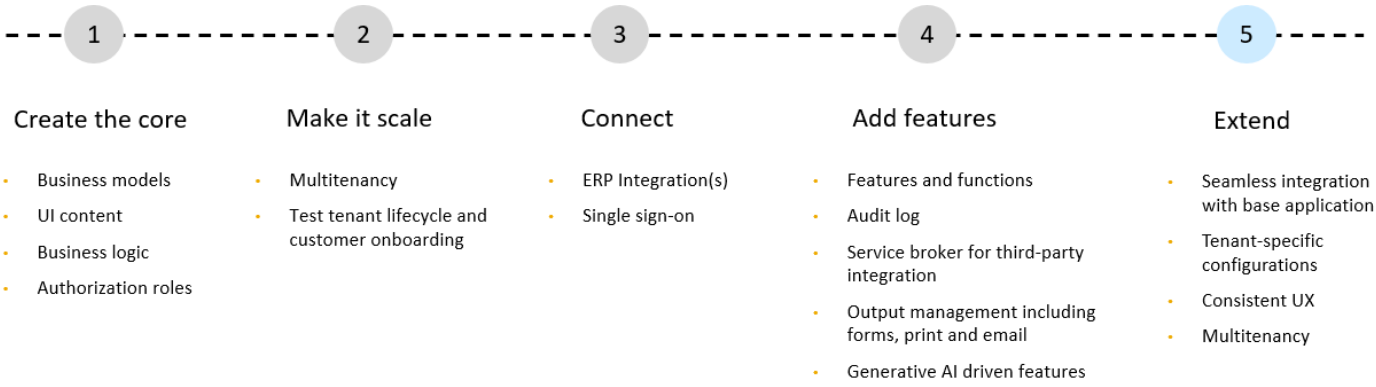
Customer-Specific Extension

The Partner Reference Application Extension demonstrates how to achieve this customer flexibility by extending the Poetry Slam Manager application with enhanced catering management capabilities. By leveraging the SAP Business Technology Platform (SAP BTP) and the SAP Cloud Application Programming Model (CAP). It enables partners to build extensions that address the unique needs of individual customers while preserving the core application's functionality for other tenants and adhering to SAP standards for scalability, security, and enterprise readiness.

The Partner Reference Application Extension showcases the platform's flexibility and extensibility with features such as:

- Seamless integration with the base application
- Granular tenant-specific configurations to deliver personalized experiences
- Consistent user experience across extended and core functionality
- Secure multitenant architecture, ensuring data isolation and performance scalability

See Partner Reference Application Extension .



Tutorials for SAP Cloud Application Programming Model

The Incident Management sample application is a showcase of the SAP Cloud Application Programming Model (CAP) and adheres to the development recommendations set out in the SAP BTP Developer's Guide. Best practices are documented in step-by-step tutorials based on the Incident Management application for learning how to develop, deploy, and operate a full-stack application using CAP and SAP Fiori. The tutorials also show how to use a set of important SAP BTP services in a CAP application.

When going through the sample application, you can choose from two options to try its services and runtimes before you purchase them:

- Free tier for SAP BTP is best for productive projects:
  - Access a productive account
  - Start with more than 30 free-tier services and upgrade from free to paid plans
  - Accelerate your ramp-up with boosters and missions that include step-by-step guidance

  The free-tier offering is available in the following consumption-based commercial models: the **SAP BTPEA** (SAP BTP Enterprise Agreement), the **CPEA** (Cloud Platform Enterprise Agreement) and **Pay-As-You-Go for SAP BTP**. See Using Free Service Plans.

- SAP BTP trial is best for learning purposes:
  - Gain instant and full access to a trial environment
  - Experiment with a diverse range of more than 30 trial services for free
  - Get started with a variety of learning materials, such as tutorials and missions

  See Start your free 90-days trial.

Before you start using the services or runtimes, you need to manage your entitlements and add quotas to your subaccounts. See Entitlements and Quotas.

Note that if you want to try out services for free, you need to select free tier service plans, if available. For a list of free services, check the Service Catalog at SAP Discovery Center.

ℹ Note

There is also a free plan for the SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime. To use the free plans for these runtimes, you have to configure the entitlements in the SAP BTP cockpit. You have one free plan for each runtime per global account and you can assign it to one subaccount at a time.

The tutorials are grouped in missions following a specific scenario. All the missions are based on the Incident Management sample application:

- Starter Mission: Develop a Full-Stack CAP Application
- Mission: Develop a Side-by-Side CAP-Based Extension Application
- Mission: Implement Observability in a Full-Stack CAP Application
- Mission: Develop an Enterprise-Grade CAP Application
- Mission: Develop a Multitenant CAP Application

Starter Mission: Develop a Full-Stack CAP Application

The starter mission is called **Develop a Full-Stack CAP Application Following the SAP BTP Developer's Guide** and contains tutorials grouped in the following way:

- The Develop a Full-Stack CAP Application group of tutorials covers the following steps for developing the Incident Management sample application:

    - Setting up the development environment.
    - Building a CAP application with SAP Fiori elements user interface and a custom logic.
    - Adding local launchpad, authorization, tests for local development and preparing for production.
- The Deploy a Full-Stack CAP Application in SAP BTP, Cloud Foundry Runtime group of tutorials covers the following steps for deploying the Incident Management application in the SAP BTP, Cloud Foundry runtime:
    - Deploying the application in a productive account in SAP BTP, Cloud Foundry runtime.
    - Using SAP Build Work Zone, standard edition.
    - Setting up continuous integration and delivery pipeline in SAP BTP.
- The Deploy a Full-Stack CAP Application in SAP BTP, Kyma Runtime group of tutorials covers the following steps for deploying the Incident Management application in the SAP BTP, Kyma runtime:
    - Deploying the application in a productive account in SAP BTP, Kyma runtime.
    - Using SAP Build Work Zone, standard edition.
    - Setting up continuous integration and delivery pipeline in SAP BTP.

See the Develop a Full-Stack CAP Application Following the SAP BTP Developer's Guide mission at SAP Discovery Center.

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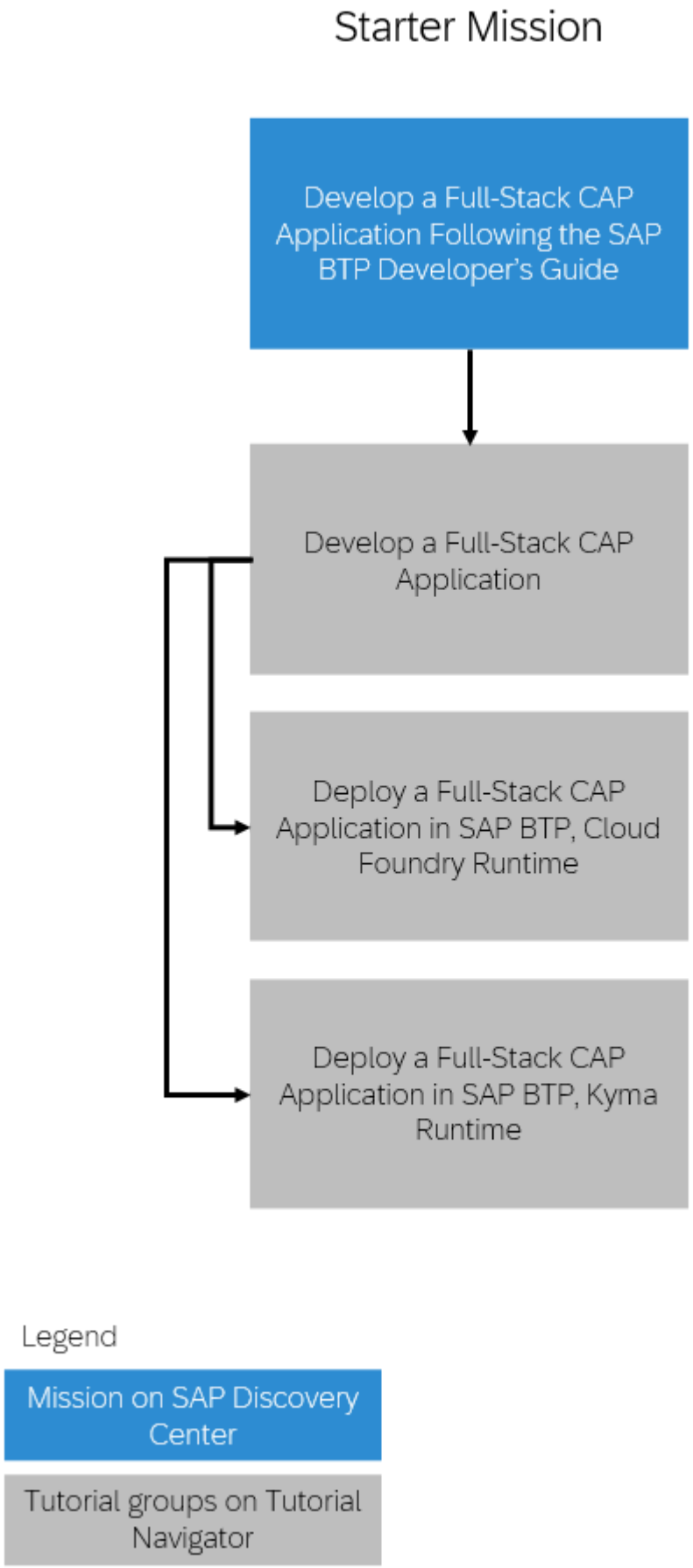

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Mission: Develop a Side-by-Side CAP-Based Extension Application

Using the SAP BTP Developer's Guide you can implement side-by-side SAP Cloud Application Programming Model (CAP)-based extension applications and deploy them in SAP BTP. The **Develop a Side-by-Side CAP-Based Extension Application Following the SAP BTP Developer's Guide** mission includes tracks for a remote service connectivity and eventing. In the scenario, an SAP S/4HANA Cloud system is used. If you don't have such a system and you still want to try out the mission, you can use a mock server instead. Depending on what you choose, there are 8 different options:

- Connect to a remote service
  - Option 1a: SAP BTP, Cloud Foundry runtime and mock server
  - Option 2a: SAP BTP, Cloud Foundry runtime and SAP S/4HANA Cloud
  - Option 3a: SAP BTP, Kyma runtime and mock server
  - Option 4a: SAP BTP, Kyma runtime and SAP S/4HANA Cloud
- Implement eventing
  - Option 1b: SAP BTP, Cloud Foundry runtime and mock server
  - Option 2b: SAP BTP, Cloud Foundry runtime and SAP S/4HANA Cloud
  - Option 3b: SAP BTP, Kyma runtime and mock server
  - Option 4b: SAP BTP, Kyma runtime and SAP S/4HANA Cloud

Out of these 8 options, you can choose one option for the remote service and the corresponding option for the eventing. If you are interested in the remote service connectivity, you don't have to finish the eventing option as well. However, if you are interested in the eventing part, you have to finish the respective remote service connectivity option.

See Develop a Side-by-Side CAP-Based Extension Application Following the SAP BTP Developer's Guide.

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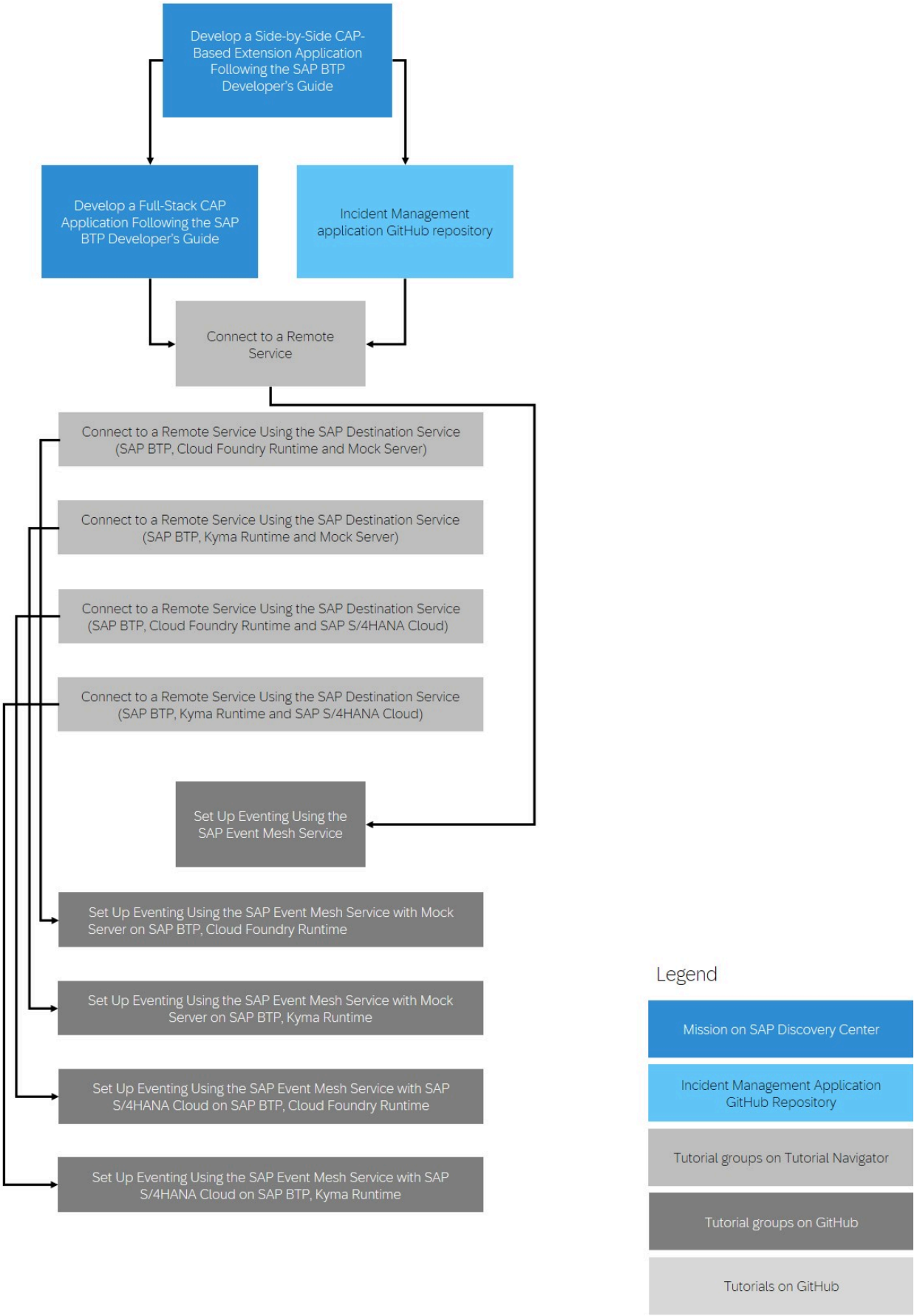
Connecting to a Remote Service Based on the Starter Mission



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Mission: Implement Observability in a Full-Stack CAP Application

You can add observability to the Incident Management application. Observability helps make informed decisions using real-time data like logs, metrics, traces, and events. It provides insights into SAP and customer systems to improve efficiency, scalability, resilience, and availability.

For detailed observability data, use SAP Cloud Logging. SAP Cloud Logging allows you to analyze logs, metrics, and traces to understand the performance of your SAP BTP workload in the SAP BTP, Cloud Foundry, and SAP BTP, Kyma environments.

See Implement Observability in a Full-Stack CAP Application Following SAP BTP Developer's Guide.

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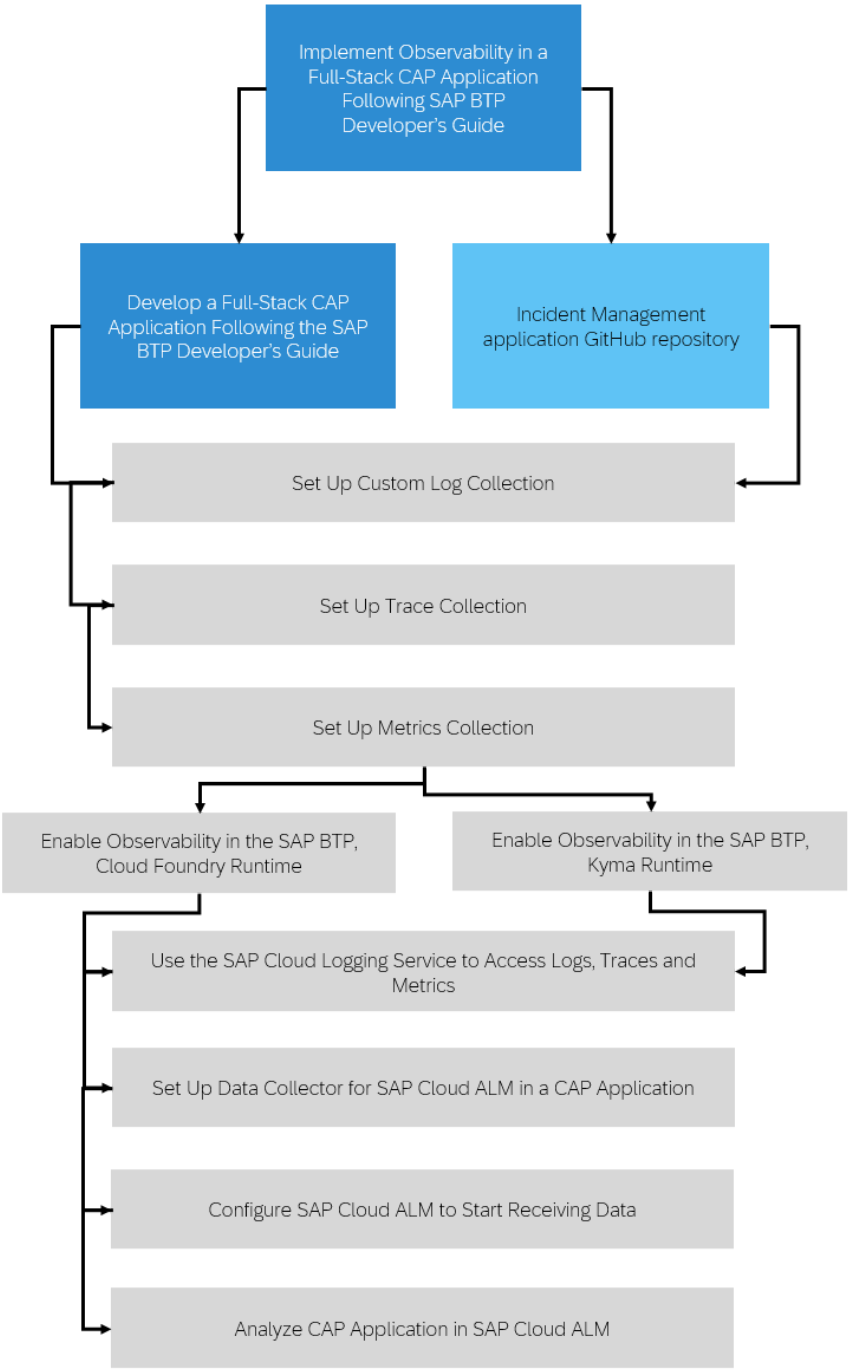
Implement Observability Based on the Starter Mission



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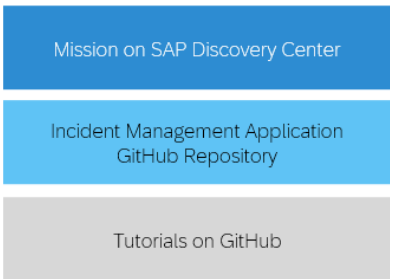

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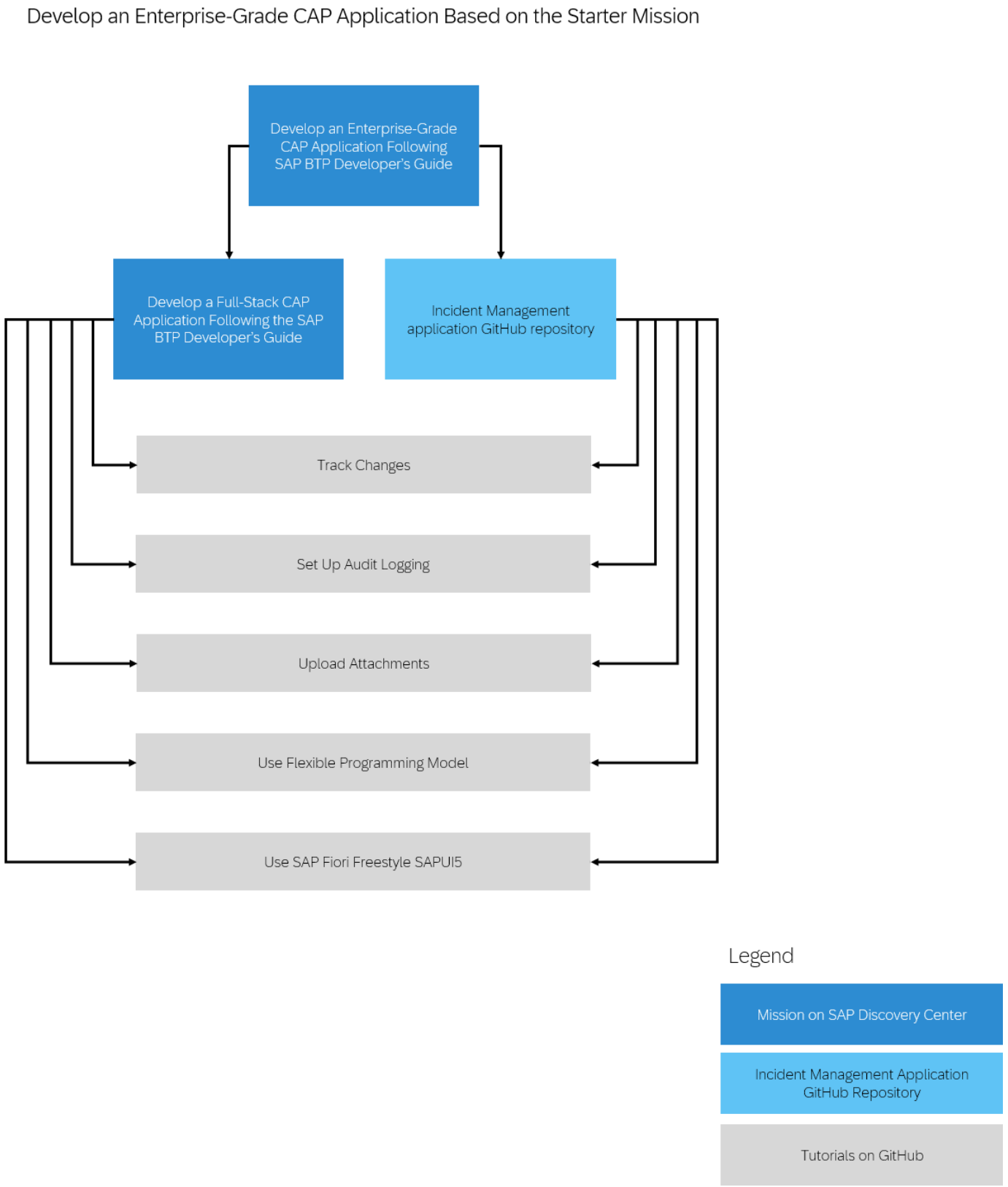
Mission: Develop an Enterprise-Grade CAP Application

You can develop enterprise-grade CAP applications and deploy them in SAP BTP. The **Develop an Enterprise-Grade CAP Application Following the SAP BTP Developer's Guide** mission includes options for tracking changes, adding audit logs, and uploading attachments.

See [Develop an Enterprise-Grade CAP Application Following the SAP BTP Developer's Guide].

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.



Please note that image maps are not interactive in PDF outputs.

Mission: Develop a Multitenant CAP Application

SAP BTP provides a multitenant functionality that allows application providers to own, deploy, and operate tenant-aware applications for multiple consumers, with reduced costs. For example, the application provider upgrades the application for all your consumers instead of performing each update individually, or share resources across multiple consumers. The application consumers launch the applications using consumer-specific URLs, and configure certain application features.

With tenant-aware applications, you can:

- Separate data securely for each tenant
- Save resources by sharing them among tenants
- Update applications efficiently, in a single step

A business solution is typically a multitenant application that consists of UI modules, business logic and a connection to a database. Unlike other applications that are integrated into SAP Build Work Zone, it also contains the site design-time definitions such as tiles, roles, spaces, and pages, and the site entity that wraps in all together. You can develop business solutions to easily share them with consumers. There are two types of business solutions you can choose from: a central or local entry point. These are the main differences between the central and local entry point:

Key Differences Between the Approaches Supporting Multitenancy in SAP Build Work Zone

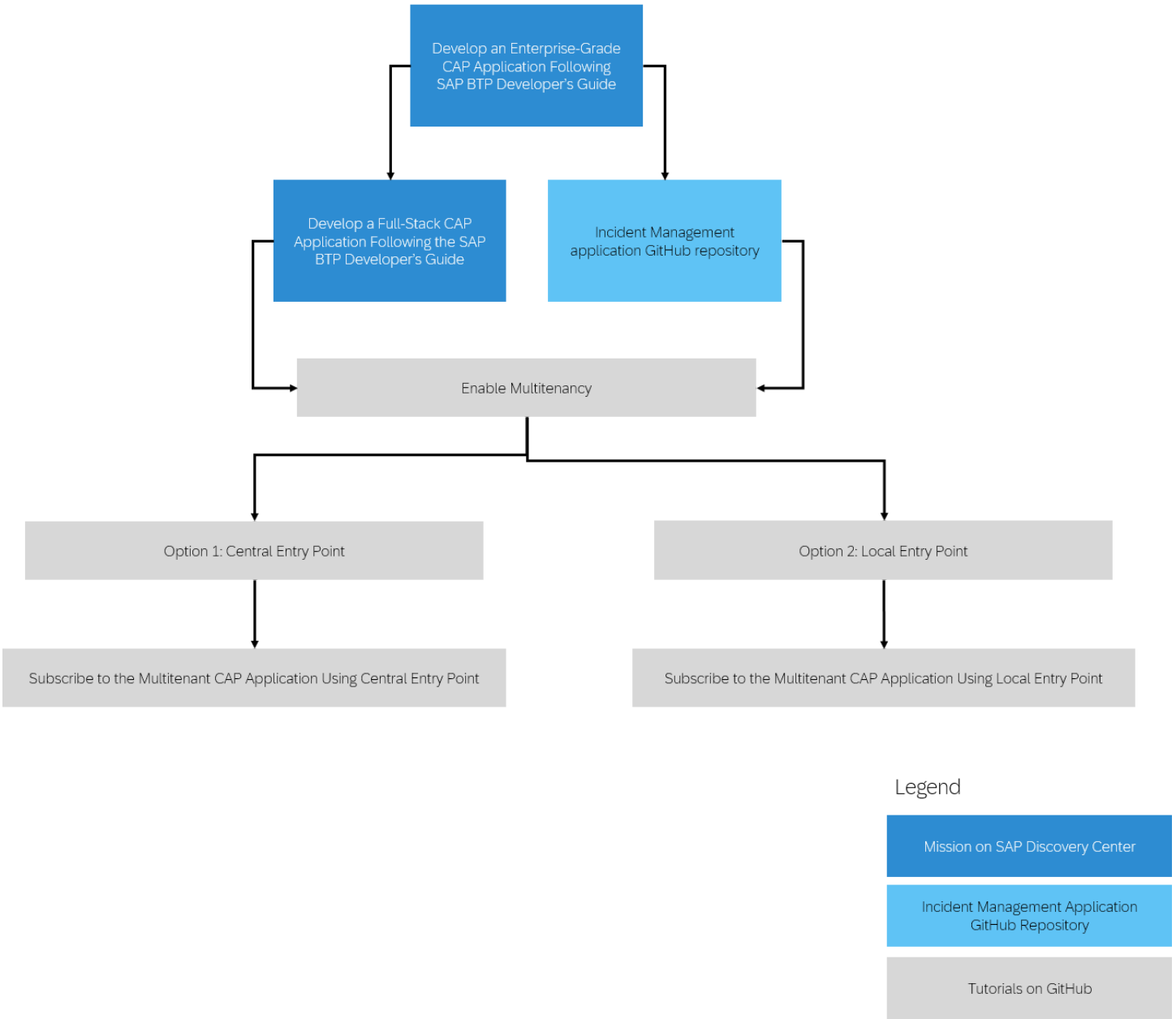
Feature	Central Entry Point	Local Entry Point
Consumer-side SAP Build Work Zone service instance	Required: standard or advanced service plan	Not required
App router management	Managed app router: you get it out of the box	Standalone app router: extensible
Site administration	Administered by the consumer: full SAP Build Work Zone service instance in the consumer subaccount	Provider-managed: not administered by the consumer
Layout customization	Customizable by the consumers: the administrator defines the spaces/pages with federated business application	Layout is fixed by the provider, groups are not supported

You can find a detailed comparison between central and local entry point at Creating Business Solutions.

To try out the multitenant functionality on the Incident Management sample application, see Develop a Multitenant CAP Application Following the SAP BTP Developer's Guide.

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.

Develop a Multitenant CAP Application Based on the Starter Mission



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Developing with ABAP Cloud in the SAP BTP ABAP Environment

Overview

ABAP Cloud is SAP's cloud-ready development model, used in the SAP BTP ABAP environment to build upgrade-stable extensions, services, and applications using released APIs and a restricted, cloud-safe ABAP language version. These are the main elements of the SAP BTP ABAP environment:

- Core Data Services (CDS) for data modeling and embedded analytics
- ABAP RESTful Application Programming Model (RAP) for service-based applications
- Cloud-ready ABAP language version (restricted to released statements and APIs)
- Released public SAP APIs and extension points for upgrade-stable extensibility and automation
- ABAP Development Tools for Eclipse as the development environment

- Reuse services and libraries, including number ranges, application jobs, the ABAP-integrated SAP Fiori launchpad, and a UI repository for SAPUI5 and SAP Fiori elements UIs

Extensibility Options

These are the extensibility options on SAP BTP:

- Low-Code / No-Code: SAP Build Apps, SAP Process Automation, SAP Build Work Zone
- Pro-Code: SAP Build Code (CAP for JavaScript/TypeScript) and SAP BTP ABAP Environment (ABAP Cloud development)

SAP BTP Developer's Guide focuses on the pro-code option on SAP BTP for building robust, cloud-ready extensions using ABAP Cloud. See Develop.

Integration in SAP Build

With SAP BTP ABAP environment integrated into SAP Build, you as a developer can create, access, and monitor ABAP Cloud projects directly from the SAP Build lobby, alongside low-code and CAP projects. This integration fosters fusion development, bridging ABAP and non-ABAP worlds and allows seamless collaboration across technologies within a single extensibility suite.

The SAP BTP ABAP environment brings the general advantages of the side-by-side approach:

- Independent lifecycle and workload management
- Separate user management and authorizations
- Integration across multiple backend systems
- Partner options to offer multitenant SaaS solutions or add-on product

Related Information

Strengths and Benefits of the Runtimes and Their Programming Models
Design
Develop
Deploy
Build and Run ABAP Applications for Partners Who Are Independent Software Vendors
Tutorials for ABAP Cloud

Design

In ABAP you decide and prepare during the design phase for each ABAP Cloud use case: transactional, analytical, and integration. ABAP Cloud follows a model-driven architecture that standardizes the programming model and tooling to deliver efficiency, scalability, and consistency from the database layer to business service exposure.

ABAP Cloud Design Principles

ABAP Cloud builds on the strengths of powerful frameworks and a standardized architecture for different use cases, aiming at saving as much implementation time as possible while providing you with flexibility - you can model your business processes with apps and services based on your business requirements along predefined technical processes. As much as possible is handled by the frameworks to decrease the probability of consistency errors during runtime. This standardized and consistent architecture across all apps and services developed with ABAP Cloud has many advantages from development perspective:

- Model-driven and standardized: A common architecture and powerful frameworks cover routine technical tasks, reducing boilerplate and runtime inconsistencies while giving you flexibility to shape business processes.
- Productivity and scale: Once you learn the ABAP Cloud development model, patterns and artifacts are reused across apps and services, decreasing effort for every additional implementation.
- Maintainability and testability: Consistent structure improves code quality. Built-in mock frameworks for data models and events support automated tests and help prevent regressions at all test levels.
- High abstraction: Low-level infrastructure concerns are handled by the frameworks. Cloud-optimized ABAP languages such as the Data Definition Language (DDL) and Entity Manipulation Language (EML) align closely with data modeling and the ABAP development process.

Typical Development Use Case Patterns

We recommend implementing the following use case patterns on SAP BTP:

- Customer-focused applications and extensions (loosely coupled)
  - Decoupled lifecycle from the core system
  - Separate user groups without SAP S/4HANA Cloud access
  - Independent workload management
  - Hub scenarios consolidating data from multiple backend systems for cloud use
- Partner solutions
  - Create and operate multitenant SaaS applications on SAP BTP
  - Develop Add-On Products that customers can install directly in their own SAP BTP ABAP environment
- Other use cases
  - Develop mobile applications or business process automation that connect multiple systems
  - Build extensions requiring scalability and elasticity independent of the SAP S/4HANA Cloud system
  - Analyze ABAP code using the ABAP Test Cockpit (ATC) with integrated SAP Code Vulnerability Analyzer (CVA) checks for static quality assurance and to evaluate the impact of custom code during the transition to a clean core

Benefits of the Model-Driven Approach

A standardized architecture approach has the following benefits:

- Faster delivery and easier reuse

  A standardized stack scales across many apps and services. Clear separation of concerns between data model behavior and service exposure makes the same domain model usable for transactional and analytical scenarios.

- Interoperability by design

  Implementations follow the same technical rules, contracts, and process flows. You can design end-to-end processes without reinventing authorization, locking, or consistency mechanisms for each step.

Basic Design-Time Architecture

Get to know the basics of the design-time architecture:

- Three-tier structure

  All programming aspects build on a three-tier architecture covering data access, domain model and implementation, and service exposure. This design is derived from the transactional aspect and applied consistently across use cases.

- Multipurpose Core Data Service (CDS) models

  The same CDS-based domain model can serve multiple purposes. With aspect-specific annotations, you can expose the model as both an OData service (transactional) and an Information Access (InA) service (analytical).

Related Information

ABAP Cloud: Background Concepts and Overview: Design

Develop

There are certain requirements and key considerations for the develop phase across the main programming model aspects you have to take into account: building new applications and services, integrating with external systems, extending existing solutions, localizing content, and documenting outcomes.

Develop Applications, Extensions and Services with ABAP Cloud

You can build new applications and services from scratch or extend existing SAP or custom services in an upgrade-safe way. To create a new application or service, implement the domain-specific model and business logic that fit your selected use case. Domain-specific data models based on the RAP architecture blueprint bring extensibility as a built-in quality. Extensibility follows an opt-in approach: the original data model or service must explicitly enable each extensibility option.

Develop an Application from Scratch

Once you select the use case, transactional, analytical, or integration, implement the domain model and logic step by step:

1. Model the domain: Define the domain-specific data model and its relationships. Align with the RAP architecture blueprint to enable upgrade-safe extensibility and consistent semantics.
2. Implement business behavior: Add the domain-specific logic for validations, determinations, actions, and consistency management according to your transactional or analytical use case.
3. Expose services: Define and bind services to make your domain capabilities consumable (for example, via OData). Ensure the service contract reflects your use case and performance needs.
4. Secure access: Configure authorization and access control for services and data. Assign the proper scopes and roles to restrict access according to business and compliance needs.
5. Build the UI: Develop SAP Fiori applications that consume your services. Use metadata-driven annotations to rapidly produce UIs and support navigation, search, and value helps.
6. Ensure quality: Add automated tests and static checks to maintain quality throughout development and delivery.

Developing a Sustainable Test Strategy

Automated software testing is essential to ensure reliability, stability, and quality. It helps detect defects early and prevents regressions, allowing you to add features with confidence that existing functionality remains intact.

- Test levels supported by ABAP Cloud: Unit tests, integration tests, and end-to-end validations to guarantee quality throughout the development process.
- Test-driven development (TDD): ABAP Cloud supports test-driven development, enabling you to validate functional correctness in all stages of development.
- Built-in capabilities: Use ABAP Unit to write executable tests for data models, behavior, and business events; use the ABAP Test Cockpit (ATC) to enforce coding standards and detect quality issues.

Develop SAP S/4HANA Cloud Extension Applications

When developing extensions for SAP S/4HANA Cloud, there are two fundamental extensibility options: on-stack and side-by-side.

On-stack extensibility refers to extensions that run directly within the SAP S/4HANA Cloud system – sharing the same database, system, and lifecycle. In this area:

- Key-User Extensibility (Low-Code): Field extensions or simple custom logic created directly in SAP S/4HANA Cloud using ABAP Cloud and in-app tools.
- Developer Extensibility (Pro-Code): Custom development directly in SAP S/4HANA Cloud using ABAP Cloud, which is cloud-ready and clean core compliant by design – this is technically ensured by the ABAP Cloud development model. In on-premise and private cloud systems, classic ABAP can still be used, but it is not clean core by default and therefore not recommended for new extensions.

Side-by-side extensibility uses SAP BTP where you deploy your extension applications. These extension applications run decoupled from the core system, with a separate runtime, lifecycle, and data center. This approach is ideal for:

- Loosely coupled customer extensions and applications
- Partner solutions, especially multitenant SaaS solutions or add-on product

You usually choose side-by-side development when:

- Extensions need to be decoupled from the SAP S/4HANA Cloud lifecycle
- New or external user groups require access without touching the core system
- Independent workload and scalability are required
- Extensions integrate multiple backend systems, also called hub scenarios
- Partners aim to deliver multitenant SaaS or add-on product solutions

On-stack extensions remain relevant for deeply integrated SAP S/4HANA Cloud scenarios.

Recommendations

We recommend that you:

- Connect your ABAP Cloud system to Eclipse with ABAP Development Tools (ADT).
- Build OData services using RAP optimized for SAP HANA Cloud.
- Perform static code analysis with ABAP Test Cloud, including Code Vulnerability Analyzer (CVA) security checks, without additional licensing fees.
- Use central ABAP Test Cloud on SAP BTP to govern custom code across side-by-side developments, private cloud, and on-premise systems.

- Leverage the Analyze Custom Code application for side-by-side code inspection, custom code migration, or clean core initiatives.

Related Information

ABAP Cloud: Background Concepts and Overview: Develop
ABAP Cloud: Background Concepts and Overview: Test

Set Up the Architecture and the System Landscape

Recommended System Landscapes

When setting up the SAP BTP ABAP environment, it's best to begin with only the systems you currently need — typically the development system (DEV).

You don't have to provision the entire landscape upfront. Additional systems like test (QAS) or production (PRD) can be added later as your development and release processes evolve.

This incremental setup approach provides better transparency, flexibility, and helps optimize costs.

Two standard landscape options are available:

- 3-system landscape (DEV, QAS, PRD): Recommended for most projects, especially when development is occasional or release cycles are less frequent. Enables testing outside the development system and verification of application behavior before go-live.
- 5-system landscape (DEV, COR, TST, QAS, PRD): Suitable for larger teams and continuous development scenarios. Enables parallel correction handling and uninterrupted development activities.

Use system hibernation to reduce costs by stopping development (DEV), correction (COR), and test (TST) systems when they are not in active use. See Use System Hibernation.

See Setting Up and Working with Your Landscape.

Sizing

For productive systems, even a single ABAP Compute Unit (ACU) can serve up to 1,000 active business users per day, showing that minimal configurations are already powerful. The recommended minimum starting point is:

- 1 ABAP Compute Unit (ACU) – 16 GB
- 2 HANA Compute Units (HCU) – 16 GB each

Both ABAP Compute Units and HANA Compute Units can be scaled up or down manually via the SAP BTP cockpit to match changing workload requirements.

With release 2402, SAP BTP ABAP environment introduces automatic runtime scaling, which dynamically adjusts ABAP Compute Units based on actual application server usage.

ℹ Note

This feature requires a consumption-based contract.

Benefit from the Elastic Scaling of ABAP Application Servers

SAP BTP ABAP environment allows dynamic adaptation of ABAP application server capacity to changing workloads. This is achieved by elastic scaling, which automatically adjusts the number of application servers based on system load.

The key points are:

- Manual scaling: ABAP Compute Units (ACUs) and HANA Compute Units (HCUs) can be increased or decreased manually via the SAP BTP Cockpit. ABAP Compute Units scale quickly with minimal downtime, while HCU adjustments are near-zero-downtime but may take longer depending on database size.
- Elastic scaling: When enabled (parameter elastic = true), the system automatically scales between 1 ACU (minimum) and the configured maximum ABAP Compute Units. Scaling is done by adding or removing application servers in increments of 0.5 ABAP Compute Units. Vertical scaling of a single server beyond 0.5 ABAP Compute Unit is not supported.
- Load metrics: Scaling decisions are based on CPU and memory utilization, as well as ABAP-specific metrics like the number of active work processes.
- Cost efficiency: Elastic scaling optimizes resource consumption, reducing costs for periods of low usage while still handling peaks.

See:

- Updating an ABAP System
- Blog Post: SAP BTP ABAP Environment – Elastic Scaling of Application Servers

Use System Hibernation

System hibernation in SAP BTP ABAP environment allows you to temporarily stop and restart a system without deleting the corresponding SAP HANA Cloud instance. This helps reduce costs for systems that are not in use, such as development, test, or analysis systems.

The key points are:

- Cost reduction:
  - Stopped systems do not consume ABAP Compute Units (ACUs) and only minimal HANA Compute Units (HCUs) for storage.
  - For consumption-based contracts, total costs can drop to less than 5%.
- Availability: SAP operations automatically start stopped systems for lifecycle management events like upgrades or hotfix imports.
- Access: Accessing a stopped system via SAP Fiori Launchpad will not start it automatically; the system must be restarted through the Landscape Portal – Manage System Hibernation application.
- Scheduling: You can manually stop/start or schedule system hibernation using the app. Free tier systems are stopped automatically each night, and scheduled start/stop is not supported.

System hibernation is a simple and effective way to save costs while keeping your SAP BTP ABAP environment up to date and ready for productive use.

See:

- Blog Post: SAP BTP ABAP Environment – Manage System Hibernation
- Blog Post: Optimize Your SAP BTP ABAP Environment Budget: A Detailed Cost Analysis for Customers

Maintain and Upgrade

SAP BTP ABAP environment provides an always up-to-date cloud system. Planned maintenance windows and major upgrades are announced in advance, and contractual Service Level Agreements (SLAs) ensure minimal disruption.

Downtime-Optimized Upgrades and Hotfix Collection Imports

The new downtime-optimized upgrade process splits the upgrade in three phases:

- Preparation: The new release is installed in a shadow layer while the system remains available. Certain lifecycle operations, such as transport imports or tenant lifecycle events, may be temporarily blocked.
- Takeover: The system switches to the new release, inducing a short downtime (typically 10–40 minutes for upgrades; about 1 minute for hotfix collection imports). Running ABAP applications are temporarily shut down.
- Postprocessing: Finalization and cleanup occur in the background while the system is fully available. Restrictions on transports and lifecycle operations are lifted.

See:

- Blog Post: SAP BTP ABAP Environment – Pre-Upgrade Option
- Blog Post: SAP BTP ABAP Environment – Maintenance Windows and Major Upgrade Windows

Pre-Upgrade Option

To ensure smooth adoption of new releases, customers and partners can nominate non-development systems for a pre-upgrade prior to the standard upgrade. This allows early regression testing of custom-built applications on the new release. Take into account the following key points:

- Only non-development systems are recommended for pre-upgrade.
- Issues detected during regression tests should be reported promptly using SAP support channels.
- Pre-upgrades do not provide early access to new features; their purpose is to validate existing custom applications.
- The pre-upgrade system receives hotfix collection imports in phases, mirroring the standard upgrade.

Nomination and management of pre-upgrade systems are done via the Landscape Portal, eliminating the need for manual requests. Exceptional processes exist for handling unresolved critical issues, ensuring minimal impact on productive environments.

We recommend to:

- Perform regression testing on pre-upgraded test systems to detect potential issues before the standard upgrade.
- Use the downtime-optimized upgrade process for non-development systems to minimize business impact.
- Monitor upgrade announcements, schedules, and hotfix collection imports to plan testing and validations efficiently.

See Blog Post: SAP BTP ABAP Environment – Pre-Upgrade Option.

Keep Clean Core Governance with ABAP Test Cockpit

Maintaining a clean core in your ABAP custom code is essential for long-term maintainability and upgrade stability. In SAP BTP ABAP environment, you can use the ABAP Test Cockpit (ATC) to govern your side-by-side developments and gradually transform your code to a clean core.

With clean core you can:

- Build upgrade-safe ABAP extensions
- Reduce technical debt when integrating classic ABAP code
- Ensure consistent governance across development branches
- Gain insights into API usage and custom enhancements

The ABAP Test Cockpit (ATC) is set on SAP BTP and provides a central, cloud-based solution to analyze, monitor, and govern ABAP development:

- Central ABAP Test Cockpit (ATC) check system
  - Create a central ABAP Test Cockpit (ATC) system in your SAP BTP ABAP environment.
  - Connect it to all development systems (DEV, QAS) for consistent governance.
- Blocking Mode
  - Configure ABAP Test Cockpit (ATC) blocking mode in DEV systems to enforce compliance for Priority 1 and 2 findings.
  - Ensure that critical issues are resolved before transports reach higher environments.

To support clean core governance in SAP BTP ABAP environment, we recommend including the following checks in your ABAP Test Cockpit (ATC) check variant:

- Allowed SAP enhancement technologies – ensures only approved enhancement approaches are used.
- Usage of APIs – governs usage of SAP classic APIs and prevents use of unauthorized internal objects.
- Critical statements – checks critical ABAP statements that may impact stability or upgrade safety.
- Code Vulnerability Analyzer (CVA) – optional security checks (requires Code Vulnerability Analyzer (CVA) license).

ℹ Note

The default ABAP Test Cockpit (ATC) variant ABAP_CLOUD_DEVELOPMENT_DEFAULT can be copied and extended with the clean core checks.

To analyze custom code in SAP BTP ABAP environment, you have to:

- Use the Analyze Custom Code App in SAP BTP ABAP environment
- Apply your clean core ABAP Test Cockpit (ATC) variant to identify non-compliant objects
- Filter findings by object type, application component, or severity to prioritize remediation
- Transform the classic ABAP to clean core standards step by step

To do that, we recommend following these steps:

1. Set up a central ABAP Test Cockpit (ATC) system in SAP BTP ABAP environment.
2. Create a clean core ABAP Test Cockpit (ATC) variant with the recommended checks
3. Analyze existing custom code using Analyze Custom Code App
4. Apply exemptions or baselines for legacy code
5. Enforce clean core practices for all new side-by-side developments

See Blog Post: ABAP Test Cockpit (ATC) Recommendations for Governance of Clean Core ABAP Development.

Use Joule for Developers: Generative AI in ABAP Cloud

Generative AI in ABAP Cloud supports developers across the end-to-end development lifecycle, improving efficiency and enabling AI-driven enhancements in custom business applications.

Joule for Developers

SAP Joule for Developers provides ABAP AI capabilities integrated into Eclipse ABAP Development Tools:

- Predictive code completion – Generate ABAP code directly in the source editor.
- Joule chat – Natural language interaction to answer development-related questions.
- Explain – Understand development objects or selected code lines in natural language.
- ABAP Unit generation – Create and enhance unit tests automatically.
- Core Data Service (CDS) test generation – Generate ABAP test classes for CDS entities.

These features simplify complex development objects and enable interactive guidance during development.

Use Cases

Generative AI is designed to support you in two main areas:

- Day-to-day development tasks – Increase efficiency and reduce total cost of development by leveraging AI capabilities directly in your ABAP Cloud projects.
- Custom business applications – Integrate generative AI features into your applications using ABAP AI SDK.

ABAP AI SDK

The ABAP AI SDK provides a standardized interface to large language models (LLMs) hosted on SAP AI Core. It allows you to:

- Integrate generative AI features into custom ABAP applications.
- Leverage Intelligent Scenario Lifecycle Management for reusable ABAP AI capabilities.

This enables a seamless approach to building AI-driven extensions in both ABAP Cloud and SAP Cloud ERP environments.

Availability and Licensing

Joule for Developers and ABAP AI capabilities are:

- Free of charge, but require an additional license activation. See SAP Note 3571857.
- Available in SAP BTP ABAP environment, SAP Cloud ERP Public Edition, and upcoming in SAP Cloud ERP Private Edition (2025 release).
- Supported in hybrid landscapes and can be connected to older SAP Cloud ERP Private Edition releases.
- Allowing continuous delivery of new AI innovations independent of release upgrades.

See RAP120 - Build SAP Fiori Apps with ABAP Cloud and Joule for Developers .

Manage Full Transport and Lifecycle

In SAP BTP ABAP environment, ABAP objects can be transported and versioned using gCTS and abapGit, enabling robust lifecycle management and collaborative development.

gCTS is the evolution of the classical Change and Transport System (CTS) and is the recommended approach for transporting objects between ABAP systems within the same global account in SAP BTP. It is fully integrated into the Manage Software Components app in the SAP Fiori launchpad, allowing for controlled transport, versioning, and lifecycle management of software components across development, test, and productive systems.

abapGit is an open-source Git client for ABAP, designed to handle scenarios such as:

- Migrating on-premise ABAP code to the cloud.
- Exporting or importing code when a system is decommissioned.
- Enabling distributed development and testing in isolated development or test systems.
- Transferring code between cloud systems, for example from a partner account to a customer account.

abapGit is ideal for cases where lightweight or distributed transport is needed, but it is not recommended as a replacement for standard transport workflows between production-aligned systems.

We recommend that you use:

- gCTS for all transports between development, test, and productive systems within the same global account in SAP BTP.
- abapGit for migration, backup, or sharing code between independent systems or accounts.
- Products in the Landscape Portal as the delivery unit for partner or multi-account scenarios.

gCTS and abapGit help maintain code quality, track changes, and enable scalable development and testing practices across ABAP landscapes.

See Transport Mechanisms.

Administrate and Operate

Effective administration and operations are essential to ensure stable, efficient, and cost-optimized ABAP system performance in the cloud.

Administrators are responsible for managing users, roles, and subaccounts, as well as monitoring performance, optimizing resources, and maintaining integrations with connected systems.

Lifecycle management includes transports, add-on updates, and tenant administration for multi-tenant applications.

Key tools such as SAP Fiori Launchpad, Landscape Portal, and the Technical Monitoring Cockpit support daily operations, while SAP Cloud ALM provides health monitoring, alerts, and proactive notifications.

Additionally, ABAP Test Cockpit (ATC) helps maintain code quality and security standards across developments.

By applying these practices, administrators can ensure system health, efficient resource utilization, and smooth operation of custom applications in a cloud-centric ABAP landscape.

As an ABAP environment administrator, your role extends beyond traditional basic tasks. Core responsibilities include:

- User and account management: Handle identity and access management, manage organizations, spaces, quotas, and roles.
- System health monitoring: Track runtime performance, SQL statements, background jobs, response times, and errors.
- System sizing and cost optimization: Monitor resource usage patterns, adjust sizing, and apply hibernation to save costs.
- Lifecycle management: Oversee custom applications, transports, add-on updates, and tenant operations.
- Integration monitoring: Ensure stable data exchange between cloud and on-premise systems.

See:

- SAP Community: How to Monitor and Support Custom Cloud Apps
- Blog Post: Monitoring the Health of the ABAP System in the Cloud

Deploy

After you have developed your application or service and have written the necessary tests, it's time to deploy and transport them to SAP BTP ABAP environment using proven ABAP lifecycle management. CI/CD, git integration, and a dedicated transport environment for key users on the SAP Fiori launchpad are supported out of the box.

Common Setup

The basic deployment unit is the software component, managed via the Manage Software Components administration application. Each software component has an automatically managed git repository via gCTS for transport management. You can import software components into test and production systems.

This common setup is sufficient for customers operating their own applications.

For more information, see:

- Software Components
- Manage Software Components

Additional Considerations for Partners

Partners can offer and deploy their solutions in two different ways within the SAP BTP ABAP environment:

- Multitenant SaaS solution

    - You offer your solution as a **cloud service** operated in your own global accout in SAP BTP
    - Each customer **subscribes** to your service
    - You are responsible for **development, operations, monitoring**, and **infrastructure costs**
    - Customers can adapt the solution using **key-user extensibility** (tenant-aware adaptations such as custom fields and logic)
- Add-on product
    - You offer your solution as an **add-on product**
    - Your customers must have their **own instance of the SAP BTP ABAP environment**
    - The **installation and lifecycle management** (upgrades, patches) are in the **customer's responsibility**
    - The installation is performed by the customer **via the Landscape Portal**, which is part of the SAP BTP ABAP environment
    - Customers can **enhance and extend** your product using **ABAP Development Tools (ADT)** and **developer extensibility**

For both delivery models, partners build their solutions based on **software components** and manage them through the **Landscape Portal**. The Landscape Portal provides a unified environment to:

- Build, register, and publish partner products,
- Manage software lifecycles such as installation, upgrades, and patches, and
- Support the setup of **multitenant SaaS** solutions or **add-on products** for customers and other partners.

For more information about managing partner solutions, see Landscape Portal.

Related Information

ABAP Cloud: Background Concepts and Overview: Deploy

Build and Run ABAP Applications for Partners Who Are Independent Software Vendors

Partner Deployment Options

As a partner, you can choose between multitenant SaaS and add-on product options for deploying your application. See the dedicated section at SAP BTP ABAP Environment with ABAP Cloud.

Landscape Portal for Partners

The Landscape Portal is the central tool for partners to manage, deploy, and operate their solutions in SAP BTP ABAP environment. It supports both multitenant SaaS applications and add-on deployment options and provides all the necessary features for lifecycle management and product delivery.

Key functions and considerations:

- Product building and deployment:

  - To offer a solution in another global account in SAP BTP, whether it's a different account you own (for example, DEV vs. PROD) or a customer account—a Product must be built using the Landscape Portal.
  - The Product serves as the deployment unit and allows controlled installation and lifecycle management.
  - Familiarity with the Landscape Portal features is essential for managing all deployment steps, including smaller intermediate steps that may be required for registration, building, and publishing.
- Lifecycle management:
  - abapGit is only recommended for importing development objects from on-premise systems to BTP or for backup purposes. It is not supported for transporting objects between BTP systems.
  - For system-to-system transports within BTP (for example, DEV → TEST), use the Manage Software Components app, which relies on gCTS.
  - Lifecycle management and tenant strategy (multitenant vs. single tenant) are independent decisions that must be defined separately.
- Technical requirements and dependencies:
  - At least one Software Component is required as a deployment unit for any solution.
  - A registered ABAP namespace is mandatory for all objects in a Product; using Y* or Z* objects is only allowed if the solution remains within the same global account in SAP BTP and only the Manage Software Components app is used.
  - Multiple global accounts in SAP BTP require building an add-on product to ensure proper deployment.

By consolidating all partner-related operations in the Landscape Portal, partners can efficiently manage solution delivery, maintain lifecycle governance, and ensure smooth deployment across multiple accounts and tenants.

Related Information

SAP BTP ABAP Environment with ABAP Cloud
Partner Reference Application 'Music Festival Manager'
SAP Community: Building Partner Products on SAP BTP ABAP Environment

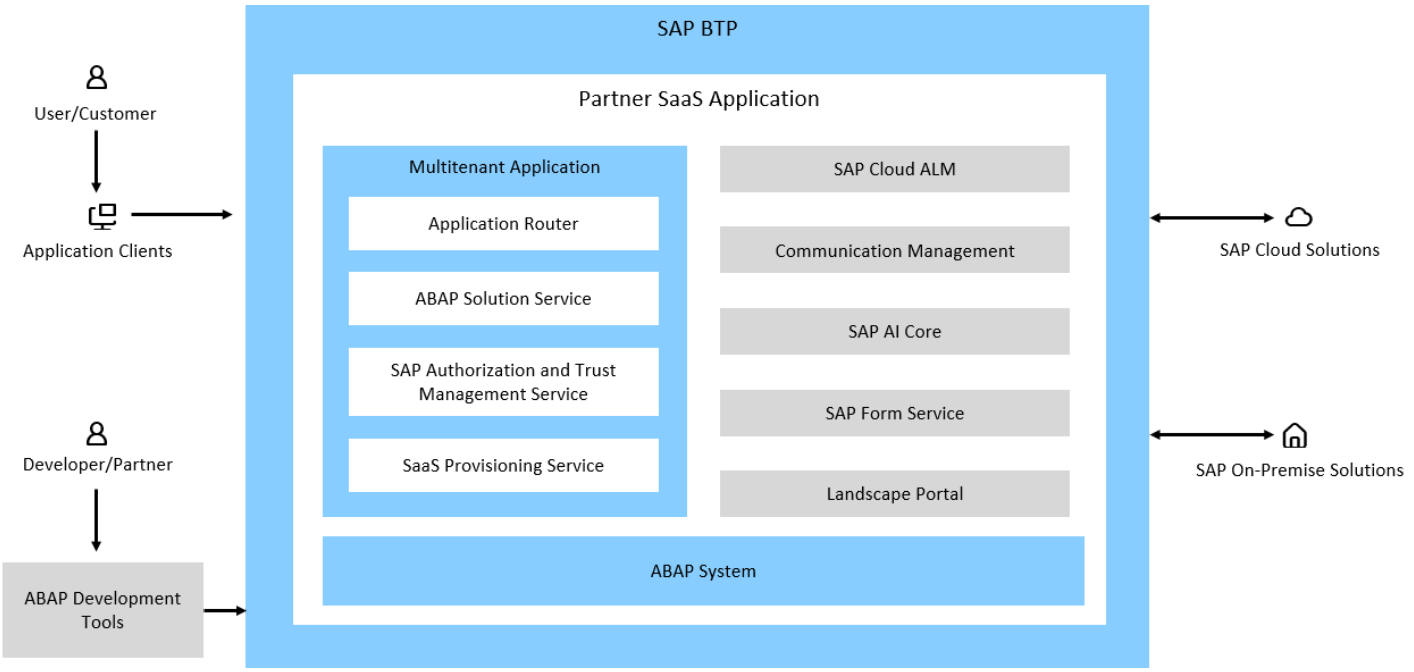
Tutorials and Reference Sample for Full-Stack ABAP Cloud Multitenant SaaS Applications for Partners

Full-Stack Multitenant SaaS Applications for Partners

The Music Festival Manager application is a reference application for SAP partners that gives you a recommended way to become a Software-as-a-Service (SaaS) provider of multitenant applications based on SAP BTP. See Partner Reference Application 'Music Festival Manager' on GitHub.

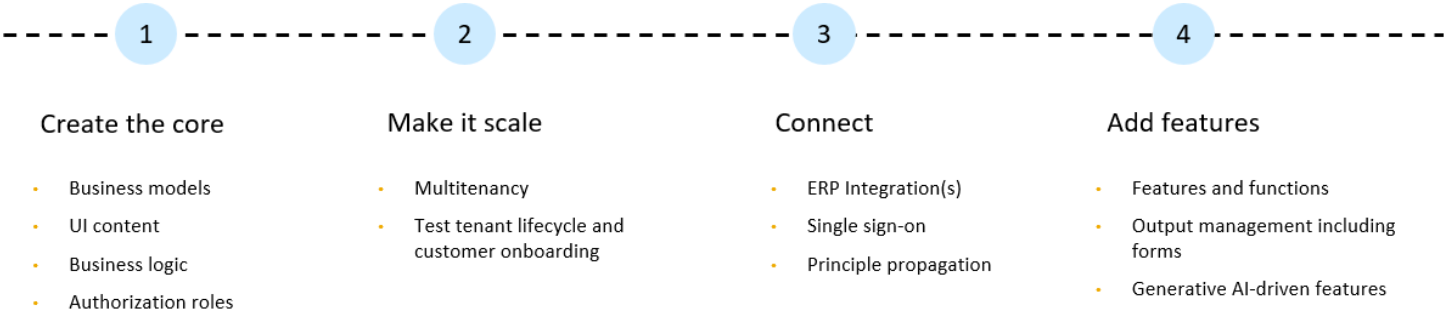
The reference application covers building, running, and integrating scalable full-stack ABAP Cloud applications using the SAP ABAP RESTful Application Programming Model (RAP) and adheres to the development best practices set out in the SAP BTP Developer's Guide. It includes an ERP-agnostic design that lets you deliver your application as a side-by-side extension to consumers using any SAP solution, such as SAP S/4HANA Cloud ERP.

By using services in SAP BTP and the ABAP RESTful Application Programming Model (RAP), your application meets SAP standards for enterprise-class business solutions and empowers you to focus on your business logic and application domain.



The tutorials in the repository follow an incremental development approach, guiding users through the development journey. They cover:

1. Developing the core application with a focus on business models, logic and UI, authentication and role-based authorization.
2. Building, deploying, and provisioning your application to consumers.
3. Integrating the application with SAP S/4HANA Cloud
4. Adding more features to the application.



Tutorials for ABAP Cloud

The ABAP Cloud best practices are documented in the form of step-by-step tutorials for learning how to develop and deploy full-stack transactional and analytical applications using the ABAP RESTful Application Programming Model (RAP) and embedded analytics.

The exercises are based on the Travel sample application of the Flight Reference Scenario that is the best practice example for developing, deploying, and operating a full-stack application using ABAP RESTful Application Programming Model (RAP) and SAP Fiori.

You can also go through the Develop a Full-Stack RAP Application Following the SAP BTP Developer's Guide that is based on the Travel sample application. This mission has three main parts. The first part contains the RAP100 basics, for building an SAP Fiori application with RAP. The second part is for creating, developing, and integrating an SAP Fiori application. The third part builds on top of RAP100 basics and contains the intermediate tutorials, with additions such as dynamic feature control and ABAP unit tests.

There are additional learning resources and certifications you can use:

- SAP Learning Journey: Practicing Clean Core Extensibility for SAP S/4HANA Cloud
- SAP Learning Journey: Acquiring Core ABAP Skills
- SAP Certification: SAP Certified Associate - Back-End Developer - ABAP Cloud

Analytical Scenario

The analytical scenario shows the end-to-end development of an analytical application. It consists of a business service exposing an analytical query and a report built with SAP Analytics Cloud. The analytical report can be used to analyze complex situations and freely interact with large amounts of data in real time.

The analytical scenario is provided in the form of tutorials with beginner and intermediate experience levels:

- Beginner: Develop and Consume Queries on SAP Analytics Cloud
- Intermediate: Develop Queries Based on Booking Supplement and Consume on SAP Analytics Cloud

Transactional Scenario

The transactional scenario shows the end-to-end development of a transactional SAP Fiori elements list report application. It consists of an OData-based business service built with RAP and an SAP Fiori elements-based UI. It also shows the deployment of the SAP Fiori application in the ABAP environment and its integration in the local SAP Fiori launchpad for SAP BTP ABAP environment.

You will develop a simple, transactional list report application, the Travel Processing application, that is used by a department responsible for processing worldwide travel for multiple agencies.

The transactional scenario built with RAP is provided as a tutorial mission, which is divided into three tutorial groups with beginner and intermediate experience levels.

The following beginner tutorial groups are available for the transactional scenario:

- Build an SAP Fiori App Using the ABAP RESTful Application Programming Model (RAP) – Basics [RAP100]
    - Get to Know the ABAP RESTful Application Programming Model
    - Create Database Table and Generate UI Service
    - Enhance the Business Object Data Model and Enable OData Streams
    - Enhance the Business Object Behavior with Unmanaged Internal Numbering
    - Enhance the Business Object Behavior with Determinations
    - Enhance the Business Object Behavior with Validations
- Create, Deploy and Integrate an SAP Fiori Elements App into SAP Fiori Launchpad for SAP BTP ABAP Environment [RAP100]
    - Create and Deploy an SAP Fiori Elements App to SAP BTP, ABAP Environment
    - Integrate an SAP Fiori Elements App into the SAP Fiori Launchpad for SAP BTP, ABAP Environment

The advanced exercises of the transactional scenario are provided as a tutorial group. They are built on top of the exercises of the beginner tutorial groups.

Build an SAP Fiori App Using the ABAP RESTful Application Programming Model (RAP) – Intermediate [RAP100]:

- Enhance the Business Object Behavior with Instance Actions
- Enhance the Business Object Behavior with Factory Action
- Enhance the Business Object Behavior with Dynamic Feature Control
- Write an ABAP Unit Test for the RAP Business Object

AI Scenario

The AI scenario includes the ABAP Cloud Generator and ABAP capabilities in SAP Joule for Developers, such as predictive code completion, unit test generation for ABAP Core Data Services (CDS) and ABAP code, and code explanation to speed up your development with ABAP Cloud

See RAP120 - Build SAP Fiori Apps with ABAP Cloud and Joule for Developers .